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(54) **APYRASE INHIBITORS**

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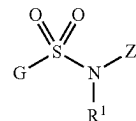
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(57)

ABSTRACT

Provided are methods of using molecules as described herein in supporting crop viability and yield, such as by protecting crops from pests. In some cases, the methods inhibit apyrase enzymes by contacting such enzymes with a compound of the formula



wherein G, Z, and R¹ are as described herein.

In further embodiments, an apyrase inhibitor as described herein is used in combination with one or more pesticide to treat a crop at risk of disease.

APYRASE INHIBITORS

CROSS REFERENCE

[0001] This application claims the benefit of the earlier filing date of U.S. Provisional Application No. 63/559,786, filed Feb. 29, 2024, the disclosure of which is incorporated herein by reference.

FIELD

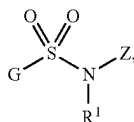
[0002] The present disclosure relates to inhibitors of apyrase and methods for their use, in particular in the treatment of crops susceptible to pathogens.

INTRODUCTION

[0003] Crops are plagued worldwide by a variety of pests. These pests, such as insects, mites, nematodes, weeds and fungi, have developed an array of mechanisms for surviving pesticides, such as by sequestering, exporting or detoxifying them. The present inventors have discovered molecules and methods for potentiating the efficacy of pesticides by blocking certain mechanisms of resistance.

SUMMARY

[0004] Provided are methods of using molecules as described herein in supporting crop viability and yield, such as by protecting crops from pests. In some cases, the methods inhibit apyrase enzymes by contacting such enzymes with a compound of the formula



wherein G, Z, and R¹ are as described herein.

[0005] In further embodiments, an apyrase inhibitor as described herein is used in combination with one or more pesticide to treat a crop at risk of disease.

DETAILED DESCRIPTION

[0006] Before the present invention is described in greater detail, it is to be understood that this invention is not limited to particular embodiments described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

[0007] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Each smaller range between any stated value or intervening value in a stated range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included or excluded in the range, and each range where either, neither or both limits are included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the

stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0008] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, some potential and exemplary methods and materials may now be described. Any and all publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. It is understood that the present disclosure supersedes any disclosure of an incorporated publication to the extent there is a contradiction.

[0009] It must be noted that as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any element, e.g., any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely”, “only” and the like in connection with the recitation of claim elements, or the use of a “negative” limitation.

[0010] The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed. To the extent the definition or usage of any term herein conflicts with a definition or usage of a term in an application or reference incorporated by reference herein, the instant application shall control.

[0011] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

Definitions

[0012] “Administering” refers to any suitable mode of administration, to control a fungal pathogen, including, treatment of an extant crop, seeds, soil or combination thereof.

[0013] “Control” with reference to a fungal pathogen, means block, inhibit and/or eradicate a fungal pathogen and/or prevent the fungal pathogen from damaging a crop. In one embodiment, control refers to the reduction of one or more fungi to undetectable levels, or to the reduction or suppression of a fungus to acceptable levels as determined by one of ordinary skill in the art (for example, a crop grower). Determinations of acceptable levels of fungus reduction are based on a number of factors, including to the crop, pathogen, severity of the pathogen, use restrictions, economic thresholds and other factors known to those of ordinary skill in the art.

[0014] As used herein, the terms “enhancer” and “potentiator”, refer to a compound or compounds disclosed herein that enhance the effects of a pesticide. Without limitation to

theory the present enhancer compounds disclosed herein may function by blocking one or more pathways by which a pathogen, such as a fungal pathogen, evades toxicity, such as by detoxifying, sequestering or transporting a pesticide. In certain embodiment, the present compounds inhibit enzymatic apyrase activity which leads to the enhancement, accentuation or potentiation of a pesticide, such as an acaricide, antimicrobial, fungicide, herbicide, insecticide, molluscicide and/or nematocide. For example, when the enhancer or potentiator is used in conjunction with a fungicide, the combination of the potentiator and the fungicide enhances the fungicidal effect of the fungicide and/or renders a fungus that has become resistant to the fungicide susceptible to the fungicide as a result of the activity of the potentiator. Most often, these enhancers or potentiators do not themselves inhibit the growth of a fungus itself, nor do they have a detrimental effect on a living organism that is (or could be) infected with a fungus.

[0015] As used herein, the term “inoculation” refers to a method used to administer or apply an effective amount of a disclosed compound or formulation thereof to a target area of a field and/or plant. The inoculation method can be, but is not limited to, aerosol spray, pressure spray, direct watering, and dipping. Target areas of a plant could include, but are not limited to, the leaves, roots, stems, buds, flowers, fruit, seed of the plant, and bulbs of the plant including bulb, corm, rhizoma, stem tuber, root tuber and rhizophore. Inoculation can include a method wherein a plant is treated in one area (for example, the root zone or foliage) and another area of the plant becomes protected (for example, foliage is inoculated when a disclosed compound is applied in the root zone or new growth when applied to foliage).

[0016] As used herein, the terms “wetable granule”, “water dispersible granule”, and “dispersible granule” refer to a solid granular formulation prepared by a granulation process, optionally containing fine particles of polymer-associated active ingredient, or aggregates of the same, a wetting agent and/or a dispersant, and optionally an inert filler. Wettable granules can be stored as a formulation, and can be provided to the market and/or end user without further processing. In some embodiments, they can be placed in a water-soluble bag for ease of use by the end user. In practical application, wettable granules are prepared for application by the end user. The wettable granules are mixed with water in the end user’s spray tank to the proper dilution for the particular application. Dilution can vary by crop, target pathogen, time of year, geography, local regulations, and intensity of infection or pathogen load among other factors. Once properly diluted, the solution can be applied by spraying.

[0017] As used herein, the terms “wetable powder”, “water dispersible powder”, and “dispersible powder”, refer to a solid powdered formulation that contains active ingredient, optionally associated with a polymer, or aggregates of the same, and optionally one or more of a dispersant, a wetting agent, and an inert filler. Wettable powders can be stored as a formulation, and can be provided to the market and/or end user without further processing. In some embodiments, they can be placed in a water-soluble bag for ease of use by the end user. In practical application, a wettable powder is prepared for application by the end user. The wettable powder is mixed with water in the end user’s spray tank to the proper dilution for the particular application. Dilution can vary by crop, fungal pathogen, time of year,

geography, local regulations, and intensity of fungal load, among other factors. Once properly diluted, the solution can be applied by spraying.

[0018] As used herein, the term “high solids liquid suspension” refers to a liquid formulation that contains fine particles of active ingredient, or fine polymer particles associated with active ingredient, or aggregates of the same, a wetting agent and/or a dispersant, an anti-freezing agent, optionally an anti-settling agent or thickener, optionally a preservative, and water or oil as a carrier. High solids liquid suspensions can be stored as a formulation, and can be provided to the market and/or end user without further processing. In practical application, high solids liquid suspensions are prepared for application by the end user. The high solids liquid suspensions are mixed with water or oil in the end user’s spray tank to the proper dilution for the particular application. Dilution can vary by crop, fungal pathogen, time of year, geography, local regulations, and intensity of infection among other factors. Once properly diluted, the solution can be applied by spraying.

[0019] As used herein, the term “phytologically acceptable” refers to compositions, diluents, excipients, and/or carriers that are generally applicable for use with any part of a plant during any part of its life cycle, including but not limited to seeds, seedlings, plant cells, plants, or flowers. The compositions can be prepared according to procedures, methods and formulas that are known to those of skill in the agricultural arts. Following the teachings of the present disclosure the artist skilled in the agricultural and/or chemical arts can readily prepare a desired composition. Most commonly, the compounds disclosed herein can be formulated to be stored, and/or applied, as aqueous or non-aqueous suspensions or emulsions prepared neat or from concentrated formulations of the compositions. Alternatively the compounds disclosed herein can be formulated for use in aerosol-generating equipment for application to agricultural produce stored in sealed chambers—an application method known as fogging. Water-soluble, water-suspendable or emulsifiable formulations comprising the presently disclosed compounds can also be converted into or formulated as solids (for example, wettable powders), which can then be diluted into a final formulation. In certain formulations, the compositions of the present disclosure can also be provided in growth media, such as in vitro media for growth of plant or other types of cells, in laboratory plant growth media, in soil, or for spraying on seeds, seedlings, roots, stems, stalks, leaves, flowers or the entire plant.

[0020] Compounds herein can include all stereoisomers, including E and Z isomers, enantiomers, diastereomers, mixtures, racemates, atropisomers, and tautomers thereof.

[0021] Non-limiting examples of optional substituents include hydroxyl groups, sulfhydryl groups, halogens, amino groups, nitro groups, nitroso groups, cyano groups, azido groups, sulfoxide groups, sulfone groups, sulfonamide groups, carboxyl groups, carboxaldehyde groups, imine groups, alkyl groups, halo-alkyl groups, alkenyl groups, haloalkenyl groups, alkynyl groups, halo-alkynyl groups, alkoxy groups, aryl groups, aryloxy groups, aralkyl groups, arylalkoxy groups, heterocyclylalkyl groups, heteroaryl groups, cycloalkyl groups, acyl groups, acyloxy groups, carbamate groups, amide groups, ureido groups, epoxy groups, and ester groups.

[0022] “Alkyl” refers to an optionally substituted straight-chain, or optionally substituted branched-chain saturated

hydrocarbon. Non-limiting examples of alkyl groups include straight, branched, and cyclic alkyl and alkylene groups. An alkyl group can be, for example, a C₁, C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, C₁₀, C₁₁, C₁₂, C₁₃, C₁₄, C₁₅, C₁₆, C₁₇, C₁₈, C₁₉, C₂₀, C₂₁, C₂₂, C₂₃, C₂₄, C₂₅, C₂₆, C₂₇, C₂₈, C₂₉, C₃₀, C₃₁, C₃₂, C₃₃, C₃₄, C₃₅, C₃₆, C₃₇, C₃₈, C₃₉, C₄₀, C₄₁, C₄₂, C₄₃, C₄₄, C₄₅, C₄₆, C₄₇, C₄₈, C₄₉, or C₅₀ group that is substituted or unsubstituted. In some cases alkyl refers to a group having from one to about ten carbon atoms, or from one to six carbon atoms, wherein an sp³-hybridized carbon of the alkyl residue is attached to the rest of the molecule by a single bond. Whenever it appears herein, a numerical range such as “C₁₋₆ alkyl” means that the alkyl group consists of 1 carbon atom, 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkyl” where no numerical range is designated. In some embodiments, the alkyl is a C₁₋₁₀ alkyl, a C₁₋₉ alkyl, a C₁₋₈ alkyl, a C₁₋₇ alkyl, a C₁₋₆ alkyl, a C₁₋₅ alkyl, a C₁₋₄ alkyl, a C₁₋₃ alkyl, a C₁₋₂ alkyl, or a C₁₋₈ alkyl.

[0023] Examples of alkyl groups include, but are not limited to, methyl, ethyl, n-propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, n-butyl, isobutyl, sec-butyl, t-butyl, n-pentyl, isopentyl, neopentyl, tert-amyl, and hexyl, and longer alkyl groups, such as heptyl, octyl, and the like.

[0024] Non-limiting examples of straight alkyl groups include methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl, octyl, nonyl, and decyl.

[0025] Branched alkyl groups include any straight alkyl group substituted with any number of alkyl groups. Non-limiting examples of branched alkyl groups include isopropyl, isobutyl, sec-butyl, and t-butyl.

[0026] Unless stated otherwise specifically in the specification, an alkyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, the alkyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, the alkyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, or —OMe. In some embodiments, the alkyl is optionally substituted with halogen. Non-limiting examples of substituted alkyl groups includes hydroxymethyl, chloromethyl, trifluoromethyl, aminomethyl, 1-chloroethyl, 2-hydroxyethyl, 1,2-difluoroethyl, and 3-carboxypropyl.

[0027] “Alkenyl” refers to an optionally substituted straight-chain, or optionally substituted branched-chain hydrocarbon having one or more carbon-carbon double bonds. The olefin or olefins of an alkenyl group can be, for example, E, Z, cis, trans, terminal, or exo-methylene. An alkenyl group can be, for example, a C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, C₁₀, C₁₁, C₁₂, C₁₃, C₁₄, C₁₅, C₁₆, C₁₇, C₁₈, C₁₉, C₂₀, C₂₁, C₂₂, C₂₃, C₂₄, C₂₅, C₂₆, C₂₇, C₂₈, C₂₉, C₃₀, C₃₁, C₃₂, C₃₃, C₃₄, C₃₅, C₃₆, C₃₇, C₃₈, C₃₉, C₄₀, C₄₁, C₄₂, C₄₃, C₄₄, C₄₅, C₄₆, C₄₇, C₄₈, C₄₉, or C₅₀ group that is substituted or unsubstituted. Non-limiting examples of alkenyl and alk- enylene groups include ethenyl, prop-1-en-1-yl, isoprop-1-en-1-yl, but-1-en-4-yl; 2-chloroethenyl, 4-hydroxybuten-1-yl, 7-hydroxy-7-methyloct-4-en-2-yl, and 7-hydroxy-7-methyl- oct-3,5-dien-2-yl.

nyl, but-1-en-4-yl; 2-chloroethenyl, 4-hydroxybuten-1-yl, 7-hydroxy-7-methyloct-4-en-2-yl, and 7-hydroxy-7-methyl- oct-3,5-dien-2-yl.

[0028] Whenever it appears herein, a numerical range such as “C₂-C₆ alkenyl” means that the alkenyl group may consist of 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms, or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkenyl” where no numerical range is designated. In some embodiments, the alkenyl is a C₂-C₁₀ alkenyl, a C₂-C₉ alkenyl, a C₂-C₈ alkenyl, a C₂-C₇ alkenyl, a C₂-C₆ alkenyl, a C₂-C₅ alkenyl, a C₂-C₄ alkenyl, a C₂-C₃ alkenyl, or a C₂ alkenyl. Unless stated otherwise specifically in the specification, an alkenyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkenyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, an alkenyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, or —OMe. In some embodiments, the alkenyl is optionally substituted with halogen.

[0029] “Alkynyl” refers to an optionally substituted straight-chain or optionally substituted branched-chain hydrocarbon. The triple bond of an alkynyl group can be internal or terminal. An alkynyl or alkynylene group can be, for example, a C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, C₁₀, C₁₁, C₁₂, C₁₃, C₁₄, C₁₅, C₁₆, C₁₇, C₁₈, C₁₉, C₂₀, C₂₁, C₂₂, C₂₃, C₂₄, C₂₅, C₂₆, C₂₇, C₂₈, C₂₉, C₃₀, C₃₁, C₃₂, C₃₃, C₃₄, C₃₅, C₃₆, C₃₇, C₃₈, C₃₉, C₄₀, C₄₁, C₄₂, C₄₃, C₄₄, C₄₅, C₄₆, C₄₇, C₄₈, C₄₉, or C₅₀ group that is substituted or unsubstituted. Non-limiting examples of alkynyl groups include ethynyl, prop-2-yn-1-yl, prop-1-yn-1-yl, and 2-methyl-hex-4-yn-1-yl; 5-hydroxy-5-methylhex-3-yn-1-yl, 6-hydroxy-6-methyl- hept-3-yn-2-yl, and 5-hydroxy-5-ethylhept-3-yn-1-yl.

[0030] Whenever it appears herein, a numerical range such as “C₂-C₆ alkynyl” means that the alkynyl group may consist of 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms, or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkynyl” where no numerical range is designated. In some embodiments, the alkynyl is a C₂-C₁₀ alkynyl, a C₂-C₉ alkynyl, a C₂-C₈ alkynyl, a C₂-C₇ alkynyl, a C₂-C₆ alkynyl, a C₂-C₅ alkynyl, a C₂-C₄ alkynyl, a C₂-C₃ alkynyl, or a C₂ alkynyl. Unless stated otherwise specifically in the specification, an alkynyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkynyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, an alkynyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, or —OMe. In some embodiments, the alkynyl is optionally substituted with halogen.

[0031] A haloalkyl group can be any alkyl group substituted with any number of halogen atoms, for example, fluorine, chlorine, bromine, and iodine atoms. A halo-alkenyl group can be any alkenyl group substituted with any number of halogen atoms. A haloalkynyl group can be any alkynyl group substituted with any number of halogen atoms.

[0032] An alkoxy group can be, for example, an oxygen atom substituted with any alkyl, alkenyl, or alkynyl group. An ether or an ether group comprises an alkoxy group.

Non-limiting examples of alkoxy groups include methoxy, ethoxy, propoxy, isopropoxy, and isobutoxy.

[0033] The term “acyl” refers to the groups HC(O)—, alkyl-C(O)—, cycloalkyl-C(O)—, cycloalkenyl-C(O)—, aryl-C(O)—, heteroaryl-C(O)— and heterocyclyl-C(O)— where alkyl, cycloalkyl, cycloalkenyl, aryl, heteroaryl, and heterocyclyl are as described herein. By way of example acyl groups include acetyl and benzoyl groups.

[0034] “Alkoxy” refers to a radical of the formula —OR^a where R^a is an alkyl radical as defined. Unless stated otherwise specifically in the specification, an alkoxy group may be optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkoxy is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, an alkoxy is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, or —OMe. In some embodiments, the alkoxy is optionally substituted with halogen.

[0035] “Aminoalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more amines. In some embodiments, the alkyl is substituted with one amine. In some embodiments, the alkyl is substituted with one, two, or three amines. Hydroxyalkyl include, for example, aminomethyl, aminoethyl, aminopropyl, aminobutyl, or aminopentyl. In some embodiments, the hydroxyalkyl is aminomethyl.

[0036] “Aryl” refers to a radical derived from a hydrocarbon ring system comprising hydrogen, 6 to 30 carbon atoms, and at least one aromatic ring. The aryl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with a cycloalkyl or heterocyclylalkyl ring, the aryl is bonded through an aromatic ring atom) or bridged ring systems. In some embodiments, the aryl is a 6- to 10-membered aryl. In some embodiments, the aryl is a 6-membered aryl. Aryl radicals include, but are not limited to, aryl radicals derived from the hydrocarbon ring systems of anthrylene, naphthylene, phenanthrylene, anthracene, azulene, benzene, chrysene, fluoranthene, fluorene, as-indacene, s-indacene, indane, indene, naphthalene, phenalene, phenanthrene, pleiadene, pyrene, and triphenylene. In some embodiments, the aryl is phenyl. Unless stated otherwise specifically in the specification, an aryl may be optionally substituted, for example, with halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the aryl is optionally substituted with halogen.

[0037] “Cycloalkyl” refers to a stable, partially or fully saturated, monocyclic or polycyclic carbocyclic ring, which may include fused (when fused with an aryl or a heteroaryl ring, the cycloalkyl is bonded through a non-aromatic ring atom), bridged, or spiro ring systems. Representative cycloalkyls include, but are not limited to, cycloalkyls having from three to fifteen carbon atoms (C₃-C₁₅ cycloalkyl), from three to ten carbon atoms (C₃-C₁₀ cycloalkyl), from three to eight carbon atoms (C₃-C₈ cycloalkyl), from three to six carbon atoms (C₃-C₆ cycloalkyl), from three to

five carbon atoms (C₃-C₅ cycloalkyl), or three to four carbon atoms (C₃-C₄ cycloalkyl). In some embodiments, the cycloalkyl is a 3- to 6-membered cycloalkyl. In some embodiments, the cycloalkyl is a 5- to 6-membered cycloalkyl. Non-limiting examples of cycloalkyl groups include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl groups. Cycloalkyl groups also include fused-, bridged-, and spiro-bicycles and higher fused-, bridged-, and spiro-systems. A cycloalkyl group can be substituted with any number of straight, branched, or cyclic alkyl groups. Non-limiting examples of cyclic alkyl groups include cyclopropyl, 2-methyl-cycloprop-1-yl, cycloprop-2-en-1-yl, cyclobutyl, 2,3-dihydroxycyclobut-1-yl, cyclobut-2-en-1-yl, cyclopentyl, cyclopent-2-en-1-yl, cyclopenta-2,4-dien-1-yl, cyclohexyl, cyclohex-2-en-1-yl, cycloheptyl, cyclooctanyl, 2,5-dimethylcyclopent-1-yl, 3,5-dichlorocyclohex-1-yl, 4-hydroxycyclohex-1-yl, 3,3,5-trimethylcyclohex-1-yl, octahydropentalenyl, octahydro-1H-indenyl, 3a,4,5,6,7,7a-hexahydro-3H-inden-4-yl, decahydroazulenyl, bicyclo-[2.1.1]hexanyl, bicyclo[2.2.1]heptanyl, bicyclo[3.1.1]heptanyl, 1,3-dimethyl[2.2.1]heptan-2-yl, bicyclo[2.2.2]octanyl, and bicyclo[3.3.3]undecanyl.

[0038] Monocyclic cycloalkyls include, for example, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl.

[0039] Polycyclic cycloalkyls or carbocycles include, for example, adamantyl, norbornyl, decalanyl, bicyclo[3.3.0]octane, bicyclo[4.3.0]nonane, cis-decalin, trans-decalin, bicyclo[2.1.1]hexane, bicyclo[2.2.1]heptane, bicyclo[2.2.2]octane, bicyclo[3.2.2]nonane, and bicyclo[3.3.2]decane, and 7,7-dimethyl-bicyclo[2.2.1]heptanyl.

[0040] Partially saturated cycloalkyls include, for example, cyclopentenyl, cyclohexenyl, cycloheptenyl, and cyclooctenyl. Unless stated otherwise specifically in the specification, a cycloalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the cycloalkyl is optionally substituted with halogen.

[0041] “Deuteroalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more deuteriums. In some embodiments, the alkyl is substituted with one deuterium. In some embodiments, the alkyl is substituted with one, two, or three deuteriums. In some embodiments, the alkyl is substituted with one, two, three, four, five, or six deuteriums. Deuteroalkyl include, for example, CD₃, CH₂D, CHD₂, CH₂CD₃, CD₂CD₃, CHDCH₂D, CH₂CH₂D, or CHCHD₂. In some embodiments, the deuteroalkyl is CD₃.

[0042] “Haloalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more halogens. In some embodiments, the alkyl is substituted with one, two, or three halogens. In some embodiments, the alkyl is substituted with one, two, three, four, five, or six halogens. Haloalkyl include, for example, trifluoromethyl, difluoromethyl, fluoromethyl, trichloromethyl, 2,2,2-trifluoroethyl, 1,2-difluoroethyl, 3-bromo-2-fluoropropyl, 1,2-dibromoethyl, and the like. In some embodiments, the haloalkyl is trifluoromethyl.

[0043] “Halo” or “halogen” refers to bromo, chloro, fluoro, or iodo. In some embodiments, halogen is fluoro or chloro. In some embodiments, halogen is fluoro.

[0044] “Heteroalkyl” refers to an alkyl group in which one or more skeletal atoms of the alkyl are selected from an atom other than carbon, e.g., oxygen, nitrogen (e.g., —NH—, —N(alkyl)—), sulfur, or combinations thereof. A heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. In one aspect, a heteroalkyl is a C₁₋₆ heteroalkyl wherein the heteroalkyl is comprised of 1 to 6 carbon atoms and one or more atoms other than carbon, e.g., oxygen, nitrogen (e.g. —NH—, —N(alkyl)—), sulfur, or combinations thereof wherein the heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. Examples of such heteroalkyl are, for example, —CH₂OCH₃, —CH₂CH₂OCH₃, —CH₂CH₂OCH₂CH₂OCH₃, or —CH(CH₃)OCH₃. Unless stated otherwise specifically in the specification, a heteroalkyl is optionally substituted for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclalkyl, heteroaryl, and the like. In some embodiments, a heteroalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heteroalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heteroalkyl is optionally substituted with halogen.

[0045] “Hydroxyalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more hydroxyls. In some embodiments, the alkyl is substituted with one hydroxyl. In some embodiments, the alkyl is substituted with one, two, or three hydroxyls. Hydroxyalkyl include, for example, hydroxymethyl, hydroxyethyl, hydroxypropyl, hydroxybutyl, or hydroxypentyl. In some embodiments, the hydroxyalkyl is hydroxymethyl.

[0046] A heterocycle can be any ring containing a ring atom that is not carbon, for example, N, O, S, P, Si, B, or any other heteroatom. A heterocycle can be substituted with any number of substituents, for example, alkyl groups and halogen atoms. A heterocycle can be aromatic (heteroaryl) or non-aromatic. Non-limiting examples of heterocycles include pyrrole, pyrrolidine, pyridine, pyrimidine, pyrazine, pyridazine, piperidine, succinimide, maleimide, morpholine, imidazole, thiophene, furan, tetrahydrofuran, pyran, and tetrahydropyran.

[0047] “Heterocyclyl” refers to a stable 3- to 24-membered heterocycle. Non-limiting examples of heterocycles include: heterocyclic units having a single ring containing one or more heteroatoms, non-limiting examples of which include, diazirinyl, aziridinyl, azetidiny, pyrazolidinyl, imidazolidinyl, oxazolidinyl, isoxazolidinyl, thiazolidinyl, isothiazolidinyl, oxathiazolidinonyl, oxazolidinonyl, hydantoinyl, tetrahydrofuranly, pyrrolidinyl, morpholinyl, piperazinyl, piperidinyl, dihydropyranly, tetrahydropyranly, piperidin-2-onyl, 2,3,4,5-tetrahydro-1H-azepinyl, 2,3-dihydro-1H-indole, and 1,2,3,4-tetrahydroquinoline; and ii) heterocyclic units having 2 or more rings one of which is a heterocyclic ring, non-limiting examples of which include hexahydro-1H-pyrroliziny, 3a,4,5,6,7,7a-hexahydro-1H-benzo[d]imidazolyl, 3a,4,5,6,7,7a-hexahydro-1H-indolyl, 1,2,3,4-tetrahydroquinolinyl, and decahydro-1H-cycloocta[b]pyrrolyl.

[0048] “Heterocyclalkyl” or “heterocycloalkyl” or “cycloheteroalkyl” refers to a stable 3- to 24-membered

partially or fully saturated ring radical comprising 2 to 23 carbon atoms and from one to 8 heteroatoms selected from the group consisting of nitrogen, oxygen, phosphorous, and sulfur. Unless stated otherwise specifically in the specification, the heterocyclalkyl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with an aryl or a heteroaryl ring, the heterocyclalkyl is bonded through a non-aromatic ring atom) or bridged ring systems; and the nitrogen, carbon, or sulfur atoms in the heterocyclalkyl radical may be optionally oxidized; the nitrogen atom may be optionally quaternized.

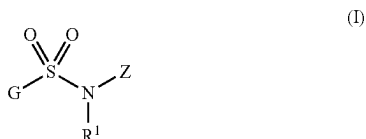
[0049] Representative heterocyclalkyls include, but are not limited to, heterocyclalkyls having from two to fifteen carbon atoms (C₂-C₁₅ heterocyclalkyl), from two to ten carbon atoms (C₂-C₁₀ heterocyclalkyl), from two to eight carbon atoms (C₂-C₈ heterocyclalkyl), from two to six carbon atoms (C₂-C₆ heterocyclalkyl), from two to five carbon atoms (C₂-C₅ heterocyclalkyl), or two to four carbon atoms (C₂-C₄ heterocyclalkyl). In some embodiments, the heterocyclalkyl is a 3- to 6-membered heterocyclalkyl or a 3- to 8-membered heterocyclalkyl. In some embodiments, the cycloalkyl is a 5- to 6-membered heterocyclalkyl. Examples of such heterocyclalkyl radicals include, but are not limited to, aziridinyl, azetidiny, dioxolanyl, thienyl[1,3]dithianyl, decahydroisquinolyl, imidazolinyl, imidazolidinyl, isothiazolidinyl, isoxazolidinyl, morpholinyl, octahydroindolyl, octahydroisindolyl, 2-oxopiperazinyl, 2-oxopiperidinyl, 2-oxopyrrolidinyl, oxazolidinyl, piperidinyl, piperazinyl, 4-piperidinyl, pyrrolidinyl, pyrazolidinyl, quinuclidinyl, thiazolidinyl, tetrahydrofuryl, trithianyl, tetrahydropyranly, thiomorpholinyl, thiamorpholinyl, 1-oxo-thiomorpholinyl, 1,1-dioxo-thiomorpholinyl, 1,3-dihydroisobenzofuran-1-yl, 3-oxo-1,3-dihydroisobenzofuran-1-yl, methyl-2-oxo-1,3-dioxol-4-yl, and 2-oxo-1,3-dioxol-4-yl. The term heterocyclalkyl also includes all ring forms of the carbohydrates, including but not limited to, the monosaccharides, the disaccharides, and the oligosaccharides. It is understood that when referring to the number of carbon atoms in a heterocyclalkyl, the number of carbon atoms in the heterocyclalkyl is not the same as the total number of atoms (including the heteroatoms) that make up the heterocyclalkyl (i.e. skeletal atoms of the heterocyclalkyl ring). Unless stated otherwise specifically in the specification, a heterocyclalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclalkyl, heteroaryl, and the like. In some embodiments, a heterocyclalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heterocyclalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heterocyclalkyl is optionally substituted with halogen.

[0050] “Heteroaryl” refers to a 5- to 14-membered ring system radical comprising hydrogen atoms, one to thirteen carbon atoms, one to six heteroatoms selected from the group consisting of nitrogen, oxygen, phosphorous, and sulfur, and at least one aromatic ring. The heteroaryl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with a cycloalkyl or heterocyclalkyl ring, the heteroaryl is bonded through an aromatic ring atom) or bridged ring

systems; and the nitrogen, carbon, or sulfur atoms in the heteroaryl radical may be optionally oxidized; the nitrogen atom may be optionally quaternized. In some embodiments, the heteroaryl is a 5- to 10-membered heteroaryl. In some embodiments, the heteroaryl is a 5- to 6-membered heteroaryl. Examples include, but are not limited to, azepinyl, acridinyl, benzimidazolyl, benzothiazolyl, benzindolyl, benzodioxolyl, benzofuranlyl, benzooxazolyl, benzothiazolyl, benzothiadiazolyl, benzo[b][1,4]dioxepinyl, 1,4-benzodioxanyl, benzonaphthofuranlyl, benzoxazolyl, benzodioxolyl, benzodioxinyl, benzopyranlyl, benzopyranonyl, benzofuranlyl, benzofuranonyl, benzothiophenyl (benzothiophenyl), benzotriazolyl, benzo[4,6]imidazo[1,2-a]pyridinyl, carbazolyl, cinnolinyl, dibenzofuranlyl, dibenzothiophenyl, furanyl, furanonyl, isothiazolyl, imidazolyl, indazolyl, indolyl, indazolyl, isoindolyl, indolyl, isoindolyl, isoquinolyl, indolizyl, isoxazolyl, naphthyridinyl, oxadiazolyl, 2-oxoazepinyl, oxazolyl, oxiranyl, 1-oxidopyridinyl, 1-oxidopyrimidinyl, 1-oxidopyrazinyl, 1-oxidopyridazinyl, 1-phenyl-1H-pyrrolyl, phenazinyl, phenothiazinyl, phenoxazinyl, phthalazinyl, pteridinyl, purinyl, pyrrolyl, pyrazolyl, pyridinyl, pyrazinyl, pyrimidinyl, pyridazinyl, quinazolinyl, quinoxalinyl, quinolyl, quinuclidinyl, isoquinolinyl, tetrahydroquinolinyl, thiazolyl, thiadiazolyl, triazolyl, tetrazolyl, triazinyl, and thiophenyl (i.e., thienyl). Unless stated otherwise specifically in the specification, a heteroaryl is optionally substituted, for example, with halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, or $-\text{NO}_2$. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, or $-\text{OMe}$. In some embodiments, the heteroaryl is optionally substituted with halogen.

Compounds

[0051] Provided are apyrase inhibitor compounds. In some cases, the compound has formula (I):



[0052] wherein:

[0053] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b ;

[0054] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C_{1-6} alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b ;

[0055] R^1 is hydrogen or C_{1-6} alkyl;

[0056] or Z and R^1 together with the nitrogen to which they are attached form a 5-, 6-, or 7-membered ring selected from heterocyclylalkyl rings optionally substituted with 1, 2 or 3 groups selected from R^a and R^b ;

[0057] each R^a is independently selected from the group consisting of C_{1-6} alkyl, C_{3-8} cycloalkyl, $\text{C}_5\text{-C}_{10}$ aryl, $\text{C}_6\text{-C}_{10}$ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

[0058] each R^a is optionally substituted with one or more groups selected from R^b and R^e ;

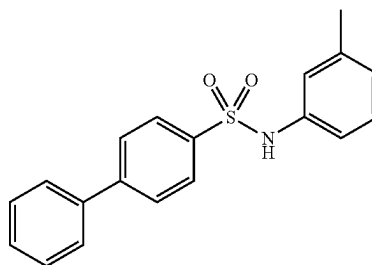
[0059] each R^b is independently selected from the group consisting of $=\text{O}$, $-\text{OR}^d$, $-\text{OCF}_3$, $-\text{OCF}_2\text{H}$, $=\text{S}$, $-\text{SR}^d$, $=\text{NR}^d$, $=\text{NOR}^d$, $-\text{NR}^c\text{R}^c$, halogen, $-\text{CF}_3$, $-\text{CF}_2\text{H}$, $-\text{CN}$, $-\text{NO}_2$, $=\text{N}_2$, $-\text{N}_3$, $-\text{S}(\text{O})\text{R}^d$, $-\text{S}(\text{O})_2\text{R}^d$, $-\text{S}(\text{O})\text{NR}^c\text{R}^c$, $-\text{S}(\text{O})_2\text{NR}^c\text{R}^c$, $-\text{NHS}(\text{O})_2\text{R}^d$, $-\text{OS}(\text{O})\text{R}^d$, $-\text{OS}(\text{O})_2\text{R}^d$, $-\text{OS}(\text{O})_2\text{OR}^d$, $-\text{OS}(\text{O})_2\text{NR}^c\text{R}^c$, $-\text{C}(\text{O})\text{R}^d$, $-\text{C}(\text{O})\text{OR}^d$, $-\text{C}(\text{O})\text{NR}^c\text{R}^c$, $-\text{C}(\text{NH})\text{NR}^c\text{R}^c$, $-\text{C}(\text{NR}^a)\text{NR}^c\text{R}^c$, $-\text{C}(\text{NOH})\text{R}^d$, $-\text{C}(\text{NOH})\text{NR}^c\text{R}^c$, $-\text{OC}(\text{O})\text{R}^d$, $-\text{OC}(\text{O})\text{OR}^d$, $-\text{OC}(\text{O})\text{NR}^c\text{R}^c$, $-\text{OC}(\text{NH})\text{NR}^c\text{R}^c$, $-\text{OC}(\text{NR}^a)\text{NR}^c\text{R}^c$, $-\text{[NHC}(\text{O})]_n\text{R}^d$, $-\text{[NR}^c(\text{O})]_n\text{R}^d$, $-\text{[NHC}(\text{O})]_n\text{OR}^d$, $-\text{[NR}^a\text{C}(\text{O})]_n\text{OR}^d$, $-\text{[NHC}(\text{O})]_n\text{NR}^c\text{R}^c$, $-\text{[NR}^a\text{C}(\text{O})]_n\text{NR}^c\text{R}^c$, $-\text{NH}\text{SO}_2\text{R}^d$, $-\text{[NHC}(\text{NH})]_n\text{NR}^c\text{R}^c$ and $-\text{[NR}^a\text{C}(\text{NR}^d)]_n\text{NR}^c\text{R}^c$;

[0060] each R^c is independently hydrogen, R^a , or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclylalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a , $=\text{O}$, halogen, and R^e ;

[0061] each R^d is independently C_{1-6} alkyl or C_{3-8} cycloalkyl;

[0062] each R^e is independently C_{1-6} alkyl, C_{3-8} cycloalkyl, $-\text{C}(\text{O})\text{R}^d$, and $-\text{C}(\text{O})\text{OR}^d$;

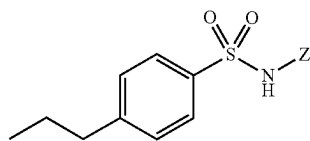
[0063] provided that the compound does not have the formula



Group G

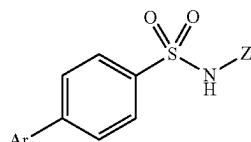
[0064] In some embodiments, G comprises a six-membered aryl or six-membered heteroaryl ring, such as phenyl or pyridine. In some embodiments, G is optionally substituted phenyl, i.e. G can be unsubstituted phenyl or it can be phenyl substituted with R^a , R^b , and R^a substituted with one or more of the same or different R^b .

[0065] In some cases, G is substituted phenyl. In some cases, G is a phenyl group substituted at its para position, such as with a C_{1-6} alkyl group. In some cases, G is a phenyl group substituted at its para position with a n-propyl group, i.e. G has formula (1a):



(Ia)

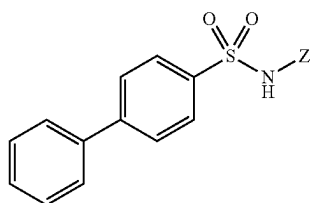
[0066] In some cases, G is a phenyl group substituted at its para position with an Ar group that is an optionally substituted aryl or heteroaryl group, as shown below in formula (Ib). As such, Ar in formula (Ib) can be an unsubstituted aryl group, a substituted aryl group, an unsubstituted heteroaryl group, or a substituted heteroaryl group.



(Ib)

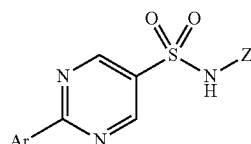
[0067] In some embodiments of formula (Ib), Ar is unsubstituted, substituted with an R^a group of C_{1-6} alkyl, or substituted with an R^b group of halogen.

[0068] In some cases, G is an optionally substituted biphenyl group, i.e. G is a phenyl group that is substituted with an R^a group of phenyl. In some cases, the compound has formula (Ib1), i.e. wherein G is an unsubstituted biphenyl group.



(Ib1)

[0069] In some cases, G is an optionally substituted pyrimidinyl. In some cases, the compound has formula (Ic), wherein the sulfonamide is at the 5-position of the pyrimidine ring, and there is an Ar group at the 2-position of the pyrimidine ring, wherein the Ar group of formula (Ic) is optionally substituted aryl or heteroaryl. If “Ar” is aryl, then it will be a C_5 - C_{10} aryl, since that is the option for R^a recited in relation to formula (I). If “Ar” is heteroaryl, then it will be a 5-10 membered heteroaryl, since that is the option for R^a recited in relation to formula (I).



(Ic)

wherein Ar is optionally substituted aryl or heteroaryl.

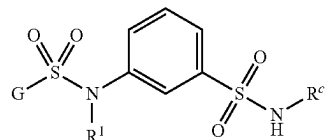
[0070] In some cases, the Ar of formula (Ic) is unsubstituted, substituted with an R^a group of C_{1-6} alkyl, or substituted with an R^b group of halogen.

Group Z

[0071] As discussed above, Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C_{1-6} alkyl.

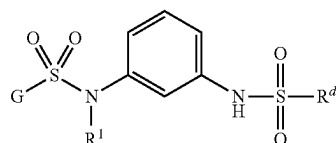
[0072] In some cases, Z is substituted phenyl. For instance, Z can be phenyl substituted with an optionally substituted aryl or heteroaryl group. In some cases, Z is phenyl substituted with an optionally substituted alkyl group. In some cases, Z is phenyl substituted with an R^b group.

[0073] In some cases, the compound has formula (Id):



(Id)

[0074] In some cases, the compound has formula (Ie):



(Ie)

[0075] In formulas (Id) and (Ie), the Z group is a phenyl group that is substituted with a sulfonamide ($-\text{NHSO}_2-$) or ($-\text{NHSO}_2-$) group, as allowed for in the description of the options for formula (I).

[0076] In some cases, Z is phenyl substituted with an R^b group of halogen or $-\text{CF}_3$.

[0077] In some cases, Z is optionally substituted monocyclic heteroaryl, such as pyridyl or pyrimidyl.

[0078] In certain cases, Z and R^1 together with the nitrogen to which they are attached form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl rings optionally substituted with 1, 2 or 3 groups selected from R^a and R^b . In one such embodiment wherein Z and R^1 together form a 5-, 6-, or 7-membered ring, the ring formed by Z and R^1 together contains one or more additional heteroatoms. In one aspect of such embodiments, the one or more additional heteroatoms are selected from the group consisting of oxygen, nitrogen and sulfur. In such embodiments wherein an additional heteroatom is nitrogen, the nitrogen atom optionally is substituted. Similarly, in such embodiments wherein an additional heteroatom is sulfur, the sulfur atom optionally is substituted.

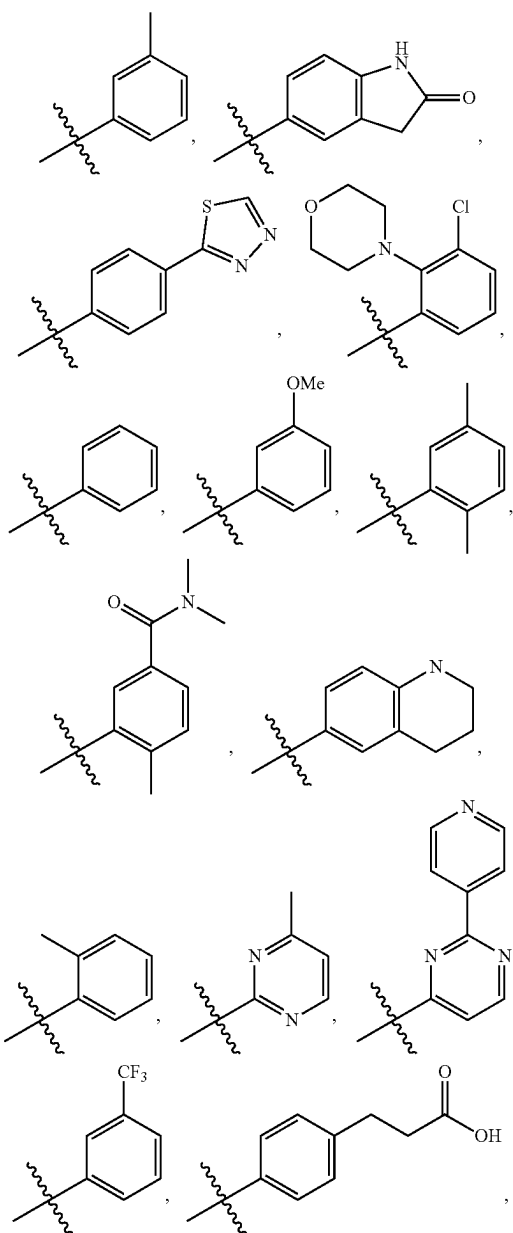
[0079] In some cases, Z and R^1 together with the nitrogen to which they are attached form a fused 5-membered or 6-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b .

[0080] In one embodiment, wherein Z and R^1 together form a 5-membered or 6-membered ring, the ring formed by Z and R^1 together contains one or more additional heteroatoms.

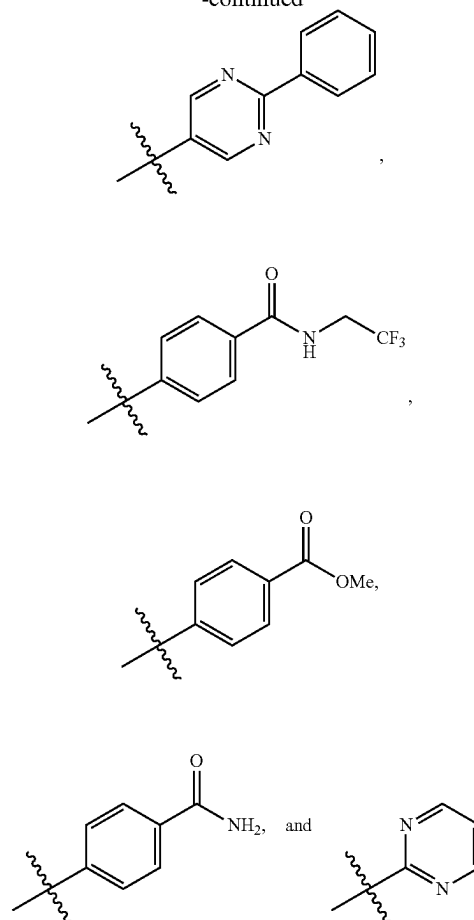
toms. In one such embodiment, the ring formed by Z and R¹ together is a morpholinyl or piperazinyl ring.

[0081] In one embodiment, wherein Z and R¹ together form a 5-membered or 6-membered ring, the ring formed by Z and R¹ together is fused to an aromatic or heteroaromatic ring, which may be substituted or unsubstituted. In one such embodiment wherein Z and R¹ together form a 5-membered or 6-membered ring, the ring formed by Z and R¹ together is fused to an aromatic or heteroaromatic ring, which is substituted with 1, 2 or 3 groups selected from R^a and R^b. In one embodiment wherein a ring formed by Z and R¹ together is fused to an aromatic or heteroaromatic ring, the resulting fused ring system is an indoline ring system.

[0082] In some embodiments, Z is selected from the group consisting of:



-continued



[0083] In some cases, features of the G groups can be combined with features of the Z groups. For instance, in some cases the G group shown in formulas (1a), (1b), (1b1), or (1c) can be combined with the Z group shown in formulas (1d) or (1e) in any possible combination.

Additional Aspects

[0084] As discussed above, each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl. In some cases, R^d is C₁₋₃ alkyl or C₃₋₄ cycloalkyl.

[0085] As discussed above, each R^e is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl. In some cases, R^e is C₁₋₃ alkyl or C₃₋₄ cycloalkyl.

[0086] As discussed above, many groups can be substituted with R^a or R^b. Additionally, R^a can be substituted with R^b. In some cases, R^b is a group that includes an R^c group, which can be R^a. In some cases, R^b is a group that includes a R^d group, which is R^a. There can be a limit on the number of times that a substituted is substituted with another substituent. For example, in some cases R^a is not substituted with R^b.

[0087] In some cases, the size of a group can be limited by specifying the number of non-hydrogen atoms in the group. For example, in some cases Z and its substituents have 6 to 20 non-hydrogen atoms, such as 6 to 16 non-hydrogen atoms or 6 to 12 non-hydrogen atoms. These numbers of non-hydrogen atoms include all levels of substitution. For

example, in Compound A-1 shown below in the next section, Z can be interpreted as a phenyl (6 atoms) substituted with an R^a of methyl (1 atom) and R^a of tetrazole (5 atoms), wherein the tetrazole is further substituted with R^b of methyl (1 atom), thereby giving a total non-hydrogen atom could of 6+1+5+1 or 13 atoms. Thus, the methyl on the tetrazole is a second-level substitution of the original Z group of phenyl, but this methyl on tetrazole is still counted towards the number of non-hydrogen atoms for Z.

[0088] Similarly, in some cases the G group can have a certain number of non-hydrogen atoms, such as 6 to 20

non-hydrogen atoms, 6 to 16 non-hydrogen atoms, or 6 to 12 non-hydrogen atoms. In Compound A-1 of the next section, G is a phenyl group (6 atoms) substituted with an R^a group of methyl (3 atoms) for a total of 9 non-hydrogen atoms in G.

Specific Embodiments of Compounds

[0089] In some cases, the compound has a structure selected from Table 1, which is shown below.

TABLE 1

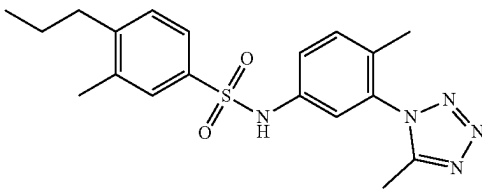
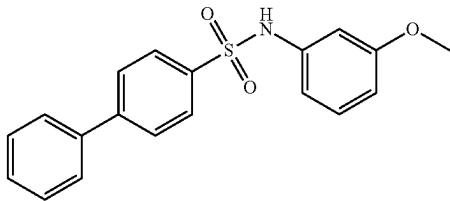
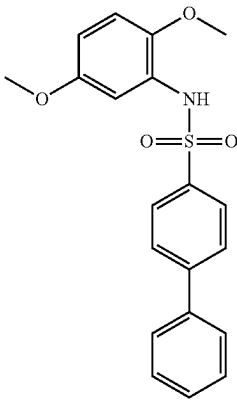
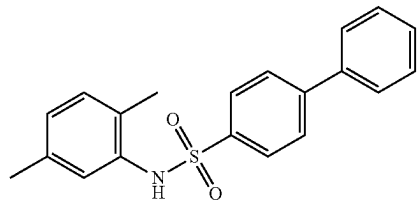
Compound	Structure	IUPAC Name
A-1		N-(4-methyl-3-(5-methyl-1H-tetrazol-1-yl)phenyl)-4-propylbenzenesulfonamide
A-3		N-(3-methoxyphenyl)-[1,1'-biphenyl]-4-sulfonamide
A-4		N-(2,5-dimethoxyphenyl)-[1,1'-biphenyl]-4-sulfonamide
A-5		N-(2,5-dimethylphenyl)-[1,1'-biphenyl]-4-sulfonamide

TABLE 1-continued

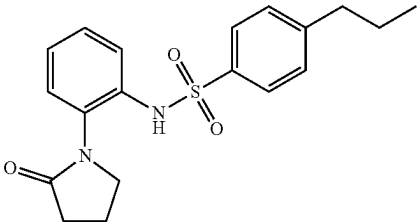
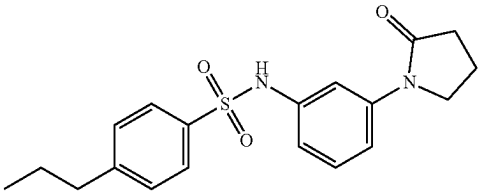
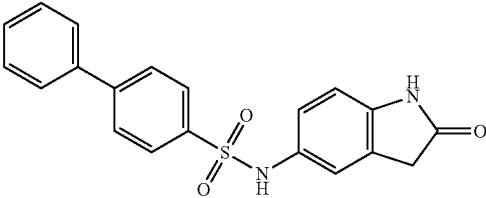
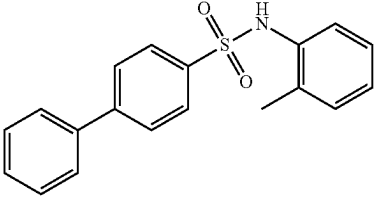
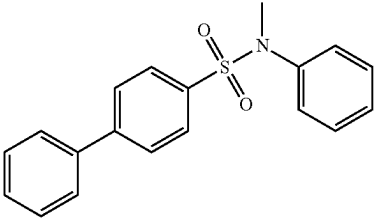
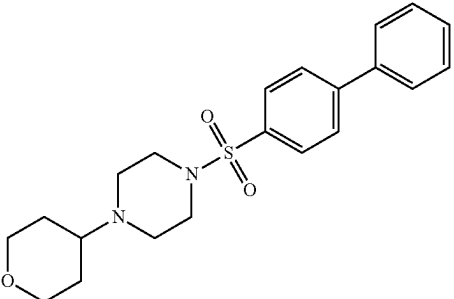
Compound	Structure	IUPAC Name
A-6		N-(2-(2-oxopyrrolidin-1-yl)phenyl)-4-propylbenzenesulfonamide
A-7		N-(3-(2-oxopyrrolidin-1-yl)phenyl)-4-propylbenzenesulfonamide
A-8		N-(2-oxoindolin-5-yl)-[1,1'-biphenyl]-4-sulfonamide
A-9		N-(o-tolyl)-[1,1'-biphenyl]-4-sulfonamide
A-10		N-methyl-N-phenyl-[1,1'-biphenyl]-4-sulfonamide
A-11		1-([1,1'-biphenyl]-4-ylsulfonyl)-4-(tetrahydro-2H-pyran-4-yl)piperazine

TABLE 1-continued

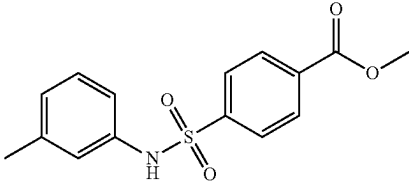
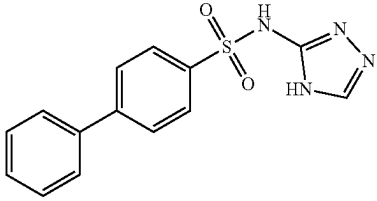
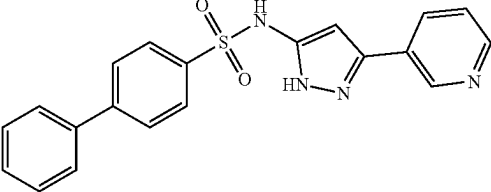
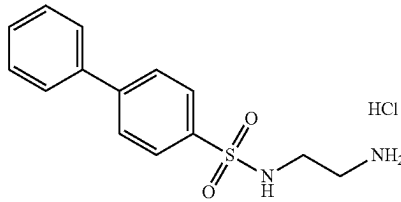
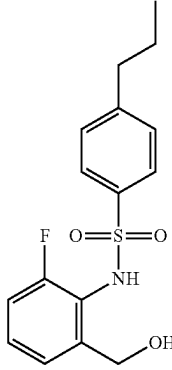
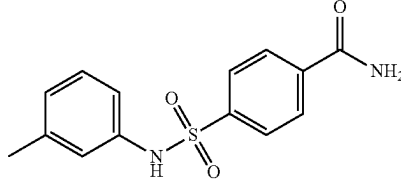
Compound	Structure	IUPAC Name
A-12		methyl 4-(N-(m-tolyl)sulfamoyl)benzoate
A-13		N-(4H-1,2,4-triazol-3-yl)-[1,1'-biphenyl]-4-sulfonamide
A-14		N-(3-(pyridin-3-yl)-1H-pyrazol-5-yl)-[1,1'-biphenyl]-4-sulfonamide
A-15		N-(2-aminoethyl)-[1,1'-biphenyl]-4-sulfonamide hydrochloride
A-16		N-(2-fluoro-6-(hydroxymethyl)phenyl)-4-propylbenzenesulfonamide
A-18		4-(N-(m-tolyl)sulfamoyl)benzamide

TABLE 1-continued

Compound	Structure	IUPAC Name
A-19		2-methyl-3-((4-(4-propylphenyl)sulfonamido)phenyl)propanoic acid
A-20		N-(pyrimidin-2-yl)-[1,1'-biphenyl]-4-sulfonamide
A-21		4-(cyanomethyl)-N-(4-methylpyrimidin-2-yl)benzenesulfonamide
A-22		N-(2-(3-methyl-4H-1,2,4-triazol-4-yl)phenyl)-4-propylbenzenesulfonamide
A-23		N-(4-methyl-3-oxo-3,4-dihydropyrazin-2-yl)-[1,1'-biphenyl]-4-sulfonamide
A-24		N-(2-chloro-5-cyanopyridin-3-yl)-[1,1'-biphenyl]-4-sulfonamide

TABLE 1-continued

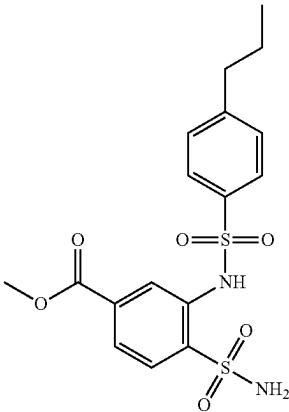
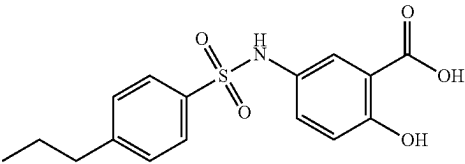
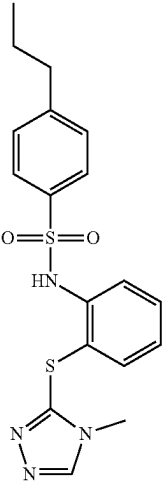
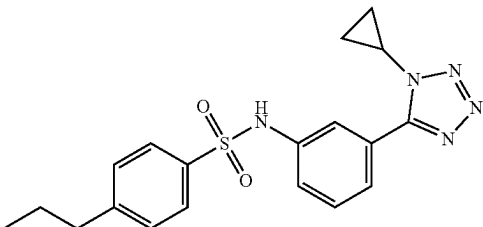
Compound	Structure	IUPAC Name
A-25		methyl 3-((4-propylphenyl)sulfonamido)-4-sulfamoylbenzoate
A-26		2-hydroxy-5-((4-propylphenyl)sulfonamido)benzoic acid
A-27		N-(2-((4-methyl-4H-1,2,4-triazol-3-yl)thio)phenyl)-4-propylbenzenesulfonamide
A-28		N-(3-(1-cyclopropyl-1H-tetrazol-5-yl)phenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

Compound	Structure	IUPAC Name
A-29		N-(4-(1,3,4-thiadiazol-2-yl)phenyl)-[1,1'-biphenyl]-4-sulfonamide
A-30		N-(4-(1,1-dioxidoisothiazolidin-2-yl)phenyl)-4-propylbenzenesulfonamide
A-31		N-(4-((4-propylphenyl)sulfonamido)phenyl)cyclopropanecarboxamide
A-32		N-(4-(ethylsulfonyl)phenyl)-4-propylbenzenesulfonamide
A-33		1-([1,1'-biphenyl]-4-ylsulfonyl)piperidin-4-amine
A-34		N-(3-methoxyphenyl)-4'-methyl-[1,1'-biphenyl]-4-sulfonamide
A-35		N-(2-fluoro-5-((4-propylphenyl)sulfonamido)phenyl)acetamide

TABLE 1-continued

Compound	Structure	IUPAC Name
A-36		N-(benzo[d][1,3]dioxol-5-ylmethyl)-[1,1'-biphenyl]-4-sulfonamide
A-37		2-phenyl-N-(3-(trifluoromethyl)phenyl)pyrimidine-5-sulfonamide
A-38		N-(2-hydroxy-5-((4-propylphenyl)sulfonamido)phenyl)acetamide
A-39		methyl 1-(2-((4-propylphenyl)sulfonamido)phenyl)-1H-1,2,3-triazole-4-carboxylate
A-40		N-(4-(2-methylpyrimidin-4-yl)phenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

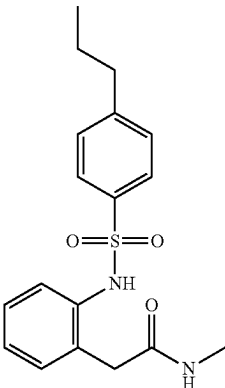
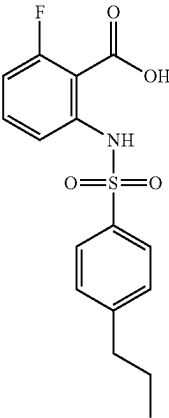
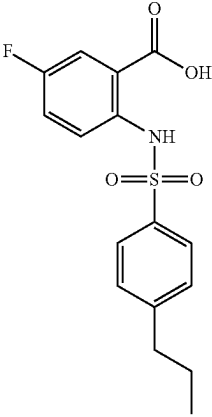
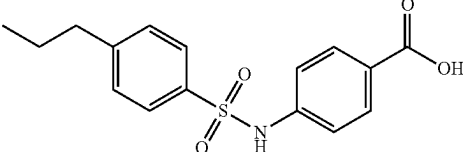
Compound	Structure	IUPAC Name
A-41		N-methyl-2-((4-propylphenyl)sulfonamido)phenylacetamide
A-42		2-fluoro-6-((4-propylphenyl)sulfonamido)benzoic acid
A-43		5-fluoro-2-((4-propylphenyl)sulfonamido)benzoic acid
A-44		4-((4-propylphenyl)sulfonamido)benzoic acid

TABLE 1-continued

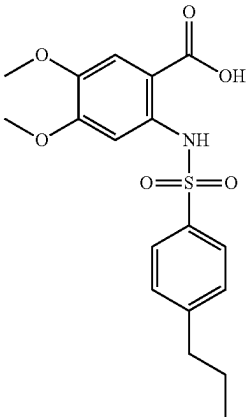
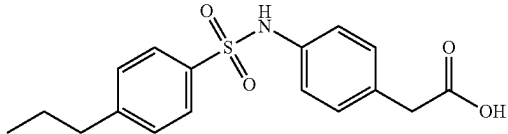
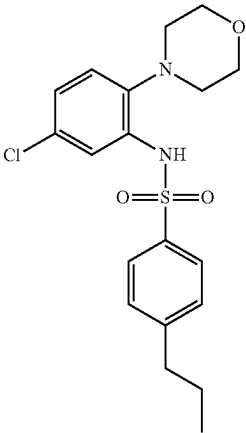
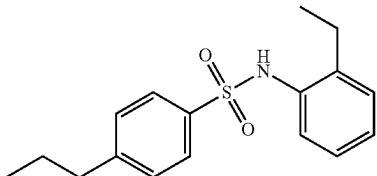
Compound	Structure	IUPAC Name
A-45		4,5-dimethoxy-2-((4-propylphenyl)sulfonamido)benzoic acid
A-46		2-(4-((4-propylphenyl)sulfonamido)phenyl)acetic acid
A-47		N-(5-chloro-2-morpholinophenyl)-4-propylbenzenesulfonamide
A-48		N-(2-ethylphenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

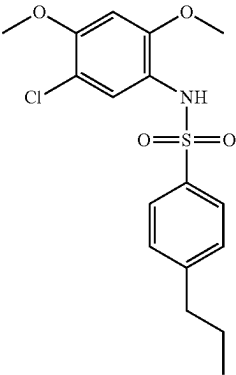
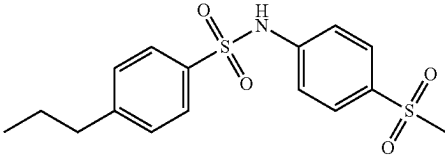
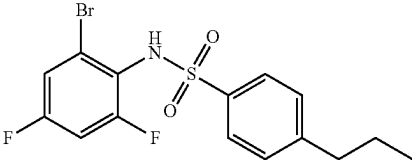
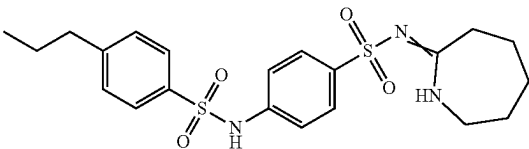
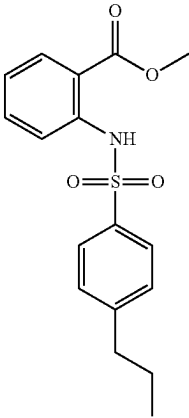
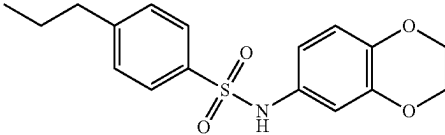
Compound	Structure	IUPAC Name
A-49		N-(5-chloro-2,4-dimethoxyphenyl)-4-propylbenzenesulfonamide
A-50		N-(4-(methylsulfonyl)phenyl)-4-propylbenzenesulfonamide
A-51		N-(2-bromo-4,6-difluorophenyl)-4-propylbenzenesulfonamide
A-52		N-(azepan-2-ylidene)-4-((4-propylphenyl)sulfonamido)benzenesulfonamide
A-53		methyl 2-((4-propylphenyl)sulfonamido)benzoate
A-54		N-(2,3-dihydrobenzo[b][1,4]dioxin-6-yl)-4-propylbenzenesulfonamide

TABLE 1-continued

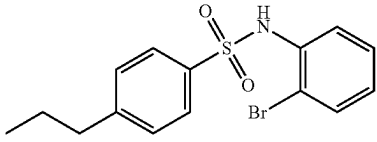
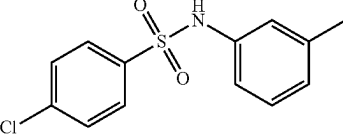
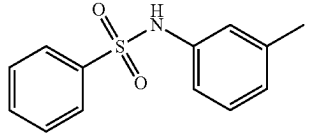
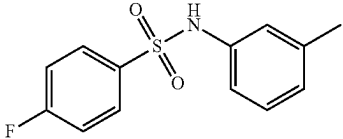
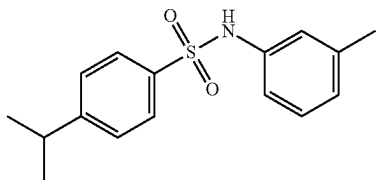
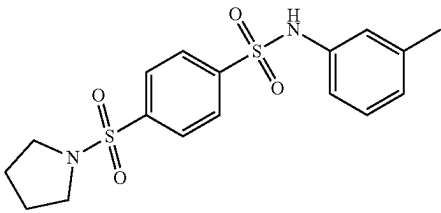
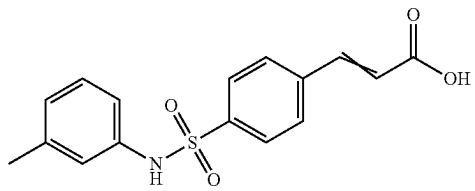
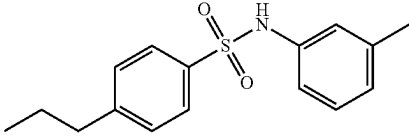
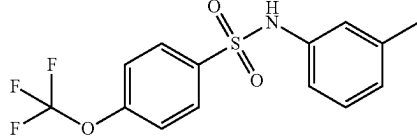
Compound	Structure	IUPAC Name
A-55		N-(2-bromophenyl)-4-propylbenzenesulfonamide
A-56		4-chloro-N-(m-tolyl)benzenesulfonamide
A-57		N-(m-tolyl)benzenesulfonamide
A-58		4-fluoro-N-(m-tolyl)benzenesulfonamide
A-59		4-isopropyl-N-(m-tolyl)benzenesulfonamide
A-60		4-(pyrrolidin-1-ylsulfonyl)-N-(m-tolyl)benzenesulfonamide
A-61		3-(4-(N-(m-tolyl)sulfamoyl)phenyl)acrylic acid
A-62		4-propyl-N-(m-tolyl)benzenesulfonamide
A-63		N-(m-tolyl)-4-(trifluoromethoxy)benzenesulfonamide

TABLE 1-continued

Compound	Structure	IUPAC Name
A-64		4-cyclohexyl-N-(m-tolyl)benzenesulfonamide
A-65		4-(N-(m-tolyl)sulfamoyl)benzoic acid
A-66		4-(N-(m-tolyl)sulfamoyl)-N-(2,2,2-trifluoroethyl)benzamide
A-67		N-(5-chloro-2-((4-methyl-4H-1,2,4-triazol-3-yl)thio)phenyl)-4-propylbenzenesulfonamide
A-68		N-(4-chloro-2,5-dimethoxyphenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

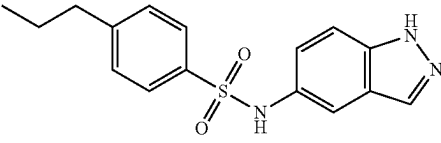
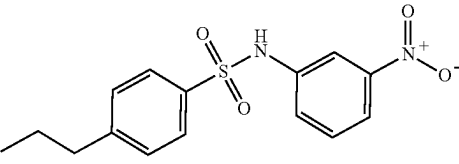
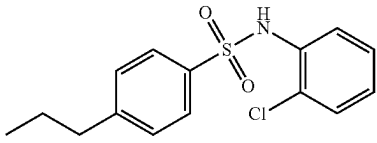
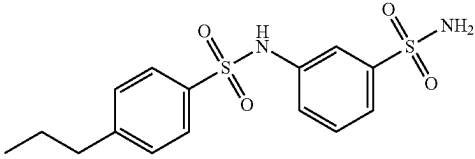
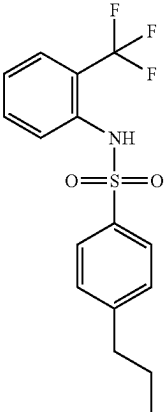
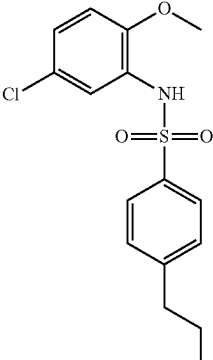
Compound	Structure	IUPAC Name
A-69		N-(1H-indazol-5-yl)-4-propylbenzenesulfonamide
A-70		N-(3-nitrophenyl)-4-propylbenzenesulfonamide
A-71		N-(2-chlorophenyl)-4-propylbenzenesulfonamide
A-72		4-propyl-N-(3-sulfamoylphenyl)benzenesulfonamide
A-73		4-propyl-N-(2-(trifluoromethyl)phenyl)benzenesulfonamide
A-74		N-(5-chloro-2-methoxyphenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

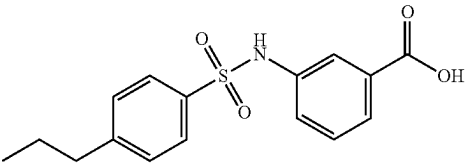
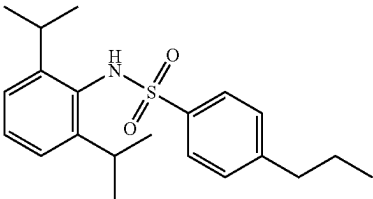
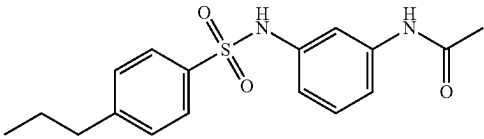
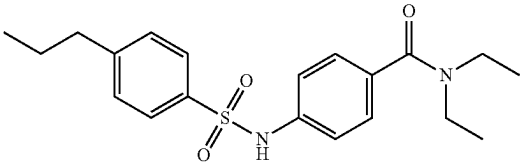
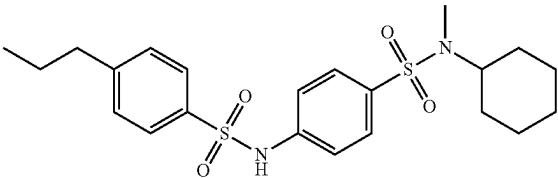
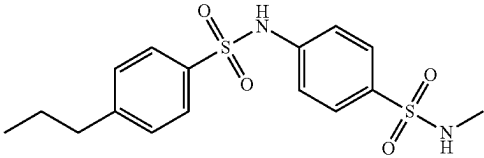
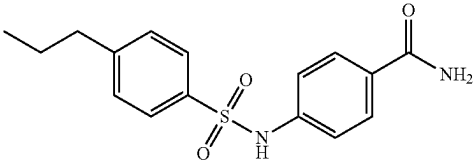
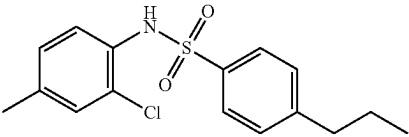
Compound	Structure	IUPAC Name
A-75		3-((4-propylphenyl)sulfonamido)benzoic acid
A-76		N-(2,6-diisopropylphenyl)-4-propylbenzenesulfonamide
A-77		N-(3-((4-propylphenyl)sulfonamido)phenyl)acetamide
A-78		N,N-diethyl-4-((4-propylphenyl)sulfonamido)benzamide
A-79		N-cyclohexyl-N-methyl-4-((4-propylphenyl)sulfonamido)benzenesulfonamide
A-80		N-methyl-4-((4-propylphenyl)sulfonamido)benzenesulfonamide
A-81		4-((4-propylphenyl)sulfonamido)benzamide
A-82		N-(2-chloro-4-methylphenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

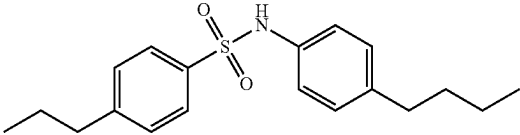
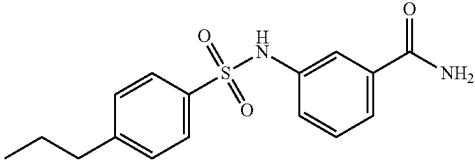
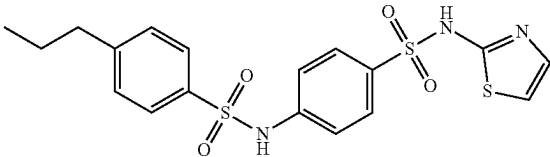
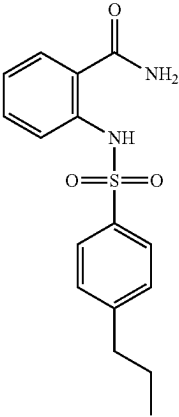
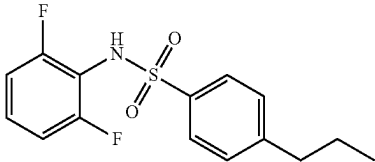
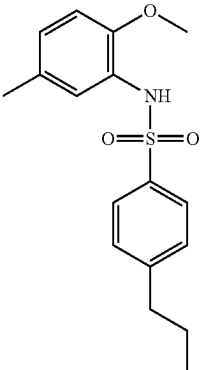
Compound	Structure	IUPAC Name
A-83		N-(4-butylphenyl)-4-propylbenzenesulfonamide
A-84		3-((4-propylphenyl)sulfonamido)benzamide
A-85		4-propyl-N-(4-(N-(thiazol-2-yl)sulfamoyl)phenyl)benzenesulfonamide
A-86		2-((4-propylphenyl)sulfonamido)benzamide
A-87		N-(2,6-difluorophenyl)-4-propylbenzenesulfonamide
A-88		N-(2-methoxy-5-methylphenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

Compound	Structure	IUPAC Name
A-89		N,N,2-trimethyl-5-((4-propylphenyl)sulfonamido)benzenesulfonamide
A-90		N-(2-methyl-5-(piperidin-1-ylsulfonyl)phenyl)-4-propylbenzenesulfonamide
A-91		N-(2-benzylphenyl)-4-propylbenzenesulfonamide
A-92		N-(2,6-dimethylphenyl)-4-propylbenzenesulfonamide
A-93		4-propyl-N-(4-(6,7,8,9-tetrahydro-5H-[1,2,4]triazolo[4,3-a]azepin-3-yl)phenyl)benzenesulfonamide
A-94		4-fluoro-N-(2-((4-propylphenyl)sulfonamido)phenyl)benzenesulfonamide

TABLE 1-continued

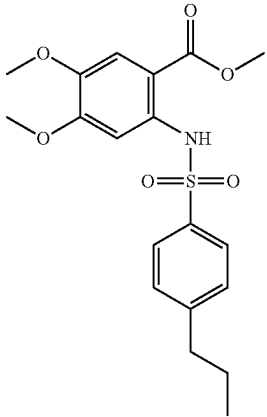
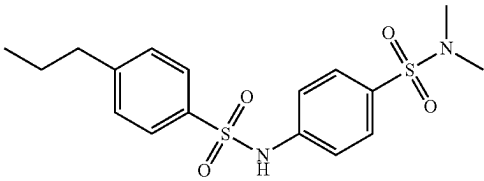
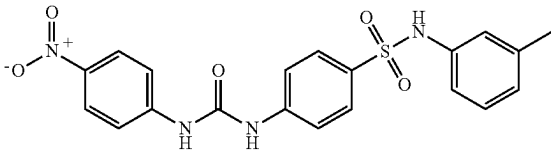
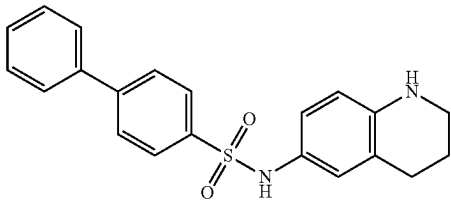
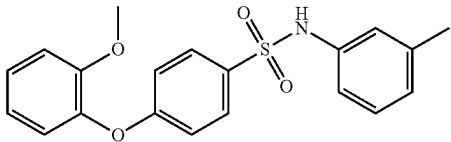
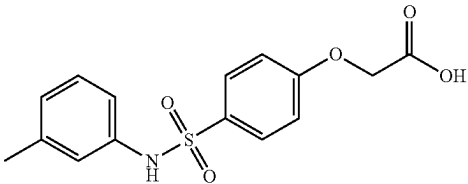
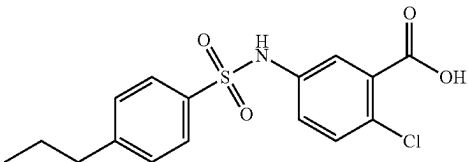
Compound	Structure	IUPAC Name
A-96		methyl 4,5-dimethoxy-2-((4-propylphenyl)sulfonamido)benzoate
A-97		N,N-dimethyl-4-((4-propylphenyl)sulfonamido)benzenesulfonamide
A-98		4-(3-(4-nitrophenyl)ureido)-N-(m-tolyl)benzenesulfonamide
A-99		N-(1,2,3,4-tetrahydroquinolin-6-yl)-[1,1'-biphenyl]-4-sulfonamide
A-100		4-(2-methoxyphenoxy)-N-(m-tolyl)benzenesulfonamide
A-101		2-(4-(N-(m-tolyl)sulfamoyl)phenoxy)acetic acid
A-102		2-chloro-5-((4-propylphenyl)sulfonamido)benzoic acid

TABLE 1-continued

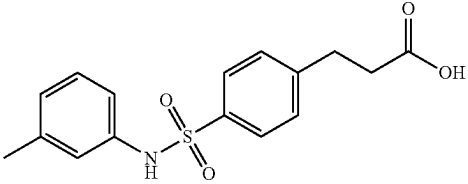
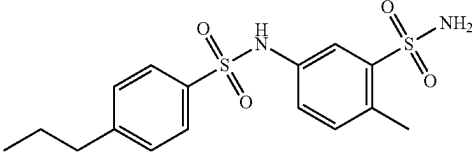
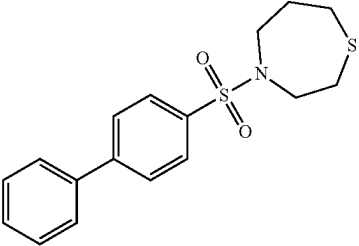
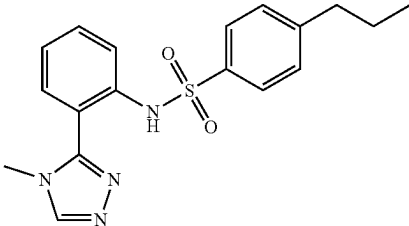
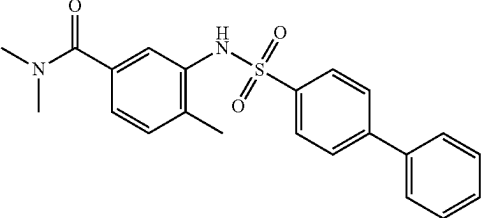
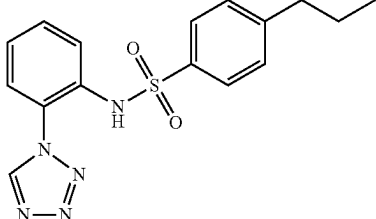
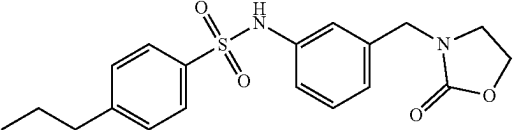
Compound	Structure	IUPAC Name
A-103		3-(4-(N-(m-tolyl)sulfamoyl)phenyl)propanoic acid
A-104		N-(4-methyl-3-sulfamoylphenyl)-4-propylbenzenesulfonamide
A-106		4-([1,1'-biphenyl]-4-ylsulfonyl)-1,4-thiazepane
A-107		N-(2-(4-methyl-4H-1,2,4-triazol-3-yl)phenyl)-4-propylbenzenesulfonamide
A-108		3-([1,1'-biphenyl]-4-sulfonamido)-N,N,4-trimethylbenzamide
A-109		N-(2-(1H-tetrazol-1-yl)phenyl)-4-propylbenzenesulfonamide
A-110		N-(3-((2-oxooxazolidin-3-yl)methyl)phenyl)-4-propylbenzenesulfonamide

TABLE 1-continued

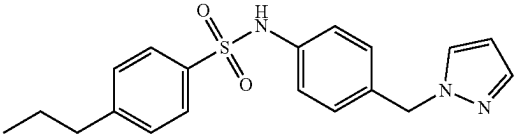
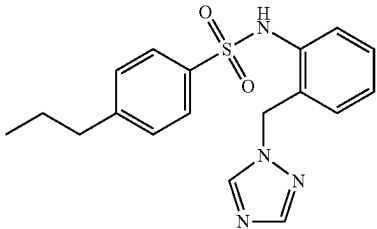
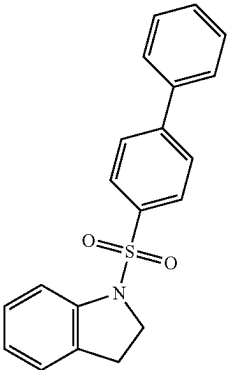
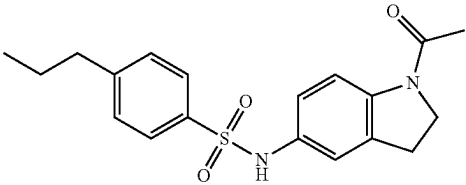
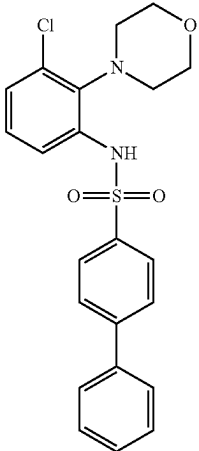
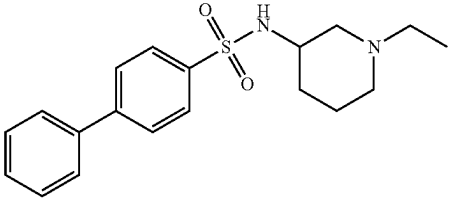
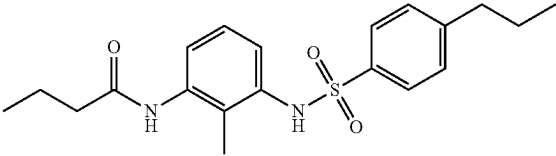
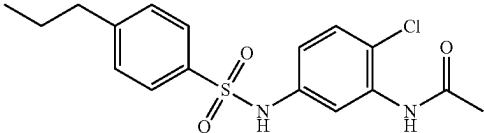
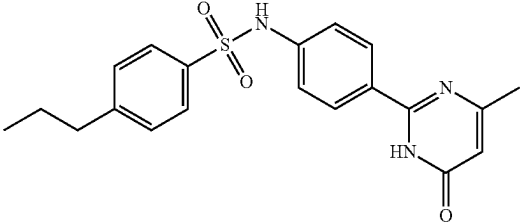
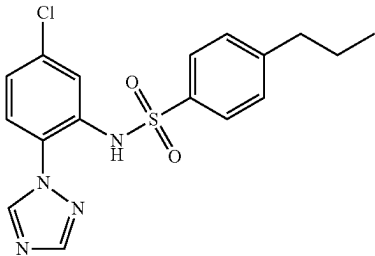
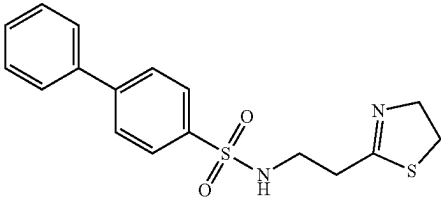
Compound	Structure	IUPAC Name
A-111		N-(4-((1H-pyrazol-1-yl)methyl)phenyl)-4-propylbenzenesulfonamide
A-113		N-(2-((1H-1,2,4-triazol-1-yl)methyl)phenyl)-4-propylbenzenesulfonamide
A-114		1-([1,1'-biphenyl]-4-ylsulfonyl)indoline
A-115		N-(1-acetylidolin-5-yl)-4-propylbenzenesulfonamide
A-116		N-(3-chloro-2-morpholinophenyl)-[1,1'-biphenyl]-4-sulfonamide

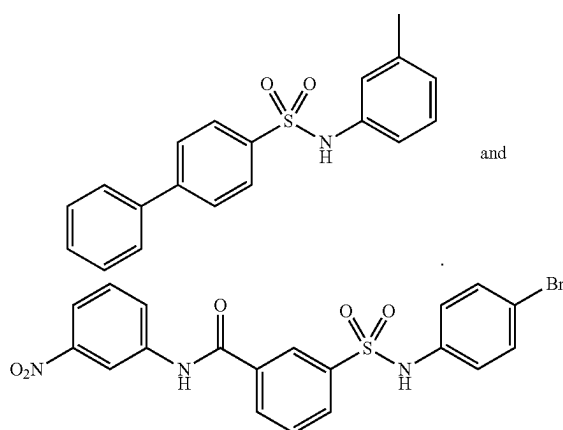
TABLE 1-continued

Compound	Structure	IUPAC Name
A-117		N-(1-ethylpiperidin-3-yl)- [1,1'-biphenyl]-4-sulfonamide
A-118		N-(2-methyl-3-((4- propylphenyl)sulfonamido) phenyl)butyramide
A-119		N-(2-chloro-5-((4- propylphenyl)sulfonamido) phenyl)acetamide
A-120		N-(4-(4-methyl-6-oxo-1,6- dihydropyrimidin-2- yl)phenyl)-4- propylbenzenesulfonamide
A-122		N-(5-chloro-2-(1H-1,2,4- triazol-1-yl)phenyl)-4- propylbenzenesulfonamide
A-123		N-(2-(thiazol-2-yl)ethyl)- [1,1'-biphenyl]-4-sulfonamide

Target Crops and their Pathogens

[0090] The present disclosure provides formulations and methods for their use in treating crops for pathogens. In one embodiment, a disclosed compound is administered in combination with an agricultural or horticultural pesticide, such as an acaricide, antimicrobial, fungicide, herbicide, insecticide, molluscicide and/or nematocide. Crops that can be treated, include those plagued by various pathogens, including without limitation, bacteria, viruses, fungal pathogens, mites, nematodes, molluscs, weeds or other pests, as is known to those of ordinary skill in the agricultural arts. By way of example, such agricultural and horticultural crops that can be treated according to the present disclosure include plants, whether genetically modified or not, including their harvested products, such as: cereals; vegetables; root crops; potatoes; trees such as fruit trees, for example banana trees, tea, coffee trees, or cocoa trees; grasses; lawn grass; or cotton.

[0091] Roux and coworkers describe the compounds:



[0092] as enhancing the ability of certain fungicides to inhibit the growth of different plant-pathogenic fungi. These compounds are referred to herein as “Roux compound 1,” and “Roux compound 13,” respectively (*Molecular Plant Pathology*, 2017, 18(7), 1012-1023; and WO 2016/123191). The present compounds surprisingly enhance the ability of a variety of pesticides against a broad variety of pathogens, including fungal pathogens. In addition, examples of the presently disclosed compound exhibit superior enhancer activity than Roux compounds 1 and 13.

[0093] The agricultural or horticultural enhancer disclosed herein may be applied to each part of plants, such as leaves, stems, patterns, flowers, buds, fruits, seeds, sprouts, roots, tubers, tuberous roots, shoots, or cuttings. The agricultural or horticultural enhancer according to the present disclosure may also be applied to improved varieties/varieties, cultivars, as well as mutants, hybrids and genetically modified embodiments of these plants.

[0094] The agricultural or horticultural treatment described herein may be used to conduct seed treatment, foliage application, soil application, or water application, so as to control various diseases occurring in agricultural or

horticultural crops, including flowers, lawns, and pastures. occurring in agricultural or horticultural crops, including flowers, lawns, and pastures.

[0095] The present compounds are useful for potentiating the effects of antimicrobial agents. For example, the present compounds can be used in combination with an antimicrobial agent to combat bacterial and viral infection.

[0096] The present compounds are useful for potentiating the effects of herbicides. For example, the present compounds can be used in combination with one or more herbicide to control weeds or other unwanted vegetation.

[0097] The present compounds are useful for potentiating the effects of insecticides. For example, the present compounds can be used in combination with one or more insecticide to control insect infestation.

[0098] The present compounds are useful for potentiating the effects of acaricides or miticides. For example, the present compounds can be used in combination with one or more acaricidal agent to control mites.

[0099] The present compounds are useful for potentiating the effects of molluscicides. For example, the present compounds can be used in combination with one or more molluscicide to prevent interference of slugs or snails with a crop.

[0100] The present compounds are useful for potentiating the effects of nematocides. For example, the present compounds can be used in combination with one or more nematocide to prevent interference of nematodes with a crop.

[0101] The present compounds are particularly useful for potentiating the effects of fungicides against plant fungal pathogens. Examples of pathogens treated according to the present disclosure include, without limitation, *Botrytis cinerea*, *Colletotrichum graminicola*, *Fusarium oxysporum*, *Sclerotinia sclerotiorum*, *Verticillium dahliae*, *Mycosphaella graminicola* and *Sphacelotheca reiliana*.

[0102] *Botrytis cinerea* is an airborne plant pathogen with a necrotrophic lifestyle attacking over 200 crop hosts worldwide. It mainly attacks dicotyledonous plant species, including important protein, oil, fiber and horticultural crops, grapes and strawberries and also *Botrytis* also causes secondary soft rot of fruits and vegetables during storage, transit and at the market. Many classes of fungicides have failed to control *Botrytis cinerea* due to its genetic plasticity.

[0103] The genus *Colletotrichum* comprises ~600 species attacking over 3,200 species of monocot and dicot plants. *Colletotrichum graminicola* primarily infects maize (*Zea mays*), causing annual losses of approximately 1 billion dollars in the United States alone (Connell et al., 2012).

[0104] *Fusarium* wilt of banana, caused by the soil-borne fungus *Fusarium oxysporum* f.sp. *cubense*, is a major threat to banana production worldwide. No fungicides are currently available to effectively control the disease once plants are infected (Peng J et al., 2014).

[0105] The white mold fungus *Sclerotinia sclerotiorum* is known to attack more than 400 host species and is considered one of the most prolific plant pathogens. The majority of the affected crop species are dicotyledonous, along with a number of agriculturally significant monocotyledonous plants. Some important crops affected by *S. sclerotiorum* include legumes (soybean), most vegetables, stone fruits and tobacco.

[0106] The ascomycete *Verticillium dahliae* is a soil-borne fungal plant pathogen that causes vascular wilt diseases in a

broad range of dicotyledonous host species. *V. dahliae* can cause severe yield and quality losses in cotton and other important crops such as vegetables, fibers, fruit, nut trees, forest trees and ornamental plants.

[0107] The ascomycete fungus *Mycosphaerella graminicola* (anamorph: *Septoria tritici*) is one of the most important foliar diseases of wheat leaves, occurring wherever wheat is grown. Yield losses attributed to this disease range from 25%-50%, and are especially high in Europe, the Mediterranean region and East Africa. Infection by *M. graminicola* is initiated by air borne ascospores produced on residues of last season's crop. Primary infection usually occurs after seedlings emerge in spring or fall. The mature disease is characterized by necrotic lesions on the leaves and stems of infected plants.

[0108] The basidiomycete fungus *Sphacelotheca reiliana* infects corn (*Zea mays*) systemically, causing Head Smut. Yield loss attributed to the disease is variable, and is directly dependent on the incidence of the disease. The fungus overwinters as diploid teliospores in crop debris or soil. Floral structures are converted to sori containing masses of powdery teliospores that resemble mature galls of common smut.

[0109] Examples of crops to be treated and plant diseases (pathogens) to be controlled using the presently disclosed compounds and compositions include, without limitation: Sugar beet: brown spot disease (*Cercospora beticola*), black root disease (*Aphanomyces cochlioides*), root rot disease (*Thanatephorus cucumeris*), leaf rot disease (*Thanatephorus cucumeris*), and the like.

[0110] Peanut: brown spot disease (*Mycosphaerella arachidis*), leaf mold (*Ascochyta* sp.), rust disease (*Puccinia arachidis*), damping-off disease (*Pythium debaryanum*), rust spot disease (*Alternaria alternata*), stem rot disease (*Sclerotium rolfsii*), black rust disease (*Mycosphaerella berkeleyi*), and the like.

[0111] Cucumber: powdery mildew (*Sphaerotheca fuliginea*), downy mildew (*Pseudoperonospora cubensis*), gummy stem blight (*Mycosphaerella melonis*), wilt disease (*Fusarium oxysporum*), sclerotinia rot (*Sclerotinia sclerotiorum*), gray mold (*Botrytis cinerea*), anthracnose (*Colletotrichum orbiculare*), scab (*Cladosporium cucumerinum*), brown spot disease (*Corynespora cassiicola*), damping-off disease (*Pythium debaryanum*, *Rhizoctonia solani* Kuhn), *Phomopsis* root rot disease (*Phomopsis* sp.), Bacterial spot (*Pseudomonas syringae* pv. *Lechrymans*), and the like.

[0112] Tomato: gray mold disease (*Botrytis cinerea*), leaf mold disease (*Cladosporium fulvum*), late blight disease (*Phytophthora infestans*), *Verticillium* wilt disease (*Verticillium albo-atrum*, *Verticillium dahliae*), powdery mildew disease (*Oidium neolyopersici*), early blight disease (*Alternaria solani*), leaf mold disease (*Pseudocercospora fuligena*), and the like.

[0113] Eggplant: gray mold disease (*Botrytis cinerea*), black rot disease (*Corynespora melongenae*), powdery mildew disease (*Erysiphe cichoracearum*), leaf mold disease (*Mycovellosiella nattrassii*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), *Verticillium* wilt disease (*Verticillium dahlia*), *Mycosphaerella* blight (*Phomopsis vexans*), and the like.

[0114] Strawberry: gray mold disease (*Botrytis cinerea*), powdery mildew disease (*Sphaerotheca humuli*), anthracnose disease (*Colletotrichum acutatum*, *Colletotrichum fragariae*), *Phytophthora* rot disease (*Phytophthora cacto-*

rum), soft rot disease (*Rhizopus stolonifer*), *fusarium* wilt disease (*Fusarium oxysporum*), *verticillium* disease dahlia), and wilt (*Verticillium* the like).

[0115] Onion: neck rot disease (*Botrytis allii*), gray mold disease (*Botrytis cinerea*), leaf blight disease (*Botrytis squamosa*), downy mildew disease (*Peronospora destructor*), *Phytophthora* porn disease (*Phytophthora porn*), and the like.

[0116] Cabbage: clubroot disease (*Plasmodiophora brassicae*), soft rot disease (*Erwinia carotovora*), black rot disease (*Xanthomonas campestris* pv. *campestris*), bacterial black spot disease (*Pseudomonas syringae* pv. *Maculicola*, P.s. pv. *alisalensis*), downy mildew disease (*Peronospora parasitica*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), black spot disease (*Alternaria brassicicola*), gray mold disease (*Botrytis cinerea*), and the like.

[0117] Common bean: *sclerotinia* rot disease (*Sclerotinia sclerotiorum*), gray mold disease (*Botrytis cinerea*), anthracnose (*Colletotrichum lindemuthianum*), angular spot disease (*Phaeoisariopsis griseola*), and the like.

[0118] Apple: powdery mildew disease (*Podosphaera leucotricha*), scab disease (*Venturia inaequalis*), Monilinia disease (*Monilinia mali*), black spot disease (*Mycosphaerella pomi*), valla canker disease (*Valsa mali*), *alternaria* blotch disease (*Alternaria mali*), rust disease (*Gymnosporangium yamadae*), ring rot disease (*Botryosphaeria berengeriana*), anthracnose disease (*Glomerella cingulata*, *Colletotrichum acutatum*), leaf rot disease (*Diplocarpon mali*), fly speck disease (*Zygophiala jamaicensis*), Sooty blotch (*Gloeodes pomigena*), violet root rot disease (*Helicobasidium mompa*), gray mold disease (*Botrytis cinerea*), and the like.

[0119] Japanese apricot: scab disease (*Cladosporium carpophilum*), gray mold disease (*Botrytis cinerea*), brown rot disease (*Monilinia mumeicola*), and the like.

[0120] Persimmon: powdery mildew disease (*Phyllactinia kakicola*), anthracnose disease (*Gloeosporium kaki*), angular leaf spot (*Cercospora kaki*), and the like.

[0121] Peach: brown rot disease (*Monilinia fructicola*), scab disease (*Cladosporium carpophilum*), *phomopsis* rot disease (*Phomopsis* sp.), bacterial shot hole disease (*Xanthomonas campestris* pv. *pruni*), and the like.

[0122] Almond: brown rot disease (*Monilinia taxa*), spot blotch disease (*Stigmata carpophila*), scab disease (*Cladosporium carpophilum*), red leaf spot disease (*Polystigma rubrum*), *alternaria* blotch disease (*Alternaria alternata*), anthracnose (*Colletotrichum gloeosporides*), and the like.

[0123] Yellow peach: brown rot disease (*Monilinia fructicola*), anthracnose disease (*Colletotrichum acutatum*), black spot disease (*Alternaria* sp.), *Monilinia kusanoi* disease (*Monilinia kusanoi*), and the like.

[0124] Grape: gray mold disease (*Botrytis cinerea*), powdery mildew disease (*Uncinula necator*), ripe rot disease (*Glomerella cingulata*, *Colletotrichum acutatum*), downy mildew disease (*Plasmopara viticola*), anthracnose disease (*Elsinoe ampelina*), brown spot disease (*Pseudocercospora vitis*), black rot disease (*Guignardia bidwellii*), white rot disease (*Coniella castaneicola*), rust disease (*Phakopsora ampelopsidis*), and the like.

[0125] Pear: scab disease (*Venturia nashicola*), rust disease (*Gymnosporangium asiaticum*), black spot disease (*Alternaria kikuchiana*), ring rot disease (*Botryosphaeria berengeriana*), powdery mildew disease (*Phyllactinia mali*), Cytospora canker disease (*Phomopsis fukushii*), brown spot

blotch disease (*Stemphylium vesicarium*), anthracnose disease (*Glomerella cingulata*), and the like.

[0126] Tea: ring spot disease (*Pestalotiopsis longiseta*, *P. theae*), anthracnose disease (*Colletotrichum theae-sinensis*), Net blister blight (*Exobasidium reticulatum*), and the like.

[0127] Citrus fruits: scab disease (*Elsinoe fawcettii*), blue mold disease (*Penicillium italicum*), common green mold disease (*Penicillium digitatum*), gray mold disease (*Botrytis cinerea*), melanose disease (*Diaporthe citri*), canker disease (*Xanthomonas campestris* pv. *Citri*), powdery mildew disease (*Oidium* sp.), and the like.

[0128] Wheat: powdery mildew (*Blumeria graminis* f. sp. *tritici*), red mold disease (*Gibberella zeae*), brown rust disease (*Puccinia recondita*), brown snow mold disease (*Pythium iwayamai*), pink snow mold disease (*Monographella nivalis*), eye spot disease (*Pseudocercospora herpotrichoides*), leaf scorch disease (*Septoria tritici*), glume blotch disease (*Leptosphaeria nodorum*), typhula snow blight disease (*Typhula incarnata*), sclerotinia snow blight disease (*Myriosclerotinia borealis*), damping-off disease (*Gaeumannomyces graminis*), ergot disease (*Claviceps purpurea*), stinking smut disease (*Tilletia caries*), loose smut disease (*Ustilago nuda*), and the like.

[0129] Barley: leaf spot disease (*Pyrenophora graminea*), net blotch disease (*Pyrenophora teres*), leaf blotch disease (*Rhynchosporium secalis*), loose smut disease (*Ustilago tritici*, *U. nuda*), and the like.

[0130] Rice: blast disease (*Pyricularia oryzae*), sheath blight disease (*Rhizoctonia solani*), bakanae disease (*Gibberella fujikuroi*), brown spot disease (*Cochliobolus miyabeanus*), damping-off disease (*Pythium graminicola*), bacterial leaf blight (*Xanthomonas oryzae*), bacterial seedling blight disease (*Burkholderia plantarii*), brown stripe disease (*Acidovorax avenae*), bacterial grain rot disease (*Burkholderia glumae*), *Cercospora* leaf spot disease (*Cercospora oryzae*), false smut disease (*Ustilaginoidea virens*), rice brown spot disease (*Alternaria alternata*, *Curvularia intermedia*), kernel discoloration of rice (*Alternaria padwickii*), pink coloring of rice grains (*Epicoccum purpurascens*), and the like.

[0131] Tobacco: sclerotinia rot disease (*Sclerotinia sclerotiorum*), powdery mildew disease (*Erysiphe cichoracearum*), phytophthora rot disease (*Phytophthora nicotianae*), and the like.

[0132] Tulip: gray mold disease (*Botrytis cinerea*), and the like.

[0133] Sunflower: downy mildew disease (*Plasmopara halstedii*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), and the like.

[0134] Bent grass: Sclerotinia snow blight (*Sclerotinia borealis*), Large patch (*Rhizoctonia solani*), Brown patch (*Rhizoctonia solani*), Dollar spot (*Sclerotinia homoeo-*

carpa), blast disease (*Pyricularia* sp.), *Pythium* red blight disease (*Pythium aphanidermatum*), anthracnose disease (*Colletotrichum graminicola*), and the like.

[0135] Orchard grass: powdery mildew disease (*Erysiphe graminis*), and the like.

[0136] Soybean: purple stain disease (*Cercospora kikuchii*), downy mildew disease (*Peronospora manshurica*), phytophthora rot disease (*Phytophthora sojae*), rust disease (*Phakopsora pachyrhizi*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), anthracnose disease (*Colletotrichum truncatum*), gray mold disease (*Botrytis cinerea*), *Sphaceloma* scab (*Elsinoe glycines*), melanoses (*Diaporthe phaseolorum* var. *sojae*), and the like.

[0137] Potato: hytophthora rot disease (*Phytophthora infestans*), early blight disease (*Alternaria solani*), scurf disease (*Thanatephorus cucumeris*), verticillium wilt disease (*Verticillium albo-atrum*, *V. dahlia*, *V. nigrescens*, and the like.

[0138] Banana: Panama disease (*Fusarium oxysporum*), Sigatoka disease (*Mycosphaerella fijiensis*, *M. musicola*), and the like.

[0139] Rapeseed: sclerotinia rot disease (*Sclerotinia sclerotiorum*), root rot disease (*Phoma lingam*), black leaf spot disease (*Alternaria brassicae*), and the like.

[0140] Coffee: rust disease (*Hemileia vastatrix*), anthracnose (*Colletotrichum coffeanum*), leaf spot disease (*Cercospora coffeicola*), and the like.

[0141] Sugarcane: brown rust disease (*Puccinia melanocephala*), and the like.

[0142] Corn: zonate spot disease (*Gloeocercospora sorghi*), rust disease (*Puccinia sorghi*), southern rust disease (*Puccinia polysora*), smut disease (*Ustilago maydis*), brown spot disease (*Cochliobolus heterostrophus*), northern leaf blight (*Setosphaeria turcica*), and the like.

[0143] Cotton: seedling blight disease (*Pythium* sp.), rust disease (*Phakopsora gossypii*), sour rot disease (*Mycosphaerella areola*), anthracnose (*Glomerella gossypii*), and the like.

Pesticides

[0144] The presently disclosed compounds are useful for enhancing the effect of a variety of agrochemicals, including fungicides, antiviral agents, bactericides, herbicides, insecticidal/acaricidal agents, molluscicides, nematocides, soil pesticides, plant control agents, synergistic agents, fertilizers and soil conditioners.

[0145] In one embodiment, the presently disclosed compounds are useful for enhancing the fungicidal effect of a variety of fungicides. Fungicides for use with the presently disclosed compounds are well known to those of skill in the art and include, without limitation, those set forth by class in Table 2, which is shown below.

TABLE 2

Family & Group #	Common Names	Trade Names (Combination Products)
Benzimidazole (Group 1)	benomyl	Benlate, Tersan 1991
	thiabendazole	Arbotect 20-S, Decco Salt No. 19, LSP
	thiophanate-methyl	Flowable Fungicide, Mertect 340-F Cavalier, Cleary's 3336, OHP 6672, Regal SysTec, Tee-Off, T-Methyl 4.5F AG, TM 85, Topsin M
Dicarboximide (Group 2)	iprodione	Epic 30, Ipro, Meteor, Nevada, OHP Chipco 26019, Rovral, (Interface)
	vinclazolin	Curalan, Ronilan

TABLE 2-continued

Family & Group #	Common Names	Trade Names (Combination Products)
Phenylpyrroles (Group 12)	fludioxonil	Cannonball, Emblem, Maxim, Medallion, Mozart, Scholar, Spirato, (Academy, Miravis Prime, Palladium, Switch)
Anilinopyrimidines (Group 9)	cyprodinil	Vanguard (Palladium, Switch, Inspire Super)
	pyrimethanil	Penbotec, Scala, (Luna Tranquility)
Hydroxyanilide (Group 17)	fenhexamid	Decree, Elevate, Judge
	fenpyrazamine	Protexio
Carboxamide (Group 7)	boscalid	Emerald, Endura, (Encartis, Honor, Pageant, Pristine)
	carboxin	Vitavax
	fluopyram	Luna Privilege, Velum Prime (Broadform, Luna Experience, Luna Sensation, Luna Tranquility, Propulse)
	flutolanil	Contrast, Moncut, ProStar
	fluxapyroxad	(Lexicon, Merivon, Orkestra)
	inpyrfluxam	Excalia
	isofetamid	Kenja
	oxycarboxin	Carboject, Plantvax
	penthiopyrad	Fontelis, Velist, Vertisan
	pydiflumetofen	Miravis, Posterity, Miravis Ace A (Miravis Neo, Miravis Prime, Miravis Duo, Miravis Top)
	solatenol (benzovindiflupyr)	Aprovia (Contend A, Elatus, Mural)
Phenylamide (Group 4)	mefenoxam	Apron, Ridomil Gold, Subdue MAXX, (Quadris Ridomil Gold, Uniform)
	metalaxyl	Acquire, Allegiance, MetaStar, Ridomil, Sebring, Subdue
	oxadixyl	Anchor
Phosphonate (Group P7)	aluminum tris Phosphorous Acid	Aliette, Flanker, Legion, Signature, Areca Agri-Fos, Alude, Appear, Fiata, Fosphite, Phospho Jet, Phostrol, Rampart, Reload
Cinnamic acid (Group 40)	dimethomorph	Forum, Stature, (Orvego, Zampro)
	mandipropamid	Micora, Revus, (Revus Top)
OSBPI (Group 49)	oxathiapiprolin	Segovis
Triazoles carboxamide (Group 22)	ethaboxam	V-10208
Group 27	cymoxanil	Curzate, (Tanos)
Carbamate (Group 28)	propamocarb	Banol, Previcur, Proplant, Tattoo
Benzamide (Group 43)	fluopicolide	Adorn, Presidio
	Demethylation-inhibiting (Group 3)	
Piperazines	triforine	Funginex, Triforine
Pyrimidines	fenarimol	Focus, Rubigan, Vintage
Imidazole	imazalil	Fungaflor, (Raxil MD Extra)
	triflumizole	Procure, Terraguard, Trionic
Triazoles	cyproconazole	Sentinel
	difenoconazole	Dividend, Inspire, (Academy, Briskway, Contend A, Inspire Super, Quadris Top, Revus Top) Miravis Duo
	fenbuconazole	Enable, Indar
	flutriafol	Topguard, (Topguard EQ)
	mefentrifluconazole	Maxtima (Navicon)
	metconazole	Quash, Tourney
	ipconazole	Rancona
	myclobutanil	Eagle, Hoist, Immunox, Laredo, Nova, Rally, Sonoma, Systhane
	propiconazole	Alamo, Banner, Break, Bumper, Infuse, Kestrel Mex, Miravis Ace B, PropiMax, ProPensity, Strider, Tilt, Topaz, (Aframe Plus, Concert, Contend B, Headway, Quilt Xcel, Stratego)
	prothioconazole	Proline (Propulse)
	tebuconazole	Bayer Advanced, Elite, Folicur, Lynx, Mirage, Orius, Raxil, Sativa, Tebucon, Tebuject, Tebusha, Tebustar, Toledo, (Absolute, Luna Experience, Unicorn), etc.
	tetraconazole	Mettle
	triadimefon	Bayleton, Strike, (Armada, Tartan, TrÍlgo)
	triadimenol	Baytan
	triticonazole	Charter, Trinity, (Pillar)
Morpholine (Group 5)	piperalin	Pipron
	spiroxamine	Accrue
Group U6	cyflufenamid	Torino
Group 50	metrafenone	Vivando
	pyriofenone	Proливо

TABLE 2-continued

Family & Group #	Common Names	Trade Names (Combination Products)
QoI Strobilurins (Group 11)	azoxystrobin	About, Aframe, Dynasty, Heritage, Prot��t��, Quadris, Quilt, (Aframe Plus, Briskway, Contend B, Dexter Max, Elatus, Headway, Mural, Quadris Top, Quilt Xcel, Renown, Topguard EQ, Uniform) (Tanos)
	femoxadone	Fenstop, Reason
	fenamidone	Aftershock, Disarm, Evito, Fame
	fluoxastrobin	Cygnus, Sovran
	kresoxim-methyl	Intuity, Pinpoint
	mandestrobin	Approach
	picoxystrobin	Cabrio, Empress, Headline, Insignia, Stamina, (Honor, Lexicon, Merivon, Navicon, Orchestra, Pageant, Pillar, Pristine)
	pyraclostrobin	Compass, Flint, Gem, (Absolute, Armada, Broadform, Interface, Luna Sensation, Stratego, Tartan, Trigo)
	trifloxystrobin	Quintec
Quinoline (Group 13)	quinoxyfen	Inorganic Compounds
Coppers (Group M1)	bordeaux	None
	copper ammonium complex	Copper Count-N
	copper hydroxide	Champ, Champion, Kalmor, Kentan, Kocide, Nu-Cop
	copper oxide	Nordox
	copper oxychloride	C—O—C—S, Oxycop
	copper sulfate	Cuprofix Disperss, many others
Sulfur (Group M2)	sulfur	Cosavet, Kumulus, Microthiol Disperss, Thiosperse
Lime sulfur	Ca polysulfides	Lime Sulfur, Sulforix
Ethylenebisdithiocarbamates (EBDC) (Group M3)	mancozeb	Dithane, Fore, Penncozeb, Protect, Manex, Manzate, Roper, Wingman, (Dexter Max, Gavel)
	maneb	Maneb
	metiram	Polyram
EBDC-like (Group M3)	ferbam	Carbamate, Ferbam
	thiram	Difiant, Spotrete, Thiram
	ziram	Ziram
Aromatic Hydrocarbon (Group 14)	dicloran (DCNA)	Allisan, Botran
	etridizole	Terrazole, Truban
	pentachloronitrobenzene	Autilus, Defend, Engage, PCNB, Terraclor, (Premion)
Chloronitrile (Group M5)	chlorothalonil	Bravo, Daconil, Docket, Echo, Ensign, Exotherm
		Termil, Funginil, Legend, Manicure, Pegasus, Terranil, (Concert, Spectro)
Phthalimides (Group M4)	captan	Captan
Guanidines (Group U12)	dodine	Syllit
QII fungicides (Group 21)	cyazofamid	Ranman, Segway
Polyoxin (Group 19)	polyoxin	Affirm, Endorse, Oso, Ph-D, Tavano, Veranda
Group 29	fluazinam	Omega, Secure
Thiazolidine (U13)	flutianil	Gatten

[0146] Fungicides are cataloged more broadly by the Fungicide Resistance Action Committee (FRAC) in the FRAC Code List 2022 and reproduced in Appendix 1 and which is incorporated herein by reference in its entirety.

[0147] In one embodiment, a presently disclosed enhancer compound is used in combination with one or more compound from the Families or Groups set forth in Table 2, Appendix 1, or both. In certain embodiments, a presently disclosed enhancer is used in combination with one or more fungicides recited in column 1 of Table 2.

[0148] In particular embodiments, a disclosed enhancer is used in combination with one or more of a fungicide selected from the benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors (OSBPI), triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piprazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles, and triazoles, such as cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, tri-

adimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam and thiazolidines.

[0149] Particular fungicides that are potentiated by use in combination with an enhancer according to the methods herein by administration of an apyrase inhibitor are coppers, such as copper octanoate, copper hydroxide and the like, myclobutanil, propiconazole, tebuconazole, epoxiconazole, difenoconazole, triticonazole, and prothioconazole.

[0150] In one embodiment, the combined treatment with a selected fungicide and an enhancer according to the present disclosure provides synergistic fungicidal activity against plant pathogenic fungi.

[0151] In one embodiment, the disclosure provides compositions and methods of treating plants or plant seeds infected with or at risk of being infected with a fungal pathogen. In one embodiment compositions of the present disclosure comprise a formulation of a fungicide, an enhancer and a phytolegically acceptable carrier. In another

embodiment, the fungicide and enhancer are administered in separate compositions. In further embodiments, an agricultural or horticultural fungicide is used in combination with other compounds in addition to the presently disclosed apyrase inhibitors. As with the apyrase inhibitors, such other compounds can be administered in the same or separate compositions as the fungicide. Examples of the other components include known carriers to be used to conduct formulation. Additional examples thereof include conventionally-known herbicides, insecticidal/acaricidal agents, nematodes, soil pesticides, plant control agents, synergistic agents, fertilizers, soil conditioners, and animal feeds. In one embodiment, the inclusion of such other components yields synergistic effects on crop growth.

[0152] In one embodiment, the presently disclosed compounds are used to potentiate the effect of a herbicide. Exemplary herbicides for use in combination with the present compounds are known to those of skill in the art and include, without limitation, those described in Appendix 2. By way of example, suitable herbicides for use in combination with the present compounds include inhibitors of acetyl COA synthase, inhibitors of acetolactate synthesis, inhibitors of microtubule assembly, inhibitors of microtubule organization, auxin mimics, photosynthesis inhibitors, deoxy-D-xylulose phosphate synthase inhibitors, enolpyruvyl shikimate phosphate synthase inhibitors, phytoene desaturase inhibitors, glutamine synthetase inhibitors, dihydropteroate synthesis inhibitors, protoporphyrinogen oxidase inhibitors, cellulose synthesis inhibitors, uncouplers, hydroxyphenyl pyruvate dioxygenase inhibitors, fatty acid thioesterase inhibitors, serine-threonine protein phosphatase inhibitors, solanesyl diphosphate synthase inhibitors, inhibitors of very long-chain fatty acid synthesis, homogentisate solanesyltransferase inhibitors, lycopene cyclase inhibitors.

[0153] In one embodiment, the presently disclosed compounds are used to potentiate the effect of an insecticide. Exemplary insecticides for use in combination with the present compounds are known to those of skill in the art and include, without limitation, those described in Appendix 3.

FORMULATIONS

Formulations

[0154] The present disclosure provides specific apyrase inhibitors to enhance the potency of pesticides to effectively restrict the growth of plant pathogenic species. In certain non-limiting embodiments, the apyrase inhibitors can be provided at: from about 0.01 to about 80% weight to weight in a final composition, or from about 25% to about 55%, such as from about 30% to about 50%, from about 35% to about 45%, such as about 0.01, 0.05, 0.1, 0.5, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10, 20, 30, 40, 50, 55, 60 or 80% weight to weight in a final composition. In one embodiment the apyrase inhibitors are provided in liquid form at from about 0.01 to about 50%, such as from about 15% to about 50%, from about 20% to about 45%, from about 25% to about 40%, such as about 0.01, 0.05, 0.1, 0.5, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10, 15, 20, 30, 40 or 50% volume to volume in a final diluted composition. The skilled artisan will recognize that the formulation of the pesticide, the apyrase inhibitor or a combination thereof can be pro-

vided in a concentrate that can be diluted prior to use, or can be provided in a diluted form ready for treatment.

[0155] The enhancer, pesticide and combinations thereof are not particularly limited by the dosage form. Examples of the dosage form include wettable powders, emulsions, emulsifiable concentrates, oil-dispersible liquids, powders, granules, water-soluble agents, suspensions, granular wettable powders, and tablets. The method for preparing formulation is not particularly limited, and conventionally-known methods may be adopted depending on the dosage form.

[0156] Several formulation examples are described below. The preparation formulations shown below are merely examples, and may be modified within a range not contrary to the essence of the present disclosure. For example, additional active and inert components may be added to the formulations below.

[0157] “Part” means “part by mass” unless otherwise specified.

Formulation Example 1: Wettable Powders

[0158] 40 parts of an enhancer disclosed herein, 53 parts of diatomaceous earth, 4 parts of ethoxylated higher alcohol sulfate ester combined with a suitable solid carrier such as magnesium sulfate, and 3 parts of alkyl naphthalene sulfonate are mixed uniformly, and then finely pulverized to obtain wettable powders containing 40 parts by mass of the enhancer.

Formulation Example 2: Emulsifiable Concentrates

[0159] 3 parts of an enhancer disclosed herein, 60 parts of mixed petroleum distillates, 27 parts of dimethyl lactamide, and 10 parts of tristyrilphenol ethoxylates are mixed and dissolved to obtain an emulsifiable concentrate containing 3% by mass of the enhancer.

Formulation Example 3: Granules

[0160] 5 parts of an enhancer disclosed herein, 10 parts of talc, 38 parts of clay, 10 parts of bentonite, 30 parts of sodium lignosulfonate and 7 parts of sodium alkyl sulfate are mixed uniformly, and then finely pulverized, followed by conducting fluidized bed granulation to make the median particle diameter thereof be 0.2 to 2.0 mm, and thus granules containing 5% by mass of an enhancer on a dry weight basis disclosed herein are obtained.

Formulation Example 4: Granules

[0161] 5 parts of an enhancer disclosed herein, 73 parts of clay, 20 parts of bentonite, 1 part of sodium dioctyl sulfosuccinate, and 1 part of potassium phosphate are mixed and then pulverized, followed by adding water thereto, and then kneading the mixture. Then, extrusion granulation is conducted, and the resultant is dried to obtain granules containing 5% by mass of the enhancer on a dry weight basis.

Formulation Example 5: Suspensions

[0162] 10 parts of an enhancer disclosed herein, 4 parts of polyoxyethylene alkyl ether, 2 parts of 3 kDa sodium polycarboxylate as dispersant, 10 parts of glycerin, 0.2 parts of xanthan gum, 0.1 parts of biocides as stabilizer, 0.1 parts of organosilicone antifoam emulsion and 73.6 parts of water

are mixed, and then wet pulverized until the particle size is 3 microns or less to obtain a suspension containing 10% by mass of the enhancer.

Formulation Example 6: Oil Dispersible Concentrates

[0163] 40 parts of an enhancer disclosed herein, 5 parts of Atlox 4914, 5 parts of organo-modified bentonite and 50 parts of methylated rapeseed oil as carrier are mixed uniformly and then wet pulverized until the median particle size is 3 microns or less to obtain an oil dispersible concentrate containing 40% by mass of the enhancer.

[0164] The skilled artisan will recognize that the various compositions are used commercially at varying concentrations and formulations. For example, it is common for fungicides to be formulated as liquids commercially at 10-40% concentrations. In one embodiment, the presently disclosed enhancers allow the use of a lower amount of a given fungicide due to the enhanced efficacy of fungicide in combination with an enhancer disclosed herein.

Methods for Assessing Enhanced Activity

Apyrase Inhibition Assay

[0165] Apyrase inhibitors useful as enhancers of pesticidal activity are assessed using an in vitro assay. The method of Windsor, *Bio Techniques* 33:1024-1030 (November 2002) was used as follows

Screen for Apyrase Inhibitors—

[0166] 96 well plates were used for the assay: (Greiner bio-one: REF-655901-96 well, PS, F-bottom, Clear, Non-binding)

Buffers:

[0167] Reaction Buffer: 60 mM Hepes; 3 mM $MgCl_2$, 3 mM $CaCl_2$ and 3 mM ATP (pH 6.5)

[0168] Development Buffer A: 2% aqueous ammonium molybdate

[0169] Development Buffer B: 11% ascorbic acid in 37.5% TCA in water

[0170] Stop buffer C: 2% trisodium citrate in 2% acetic acid solution in water

Steps:

[0171] Add 100 μ l of reaction buffer to each well.

[0172] Add 10 μ l of DMSO (control) or inhibitor/compound or compounds such as N1915 or orthovanadate to each well. (use inhibitor conc at 1 mM; orthovanadate at 2 mM and N1915 at 1 mM)

[0173] Add 10 μ l of apyrase (concentration based on optimization—Dilute 1 U/ μ l enzyme to different concentrations such as 0.1 U, 0.05 U, 0.0025, 0.001 U, 0.0005 U—to find a good range)

[0174] Incubate plate at room temperature for 1 hr

[0175] Mix development buffer A and B in the ratio of 1:1.5 (just before use).

[0176] Add 50 μ l of A: B mix in each well (incubate for 2 mins)

[0177] Add 50 μ l of C in each well

[0178] Measure/Read Absorbance of plate @630 nm

[0179] Inhibitory data for the apyrase assay described above are provided for selected compounds in Table 3. D is inhibition of 0% to 10%, C is inhibition from 10% to 20%, B is inhibition from 20% to 30%, and A is inhibition above 30%. Blank entries signify compounds that were not tested.

TABLE 3

Compound	Inhibition
A-1	C
A-3	A
A-4	B
A-5	A
A-6	D
A-7	C
A-8	A
A-9	A
A-10	A
A-11	D
A-12	C
A-13	D
A-14	D
A-15	D
A-16	C
A-18	C
A-19	B
A-20	D
A-21	A
A-22	D
A-23	B
A-24	C
A-25	D
A-26	B
A-27	D
A-28	D
A-29	A
A-30	C
A-31	D
A-32	C
A-33	B
A-34	A
A-35	D
A-36	A
A-37	B
A-38	C
A-39	D
A-40	D
A-41	A
A-42	B
A-43	C
A-44	D
A-45	C
A-46	A
A-47	B
A-48	B
A-49	D
A-50	C
A-51	D
A-52	C
A-53	C
A-54	D
A-55	D
A-56	C
A-57	D
A-58	D
A-59	A
A-60	D
A-61	D
A-62	A
A-63	C
A-64	A
A-65	B
A-66	C
A-67	C
A-68	D
A-69	D
A-70	C

TABLE 3-continued

Compound	Inhibition
A-71	D
A-72	C
A-73	D
A-74	C
A-75	D
A-76	C
A-77	D
A-78	D
A-79	A
A-80	D
A-81	D
A-82	C
A-83	B
A-84	D
A-85	D
A-86	D
A-87	C
A-88	D
A-89	D
A-90	A
A-91	B
A-92	B
A-93	D
A-94	A
A-96	C
A-97	C
A-98	A
A-99	A
A-100	D
A-101	B
A-102	D
A-103	B
A-104	D
A-106	D
A-107	C
A-108	A
A-109	C
A-110	C
A-111	D
A-113	D
A-114	D
A-115	A
A-116	A
A-117	D
A-118	D
A-119	D
A-120	B
A-122	C
A-123	D

Method 2: In Vitro Assessment of Combination Activity

[0180] Selected compounds are assessed in combination with fungicides against a range of commercially important plant pathogenic fungi.

[0181] The test is conducted as follows. A fungicide is applied to a fungal plant pathogen at a rate slightly below that at which it gave any control, in combination with a suitable dose of the test compound. The test compounds are recorded as active if control of the pathogen was observed.

[0182] In more detail, the test is conducted as follows. For each combination of fungicide, pathogen and test compound, the following wells are used. Well 1 contains a fungal pathogen growing on agar, and a fungicide at a rate just below that at which it gave any control of the pathogen. Well 2 is the same as Well 1, except that the test compound is also added at Rate 1. Well 3 is the same as Well 2, except that the test compound is added at Rate 2, where Rate 2 is higher than Rate 1. Finally, as a benchmark, Well 4 is the same as

Well 1, except that it contained the fungicide at a higher rate, at which it gave partial control of the pathogen. Each of the Wells 1 to 4 are run in duplicate, giving a total of 8 wells for each combination of fungicide, pathogen and test compound. For each well, after a suitable period of incubation, a visual assessment of the % control of the pathogen by the fungicide is made. Test compounds are scored as inactive, active or highly active.

[0183] The following fungicides are used in this assay: azoxystrobin, fluxapyroxad, and desthio prothioconazole. The following fungal pathogens are used in this assay: First, a strain of *Zymoseptoria tritici* with a reduced susceptibility to strobilurin fungicides; second a strain of *Zymoseptoria tritici* with a reduced susceptibility to SDHI fungicides (i.e., those that inhibit succinate dehydrogenase); and third, *Microdochium nivale*. In this assay, Roux Compound 13 exhibited activity in only one combination, enhancing the activity of azoxystrobin against *Microdochium nivale*, but failing to enhance activity of any fungicide against either of the *Zymoseptoria tritici* strains. In contrast, compounds of the present disclosure are effective in the combination assay, enhancing fungicidal activity of one or more fungicides against at least one strain.

Method 3: Greenhouse Crop Tests

[0184] In this method, exemplary compounds were evaluated for their ability to control Brown Rust (*Puccinia recondita*) on wheat in a controlled greenhouse environment in combination with one of four fungicides, Amistar, Imtrex, Proline or Balaya. In these studies, wheat plants (JB Diego) were used. Seeds were sown in 9 cm diameter pots to a depth of 1 to 2 cm using Petersfield potting compost (75% medium grade peat, 12% screened sterilized loam, 3% medium grade vermiculite, 10% grit (5 mm screened, lime free), 1.5 kg PG mix per m³, lime to pH5.5-6.0 and wetting agent (Vitax Ultrawet 200 ml per m³) and germinated/grown at 23 °C under a 16 h day/8 h night light regime. Plants were treated two to three weeks after sowing when they were at the BBCH 11 growth stage (first pair of true leaves (unifoliate) unfolded). Wheat plants were inoculated with *Puccinia triticina* (Brown rust) 24 hours after treatment. A track sprayer was used to treat the plants with the mixture of commercial fungicide and test compound using a water volume of 200 L/ha. Four replicates were used for each combination of fungicide, pathogen and test compound. Each plant was evaluated at fourteen days (once the disease symptoms were fully expressed) for % control of the disease. Appropriate controls were used for all experiments, including an 'inoculation check' wherein plants were inoculated with their specific pathogen to assess disease levels. Also, each commercial fungicide was tested on its own as a part of each treatment, this being benchmark against which the experimental compounds were evaluated. Exemplary compounds demonstrated enhanced disease control in combination with fungicides as compared to disease control observed with fungicide alone. That is, the present compounds, although not fungicidal by themselves, enhance the activity of fungicides, thus the enhancer compounds work synergistically in combination with fungicides to control disease.

[0185] Amistar (0.04 L/ha)

[0186] Imtrex (0.45 L/ha)

[0187] Proline (0.15 L/ha)

[0188] Balaya (0.25 L/ha)

TABLE 4

Compound	Amistar	Imtrex	Proline	Balaya
A-3	81.4%	47.6%	3.8%	38.9%
A-5	89.2%	24.3%	15.1%	13.0%
A-8	72.9%	47.8%	13.3%	77.2%
A-9	65.5%	-13.9%	12.4%	54.9%
A-10	87.4%	15.0%	4.2%	56.5%
A-21	48.5%	47.7%	1.2%	39.0%
A-29	67.0%	60.1%	18.7%	59.5%
A-34	51.9%	34.5%	3.8%	48.3%
A-37	58.7%	12.7%	-1.2%	78.0%
A-59	95.0%	30.9%	15.6%	50.0%
A-62	71.7%	38.8%	1.4%	21.7%
A-64	69.5%	-15.4%	2.9%	-75.9%
A-99	71.4%	16.1%	5.7%	84.1%
A-103	37.0%	13.2%	4.4%	67.2%
A-108	92.1%	40.5%	-1.2%	31.7%
A-116	68.0%	48.4%	14.6%	24.2%

Method 4: Enhancement of Herbicide Activity

[0189] This method demonstrates the enhancement of herbicidal activity provided by the present compounds. Specifically, *Amaranthus retroflexus* (pigweed) was treated pre-emergence with metribuzin alone or in combination with an enhancer compound disclosed herein. Fifteen seeds of *Amaranthus retroflexus* were sown in soil in each pot and then sprayed with a mixture of metribuzin at the appropriate rate shown in Table 5 and the compound of this invention at 100 g/ha using a track sprayer at a water volume of 200 L/hectare. Small amounts of acetone were used to aid solubility. The pots were watered immediately after sowing and treatment and then allowed to stand in a glasshouse. Assessment of kill was made seven and fourteen days after treatment. The test included four replicates for each treatment and the control. Results are shown in Table 5. Exemplary compounds A-62, A-8, A-29, A-3, A-108, A-34, A-21, A-59 and A-103 demonstrated significant enhancement of metribuzin herbicidal activity at seven and fourteen days.

TABLE 5

Pre-Emergence Amaranthus Kill Percentage								
Enhancer	Assessed 7 DAT Metribuzin rate (g/ha)				Assessed 14 DAT Metribuzin rate (g/ha)			
	0	50	100	200	0	50	100	200
Compound								
None (metribuzin only)	0.0	0.0	15.2	60.9	4.8	7.7	38.3	74.0
A-62	—	23.5	58.9	95.8	—	35.7	61.0	96.0
A-8	—	12.8	28.6	100	—	18.8	35.67	100
A-29	—	13.2	66.7	87.7	—	17.1	71.4	90.0
A-3	—	15.6	68.9	34.0	—	21.7	72.0	38.5
A-108	—	6.3	76.5	100	—	13.5	78.8	100
A-34	—	28.3	70.5	100	—	37.3	77.8	100

TABLE 5-continued

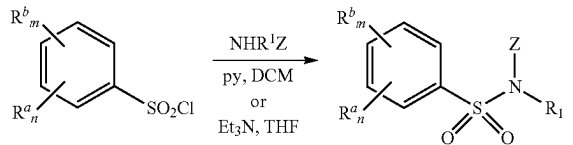
Pre-Emergence Amaranthus Kill Percentage								
Enhancer	Assessed 7 DAT Metribuzin rate (g/ha)				Assessed 14 DAT Metribuzin rate (g/ha)			
	0	50	100	200	0	50	100	200
Compound								
A-21	—	0.0	62.3	100	—	4.4	63.0	100
A-59	—	14.3	64.6	95.0	—	25.9	67.3	100
A-103	—	5.1	54.7	91.5	—	30.6	43.1	98.4

Methods of Making Enhancer Compounds

[0190] Exemplary enhancer compounds were purchased from commercial suppliers, such as Enamine, located at Industriepark Hoechst, G837. 65926 Frankfurt am Main Germany. Other compounds were prepared consistent with the methods set forth below:

Synthesis of Compounds

[0191] Compounds of the present disclosure can be synthesized according to the scheme shown below. Specifically, an aryl sulfonyl chloride can be allowed to react with the amine NHR¹R² to give the required sulfonamide. The R group on the aryl ring can have various forms and variations. In some cases, the aryl ring can be substituted with an R group at the ortho or meta position, or substituted with multiple R groups at multiple positions.



[0192] wherein:

[0193] n and m are each independently 0, 1, 2, or 3, provided that n plus m is less than or equal to 3.

[0194] R^a and R^b are as described herein. In some cases R^a is phenyl, alkyl, heteroaryl, or cycloalkyl.

[0195] Such syntheses are reported in Journal of Medicinal Chemistry 2022, 65, 15710-15724; Bioorganic & Medicinal Chemistry Letters 2020, 30, 127650; and Journal of Medicinal Chemistry 2004, 47, 4979-4982.

[0196] Many sulfonyl chloride (—SO₂Cl) starting compounds that can be used in the above scheme are commercially available, or can be chemically synthesized from commercially available compounds. For example, the compound 4-biphenylsulfonyl chloride (CAS number 1623-93-4) is an exemplary starting material where m is 0, n is 1, and the one R^a group is phenyl. 4-biphenylsulfonyl chloride is commercially sold by more than 60 chemical suppliers.

Appendix 1

[0197] Shown below is Appendix 1

MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
A②	A1 RNA polymerase I	PA - fungicides (PhenylAmides)	acylalanines	benalaxyl benalaxyl-M (=kiralaxyl) furalaxyl	Resistance and cross resistance well known in various Oomycetes but mechanism unknown.	4

-continued

MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
B: Cytoskeleton and motor protein	A2 adenosin- deaminase	hydroxy- (2-amino-) pyrimidines	oxazolidinones butyrolactones hydroxy- (2-amino-) pyrimidines	metalaxyl metalaxyl-M (=mefenoxam) oxadixyl ofurace bupirimate dimethirimol ethirimol	High risk. See FRAC Phenylamide Guidelines for resistance management	8
	A3 DNA/RNA synthesis (proposed)	heteroaromatics	isoxazoles isothiazolones	hymexazole oethilnone	Medium risk. Resistance and cross resistance known in powdery mildews. Resistance management required.	32
	A4 DNA topoisomerase type II (gyrase)	carboxylic acids	carboxylic acids	oxolinic acid	Resistance not known.	31
	A5 inhibition of dihydroorotate dehydrogenase within de novo pyrimidine biosynthesis	DHODHI- fungicides	phenyl-propanol	ipflufenquin	Bactericide. Resistance known. Risk in fungi unknown. Resistance management required. Medium to high risk.	52
	B1 tubulin polymerization	MBC - fungicides (Methyl Benzimidazole Carbamates)	benzimidazoles thiophanates	benomyl carbendazim fuberidazole thiabendazole thiophanate thiophanate-methy	Resistance common in many fungal species. Several target site mutations, mostly E198A/G/K, F200Y in β -tubulin gene. Positive cross resistance between the group members. Negative cross resistance to N-phenyl carbamates. High risk. See FRAC Benzimidazole Guidelines for resistance management.	1
	B2 tubulin polymerization	N-phenyl carbamates	N-phenyl carbamates	diethofencarb	Resistance known. Target site mutation E198K. Negative cross resistance to benzimidazoles. High risk. Resistance management required.	10
	B3 tubulin polymerization	benzamides thiazole carboxamide phenylureas	toluamides ethylamino-thiazole- carboxamide phenylureas	zoxamide ethaboxam	Low to medium risk. Resistance management required.	22
	B4 cell division (unknown site)			pencycuron	Resistance not known.	20
	B5 delocalisation of spectrin-like proteins	benzamides	pyridinylmethyl- benzamides	fluopicolide fluopimomide	Resistant isolates detected in grapevine downy mildew. Medium risk. Resistance management required	43
	B6 actin/myosin/fimbrin function	cyanoacrylates	aminocyanoacrylates	phenamacril	Resistant isolates detected in grapevine downy mildew. Medium risk. Resistance management required	47
		aryl-phenyl- ketones	benzophenone benzoylpyridine	metrafenone pyriofenone	Resistance known in <i>Fusarium graminearum</i> . Target site mutations in the gene coding for myosin- 5 found in lab studies. Medium to high risk. Resistance management required. Less sensitive isolates detected in powdery mildews (<i>Blumeria</i> and <i>Sphaerotheca</i>) Medium risk. Resistance management required. Reclassified from U8 in 2018	50

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
C. respiration	B7 tubulin dynamics modulator	pyridazine	pyridazine	pyridachlometyl	High risk.	53
	C1 complex I NADH oxido-reductase	pyrimidinamines pyrazole-MET1	pyrimidinamines pyrazole-5-carboxamides	diflumerim tolfeprad	Resistance not known.	39
	C2 complex II: succinate-dehydrogenase	Quinazoline SDHI (Succinate-dehydrogenase inhibitors)	quinazoline phenyl-benzamides phenyl-oxo-ethyl thiophene amide pyridinyl-ethyl-benzamides phenyl-cyclobutyl-pyridineamide furan- carboxamides oxathiin-carboxamides thiazole-carboxamides pyrazole-4-carboxamides N-cyclopropyl-N-benzyl-pyrazole-carboxamides N-methoxy-(phenyl-ethyl)-pyrazole-carboxamides pyridine-carboxamides pyrazine-carboxamides methoxy-acrylates	fenazaquin benodanil flutolanil mepronil isofetamid fluopyram cyclobutrifluram fenfuram carboxin oxycarboxin thifluzamide benzovindiflupyr bixafen fluindapyr fluxapyroxad furametpyr inpyrfluxam isopyrazam penflufen penthiopyrad sedaxane isoeflucypram pydiflumetofen boscalid pyraziflumid	Resistance known for several fungal species in field populations and lab mutants. Target site mutations in sdh gene, e.g. H/Y (or H/L) at 257, 267, 272 or P225L, dependent on fungal species. Resistance management required. Medium to high risk. See FRAC SDHI Guidelines for resistance management.	7
C. respiration	C3 complex III: cytochrome bc1 (ubiquinol oxidase) at Qo site (cyt b gene)	QoI-fungicides (Quinone outside Inhibitors)	methoxy-acetamide methoxy-carbamates oximino-acetates oximino-acetamides oxazolidine-diones dihydro-dioxazines imidazolinones benzyl-carbamates tetrazolinones	azoxystrobin coumoxystrobin enoxastrobin flufenoxystrobin picoxystrobin pyraoxystrobin mandestrobin pyraclostrobin pyrametostrobin triclopyricarb kresoxim-methyl trifloxystrobin dimoxystrobin fenaminstrobin metominostrobin orysastrobin famoxadone fluoxastrobin fenamidone pyribencarb metyltetraprole	Resistance known in various fungal species. Target site mutations in cyt b gene (G143A, F129L) and additional mechanisms. Cross resistance shown between all members of the Code 11 fungicides. High risk. See FRAC QoI Guidelines for resistance management.	11
		QoI-fungicides (Quinone outside Inhibitors; Subgroup A)			Resistance not known. Not cross resistant with Code 11 fungicides on G143A mutants. High risk. See FRAC QoI Guidelines for resistance management.	11A

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
C: respiration (continued)	C4 complex III: cytochrome bc1 (ubiquinone reductase) at Qi site	Qil - fungicides (Quinone inside Inhibitors)	cyano-imidazole sulfamoyl-triazole picolinamides	cyazofamid amisulbrom fepicoxamid florylpicoxamid	Resistance risk unknown but assumed to be medium to high (mutations at target site known in model organisms). Resistance management required. No spectrum overlap with the Oomycete-fungicides cyazofamid and amisulbrom Resistance not known. Also acaricidal activity.	21
	C5 uncouplers of oxidative phos- phorylation		dinitrophenyl- crotonates 2,6-dinitro-anilines	binapacryl meptyldinocap dinocap fluazinam	Resistance risk unknown but assumed to be medium to high (mutations at target site known in model organisms). Resistance management required. No spectrum overlap with the Oomycete-fungicides cyazofamid and amisulbrom Resistance not known. Also acaricidal activity.	29
	C6 inhibitors of oxidative phos- phorylation, ATP synthase	organo tin compounds	(pyr.-hydrazones) tri-phenyl tin compounds	(ferimzone) fentin acetate fentin chloride fentin hydroxide	Resistance risk unknown but assumed to be medium to high (mutations at target site known in model organisms). Resistance management required. No spectrum overlap with the Oomycete-fungicides cyazofamid and amisulbrom Resistance not known. Also acaricidal activity.	30
	C7 ATP transport (proposed)	thiophene- carboxamides	thiophene- carboxamides	silthiofam	Resistance reported. Risk low.	38
	C8 complex III: cytochrome bc1 (ubiquinone reductase) at Qo site, stigmatellin binding sub-site	QoSI fungicides (Quinone outside Inhibitor, stigmatellin binding type)	triazolo- pyrimidylamine	ametotradin	Not cross resistant to QoI fungicides. Resistance risk assumed to be medium to high (single site inhibitor). Resistance management required.	45
	D1 methionine biosynthesis (proposed) (cgs gene)	AP - fungicides (Anilino- Pyrimidines)	anilino-pyrimidines	cyprodinil mepaniprim pyrimethanil	Resistance known in <i>Botrytis</i> and <i>Venturia</i> , sporadically in <i>Oculimacula</i> . Medium risk. See FRAC Anilinopyrimidine Guidelines for resistance management.	9
	D2 protein synthesis (ribosome, termination step)	enopyranuronic acid antibiotic	enopyranuronic acid antibiotic	blastidicin-S	Low to medium risk. Resistance management required.	23
	D3 protein synthesis (ribosome, initiation step)	hexopyranosyl antibiotic	hexopyranosyl antibiotic	kasugamycin	Resistance known in fungal and bacterial (<i>P. glumae</i>) pathogens. Medium risk. Resistance management required.	24
D: amino acids and protein ②	D4 protein synthesis (ribosome, initiation step)	glucopyranosyl antibiotic	glucopyranosyl antibiotic	streptomycin	Bactericide. Resistance known. High risk. Resistance management required.	25
	D5 protein synthesis (ribosome, elongation step)	tetracycline antibiotic	tetracycline antibiotic	oxytetracycline	Bactericide. Resistance known. High risk. Resistance management required.	41
	E1 signal transduction (mechanism unknown)	aza- naphthalenes	aryloxyquinoline quinazolinone	quinoxifen proquinazid	Resistance to quinoxifen known. Medium risk. Resistance management required. Cross resistance found in <i>Erysiphe (Uncinula)</i> <i>necator</i> but not in <i>Blumeria</i> <i>graminis</i> .	13
	E2 MAP/Histidine- Kinase in osmotic signal transduction (os-2, HOG1)	PP-fungicides (PhenylPyrroles)	phenylpyrroles	fepiclonil fludioxonil	Resistance found sporadically, mechanism speculative. Low to medium risk. Resistance management required.	12

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
F: lipid synthesis or transport/membrane integrity or function	E3 MAP/Histidine-Kinase in osmotic signal transduction (os-1, Daf1)	dicarboximides	dicarboximides	chlozolate dimethachlone iprodione procymidone vinclozolin	Resistance common in <i>Botrytis</i> and some other pathogens. Several mutations in OS-1, mostly I365S. Cross resistance common between the group members. Medium to high risk. See FRAC Dicarboximide Guidelines for resistance management	2
	F1			formerly dicarboximides		
	F2 phospholipid biosynthesis, methyltransferase	phosphoro-thiolates	phosphoro-thiolates	edifenphos iprobenfos (IBP) pyrazophos isoprothiolane	Resistance known in specific fungi. Low to medium risk. Resistance management required if used for risky pathogens.	6
	F3 cell peroxidation (proposed)	AH-fungicides (Aromatic Hydrocarbons) (chlorophenyls, nitroanilines)	aromatic hydrocarbons	biphenyl chloroneb dicloran quintozone (PCNB) tecnazene (TCNB) tolclofos-methyl	Resistance known in some fungi. Low to medium risk. Cross resistance patterns complex due to different activity spectra.	14
	F4 cell membrane permeability, fatty acids (proposed)	heteroaromatics Carbamates	1,2,4-thiadiazoles carbamates	etridiazole iodocarb propamocarb prothiocarb	Low to medium risk. Resistance management required.	28
	F5			formerly CAA-fungicides		
	F6 microbial disrupters of pathogen cell membranes			formerly <i>Bacillus amyloliquefaciens</i> strains (FRAC Code 44); reclassified to BM02 in 2020		
	F7 cell membrane disruption			formerly extract from <i>Melaleuca alternifolia</i> (tea tree oil) and plant oils (eugenol, geraniol, thymol)		
	F8 ergosterol binding	Polyene	FRAC Code 46, reclassified to BM01 in 2021 amphoteric macrolide antifungal antibiotic from <i>Streptomyces natalensis</i> or <i>S. chattanoogensis</i>	natamycin (pimaricin)	Resistance not known. Agricultural, food and topical medical uses.	48
	F9 lipid homeostasis and transfer/storage	OSBPI oxysterol binding protein homologue inhibition	piperidinyl-thiazole- isoxazolines	oxathiapiroline fluoxapiroline	Resistance risk assumed to be medium to high (single site inhibitor). Resistance management required. (Previously U15).	49
G: sterol biosynthesis in ②	F10 interaction with lipid fraction of the cell membrane, with multiple effects on cell membrane integrity	protein fragment	polypeptide	polypeptide ASFBIOF01-02	Resistance not known.	51
	G1 C14- demethylase in sterol biosynthesis (erg11/cyp51)	DMI-fungicides (DeMethylation Inhibitors) (SBI: Class I)	piperazines pyridines pyrimidines imidazoles triazoles triazolinthiones	triforine pyrifeno pyrisoxazole fenarimol nuarimol imazalil oxpoconazole pefurazoate prochloraz triflumizole azaconazole bitertanol bromuconazole cyproconazole difenoconazole diniconazole epoxiconazole etaconazole	There are big differences in the activity spectra of DMI fungicides. Resistance is known in various fungal species. Several resistance mechanisms are known incl. target site mutations in cyp51 (erg 11) gene, e.g. V136A, Y137F, A379G, I381V; cyp51 promotor; ABC transporters and others. Generally wise to accept that cross resistance is present between DMI fungicides active against the same fungus. DMI fungicides are Sterol Biosynthesis Inhibitors (SBIs),	3

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
				fenbuconazole fluquinconazole flusilazole flutriafol hexaconazole imibenconazole ipconazole mefentrifluconazole metconazole myclobutanil penconazole propiconazole simeconazole tebuconazole tetraconazole triadimefon triadimenol triticonazole prothioconazole	but show no cross resistance to other SBI classes. Medium risk. See FRAC SBI Guidelines for resistance management.	
	G2 Δ^{14} -reductase and $\Delta^8 \rightarrow \Delta^7$ - isomerase in sterol biosynthesis (erg24, erg2)	amines ("morpholines") (SBI: Class II)	morpholines	aldimorph dodemorph fenpropimorph tridemorph fenpropidin piperalin spiroketal-amines spiroxamine	Decreased sensitivity for powdery mildews. Cross resistance within the group generally found but not to other SBI classes. Low to medium risk. See FRAC SBI Guidelines for resistance management	5
	G3 3-keto reductase, C4- de-methylation (erg27)	KRI fungicides (KetoReductase Inhibitors) (SBI: Class III)	hydroxyanilides amino-pyrazolinone	fenhexamid fenpyrazamine	Low to medium risk. Resistance management required.	17
	G4 squalene-epoxidase in sterol biosynthesis (erg1)	(SBI class IV)	thiocarbamates allylamines	pyributicarb naftifine terbinafine	Resistance not known, fungicidal and herbicidal activity. Medical fungicides only.	18
H: cell wall biosynthesis	H3		Formerly glucopyranosyl antibiotic (validamycin)		reclassified to U18	26
	H4 chitin synthase	polyoxins	peptidyl pyrimidine nucleoside	polyoxin	Resistance known. Medium risk. Resistance management required.	19
	H5 cellulose synthase	CAA-fungicides (Carboxylic Acid Amides)	cinnamic acid amides valinamide carbamates mandelic acid amides	dimethomorph flumorph pyrimorph benthiavalicarb iprovalicarb valifenalate mandipropamid	Resistance known in <i>Plasmopara viticola</i> but not in <i>Phytophthora infestans</i> . Cross resistance between all members of the CAA group. Low to medium risk. See FRAC CAA Guidelines for resistance management.	40
I: melanin synthesis in cell wall	I1 reductase in melanin biosynthesis	MBI-R (Melanin Biosynthesis Inhibitors - Reductase)	isobenzo-furanone pyrrolo-quinolinone triazolobenzo- thiazole	ftalide pyroquilon tricyclazole	Resistance not known.	16.1
	I2 dehydratase in melanin biosynthesis	MBI-D (Melanin Biosynthesis Inhibitors - Dehydratase)	cyclopropane- carboxamide carboxamide propionamide	carpropamid diclocymet fenoxanil	Resistance known. Medium risk. Resistance management required.	16.2
	I3 polyketide synthase in melanin biosynthesis	MBI-P (Melanin Biosynthesis Inhibitors - Polyketide synthase)	trifluoroethyl- carbamate	tolprocarb	Resistance not known. Additional activity against bacteria and fungi through induction of host plant defence	16.3
P: host plant	P 01 salicylate-related	benzo- thiadiazole (BTH)	benzo-thiadiazole (BTH)	acibenzolar-S-methyl	Resistance not known.	P 01

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
defence induction	P 02 salicylate-related	benzisothiazole	benzisothiazole	probenazole (also antibacterial and antifungal activity)	Resistance not known.	P 02
	P 03 salicylate-related	thiadiazole-carboxamide	thiadiazole-carboxamide	tiadinil	Resistance not known.	P 03
	P 04 polysaccharide elicitors	natural compound	polysaccharides	isotianil laminarin	Resistance not known.	P 04
	P 05 anthraquinone elicitors	plant extract	complex mixture, ethanol extract (anthraquinones, resveratrol)	extract from <i>Reynoutria sachalinensis</i> (giant knotweed)	Resistance not known.	P 05
	P 06 microbial elicitors	microbial	bacterial <i>Bacillus</i> spp. fungal <i>Saccharomyces</i> spp.	<i>Bacillus mycoides</i> isolate J cell walls of <i>Saccharomyces cerevisiae</i> strain LAS117	Resistance not known.	P 06
	P 07 phosphonates	phosphonates	ethyl phosphonates	fosetyl-Al phosphorous acid and salts	Few resistance cases reported in few pathogens. Low risk. Reclassified from U33 in 2018	P07
U: Unknown mode of action (U numbers not appearing in the list derive from reclassified fungicides)	P 08 salicylate-related	isothiazole	isothiazolylmethyl ether	dichlobentiazox	activates SAR both up- and downstream of SA. Resistance not known.	P 08
	unknown	cyanoacetamide-oxime	cyanoacetamide-oxime	cymoxanil	Resistance claims described. Low to medium risk. Resistance management required.	27
	formerly phosphonates (FRAC code 33), reclassified to P 07 in 2018					
	unknown	phthalamic acids	phthalamic acids	tecloftalam (Bactericide)	Resistance not known.	34
	unknown	benzotriazines	benzotriazines	triazoxide	Resistance not known.	35
	unknown	benzene-sulfonamides	benzene-sulphonamides	flusulfamide	Resistance not known.	36
	unknown	pyridazinones	pyridazinones	diclomezine	Resistance not known.	37
	formerly methasulfocarb (FRAC code 42), reclassified to M 12 in 2018					
	unknown	phenyl-acetamide	phenyl-acetamide	cyflufenamid	Resistance in <i>Sphaerotheca</i> . Resistance management required	U 06
	cell membrane disruption (proposed)	guanidines	guanidines	dodine	Resistance known in <i>Venturia inaequalis</i> . Low to medium risk. Resistance management recommended.	U 12
	unknown	thiazolidine	cyano-methylene-thiazolidines	flutianil	Resistance in <i>Sphaerotheca</i> and <i>Podosphaera xanthii</i> . Resistance management required.	U 13
	unknown	pyrimidinone-hydrazones	pyrimidinone-hydrazones	ferimzone	Resistance not known (previously C5).	U 14
	complex III: cytochrome bc1, unknown binding site (proposed)	4-quinolyl-acetate	4-quinolyl-acetates	tebufloquin	Not cross resistant to Qol. Resistance risk unknown but assumed to be medium. Resistance management required.	U 16
Not specified	Unknown	tetrazoloxime	tetrazoloximes	picarbutrazox	Resistance not known. Not cross resistant to PA, Qol, CAA.	U 17
	Unknown (Inhibition of trehalase)	glucopyranosyl antibiotic	glucopyranosyl antibiotics	validamycin	Resistance not known. Induction of host plant defense by trehalose proposed (previously H3).	U 18
	Unknown	diverse	diverse	mineral oils, organic oils, inorganic salts, material of biological origin	Resistance not known.	NC

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
M: Chemicals with multi-site activity	multi-site contact activity	inorganic (electrophiles)	inorganic	copper (different salts)	Also applies to organic copper complexes generally considered as a low risk group without any signs of resistance developing to the fungicides. reclassified from U42 in 2018	M 01
		inorganic (electrophiles)	inorganic	sulphur		M 02
		dithiocarbamates and relatives (electrophiles)	dithio-carbamates and relatives	amobam ferbam mancozeb maneb metiram propineb thiram zinc thiazole zineb ziram		M 03
		phthalimides (electrophiles)	phthalimides	captan captafol folpet		M 04
		chloronitriles (phthalonitriles) (unspecified mechanism)	chloronitriles (phthalonitriles)	chlorothalonil		M 05
		sulfamides (electrophiles)	sulfamides	dichlofluanid tolylfluanid		M 06
		bis-guanidines (membrane disruptors, detergents)	bis-guanidines	guazatine iminocetadine		M 07
		triazines (unspecified mechanism)	triazines	anilazine		M 08
		quinones (anthraquinones) (electrophiles)	quinones (anthraquinones)	dithianon		M 09
		quinoxalines (electrophiles)	quinoxalines	chinomethionat/ quinomethionate		M 10
		maleimide (electrophiles)	maleimide	fluoroimide		M 11
		thiocarbamate (electrophiles)	thiocarbamate	methasulfocarb		M 12
		plant extract	polypeptide (lectin)	extract from the cotyledons of lupine plantlets ("BLAD")	Resistance not known. (previously M12).	BM 01
		plant extract	phenols, sesquiterpenes, triterpenoids, coumarins	extract from <i>Swinglea glutinosa</i>	Resistance not known.	
		plant extract	terpene hydrocarbons, terpene alcohols and terpene phenols	extract from <i>Melaleuca alternifolia</i> (tea tree oil) plant oils (mixtures); eugenol, geraniol, thymol	Resistance not known. (previously F7)	
BM: Biologicals with multiple modes of action: Plant extracts	multiple effects on ion membrane transporters; chelating effects affects fungal spores and germ tubes, induced plant defense					
BM: Biologicals with multiple modes of action: Microbial (living microbes, extracts or metabolites)	multiple effects described (examples, not all apply to all biological groups): competition, mycoparasitism, antibiosis, membrane disruption by fungicidal lipopeptides, lytic enzymes, induced plant defence	microbial (strains of living microbes or extract, metabolites)	fungus <i>Trichoderma</i> spp.	<i>T. atroviride</i> strain I-1237 strain LU132 strain SC1 strain SKT-1 strain 77B <i>T. asperellum</i> strain T34 strain kd <i>T. harzianum</i> strain T-22 <i>T. virens</i> strain G-41 <i>C. rosea</i> strain J1446 strain CR-7	nomenclature change from <i>Gliocladium catenulatum</i> to <i>Clonostachys rosea</i> Resistance not known. <i>Bacillus amyloliquefaciens</i> reclassified from F6, Code 44 in 2020 synonyms for <i>Bacillus amyloliquefaciens</i> are <i>Bacillus subtilis</i> and <i>B. subtilis</i> var. <i>amyloliquefaciens</i> (previous taxonomic classification).	BM 02
			fungus <i>Clonostachys</i> spp.			

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MOA	TARGET SITE AND CODE	GROUP NAME	CHEMICAL OR BIOLOGICAL GROUP	COMMON NAME	COMMENTS	FRAC CODE
			fungal	<i>C. minitans</i>		
			<i>Coniothyrium</i> spp.	strain CON/M/91-08		
			fungal	<i>H. uvarum</i>		
			<i>Hanseniaspora</i> spp.	strain BC18Y		
			fungal	<i>T. flavus</i>		
			<i>Talaromyces</i> spp.	strain SAY-Y-94-01		
			fungal	<i>S. cerevisiae</i>		
			<i>Saccharomyces</i> spp.	strain LAS02		
				strain DDSF623		
			bacterial	<i>B. amyloliquefaciens</i>		
			<i>Bacillus</i> spp.	strain QST713		
				strain FZB24		
				strain MBI600		
				strain D747		
				strain F727		
				strain AT-332		
				<i>B. subtilis</i>		
				strain AFS032321		
				strain Y1336		
				strain HAI-0404		
				PHC25279		
			bacterial			
			<i>Erwinia</i> spp.			
			(peptide)			
			bacterial	<i>G. cerinus</i>		
			<i>Gluconobacter</i> spp.	strain BC18B		
			bacterial	<i>P. chlororaphis</i>		
			<i>Pseudomonas</i> spp.	strain AFS009		
			bacterial	<i>S. griseovirides</i>		
			<i>Streptomyces</i> spp.	strain K61		
				<i>S. lydicus</i>		
				strain WYEC108		

② indicates text missing or illegible when filed

Appendix 2

[0198] Shown below is Appendix 2.

MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Alloxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Butroxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Clethodim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Cloproxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Cycloxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Profoxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Sethoxydim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Tepraloxymdim
Inhibition of Acetyl CoA Carboxylase	Cyclohexanediones (DIMs)	Tralkoxydim
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Clodinafop-propargyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Clofop
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Cyhalofop-butyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Diclofop-methyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Fenoxaprop-ethyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Fenthiaprop

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Fluazifop-butyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Haloxypop-methyl
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Isoxapyrifop
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Metamifop
Inhibition of Acetyl CoA Carboxylase	Aryloxyphenoxy-propionates (FOPs)	Quizalofop-ethyl
Inhibition of Acetyl CoA Carboxylase	Phenylpyrazoline	Pinoxaden
Inhibition of Acetolactate Synthase	Pyrimidinyl benzoates	Bispyribac-sodium
Inhibition of Acetolactate Synthase	Pyrimidinyl benzoates	Pyribenzoxim (prodrug of bispyribac)
Inhibition of Acetolactate Synthase	Pyrimidinyl benzoates	Pyrifitalid
Inhibition of Acetolactate Synthase	Pyrimidinyl benzoates	Pyriminobac-methyl
Inhibition of Acetolactate Synthase	Pyrimidinyl benzoates	Pyriothiobac-sodium
Inhibition of Acetolactate Synthase	Sulfonanilides	Pyrimisulfan
Inhibition of Acetolactate Synthase	Sulfonanilides	Triafamone
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 1	Cloransulam-methyl
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 1	Diclosulam
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 1	Florasulam
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 1	Flumetsulam
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 1	Metosulam
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 2	Penoxsulam
Inhibition of Acetolactate Synthase	Triazolopyrimidine - Type 2	Pyroxsulam
Inhibition of Acetolactate Synthase	Sulfonylureas	Amidosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Azimsulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Bensulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Chlorimuron-ethyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Chlorsulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Cinosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Cyclosulfamuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Ethametsulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Ethoxysulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Flazasulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Flucetosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Flupyralsulfuron-methyl-Na
Inhibition of Acetolactate Synthase	Sulfonylureas	Foramsulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Halosulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Imazosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Iodosulfuron-methyl-Na
Inhibition of Acetolactate Synthase	Sulfonylureas	Mesosulfuron-methyl

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Acetolactate Synthase	Sulfonylureas	Metazosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Metsulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Nicosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Orthosulfamuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Oxasulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Primisulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Propyrisulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Prosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Pyrazosulfuron-ethyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Rimsulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Sulfometuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Sulfosulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Triasulfuron
Inhibition of Acetolactate Synthase	Sulfonylureas	Tribenuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Thifensulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Trifloxysulfuron-Na
Inhibition of Acetolactate Synthase	Sulfonylureas	Triflusulfuron-methyl
Inhibition of Acetolactate Synthase	Sulfonylureas	Tritosulfuron
Inhibition of Acetolactate Synthase	Imidazolinones	Imazamethabenz-methyl
Inhibition of Acetolactate Synthase	Imidazolinones	Imazamox
Inhibition of Acetolactate Synthase	Imidazolinones	Imazapic
Inhibition of Acetolactate Synthase	Imidazolinones	Imazapyr
Inhibition of Acetolactate Synthase	Imidazolinones	Imazaquin
Inhibition of Acetolactate Synthase	Imidazolinones	Imazethapyr
Inhibition of Acetolactate Synthase	Triazolinones	Flucarbazone-Na
Inhibition of Acetolactate Synthase	Triazolinones	Propoxycarbazone-Na
Inhibition of Acetolactate Synthase	Triazolinones	Thiencarbazone-methyl
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Atraton
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Atrazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Ametryne
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Aziprotryne = aziprotryn
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Chlorazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	CP 17029
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Cyanazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Cyprazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Desmetryne
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Dimethametryn

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Dipropetryn
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Eglinazine-ethyl
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Ipazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Methoprotryne = methoprotryn
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	procyzazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Proglinazine-ethyl
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Prometon
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Prometryne
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Propazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Sebuthylazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Secbumeton
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Simetryne
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Simazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Terbumeton
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Terbuthylazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Terbutryne
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazines	Trietazine
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazolinone	Amicarbazone
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazinones	Ethiozin
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazinones	Hexazinone
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazinones	Isomethiozin
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazinones	Metamitron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Triazinones	Metribuzin
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Uracils	Bromacil
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Uracils	Isocil
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Uracils	Lenacil
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Uracils	Terbacil
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Phenylcarbamates	Chlorprocarb
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Phenylcarbamates	Desmedipham
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Phenylcarbamates	Phenisopham
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Phenylcarbamates	Phenmedipham
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Pyridazinone	Chloridazon (=pyrazon)
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Pyridazinone	Brompyrazon
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Benzthiazuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Bromuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Buturon
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Chlorbromuron

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Chlorotoluron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Chloroxuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Difenoxuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Dimefuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Diuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Ethidimuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Fenuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Fluometuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Fluothiuuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Isoproturon
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Isouron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Linuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Metobenzuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Metobromuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Methabenzthiazuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Metoxuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Monolinuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Monuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Neburon
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Parafluron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Siduron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Tebuthiuron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Ureas	Thiazafluron
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Amides	Chloranocryl = dicryl
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Amides	Pentanochlror
Inhibition of Photosynthesis at PSII - Serine 264 Binders	Amides	Propanil
Inhibition of Photosynthesis at PSII - Histidine 215 Binders	Nitriles	Bromofenoxim
Inhibition of Photosynthesis at PSII - Histidine 215 Binders	Nitriles	Bromoxynil
Inhibition of Photosynthesis at PSII - Histidine 215 Binders	Nitriles	Ioxynil
Inhibition of Photosynthesis at PSII - Histidine 215 Binders	Phenyl-pyridazines	Pyridate
Inhibition of Photosynthesis at PSII - Histidine 215 Binders	Benzothiadiazinone	Bentazon
PS I Electron Diversion	Pyridiniums	Cyperquat
PS I Electron Diversion	Pyridiniums	Diquat
PS I Electron Diversion	Pyridiniums	Morfamquat
PS I Electron Diversion	Pyridiniums	Paraquat
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Lactofen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Acifluorfen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Bifenox
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Chlornitrofen

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Fomesafen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Fluorodifen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Fluoroglycofen-ethyl
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Fluoronitrofen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Nitrofen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Oxyfluorfen
Inhibition of Protoporphyrinogen Oxidase	Diphenyl ethers	Chlomethoxyfen
Inhibition of Protoporphyrinogen Oxidase	Phenylpyrazoles	Pyraflufen-ethyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-oxadiazolones	Oxadiazyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-oxadiazolones	Oxadiazon
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-triazolinones	Azafenidin
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-triazolinones	Carfentrazone-ethyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-triazolinones	Sulfentrazone
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides (procide active form)	Fluthiacet-methyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Butafenacil
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Saflufenacil
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Pentoxazone
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Chlorphthalim
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Cinidon-ethyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Flumiclorac-pentyl
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Flumioxazin
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Flumipropyn
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Trifludimoxazin
Inhibition of Protoporphyrinogen Oxidase	N-Phenyl-imides	Tiafenacil
Inhibition of Protoporphyrinogen Oxidase	Other	Pyraclonil
Inhibition of Phytoene Desaturase	Phenyl ethers	Beflubutamid
Inhibition of Phytoene Desaturase	Phenyl ethers	Diflufenican
Inhibition of Phytoene Desaturase	Phenyl ethers	Picolinafen
Inhibition of Phytoene Desaturase	N-Phenyl heterocycles	Flurochloridone
Inhibition of Phytoene Desaturase	N-Phenyl heterocycles	Norflurazon
Inhibition of Phytoene Desaturase	Diphenyl heterocycles	Fluridone
Inhibition of Phytoene Desaturase	Diphenyl heterocycles	Flurtamone
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Mesotrione
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Sulcotrione
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Tembotrione
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Tefuryltrione
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Bicyclopyrone

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones	Fenquinotriene
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Triketones (procide)	Benzobicyclon
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles (procide)	Benzofenap
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles	Pyrasulfotole
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles	Topramezone
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles (procide)	Pyrazolynate
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles (procide)	Pyrazoxyfen
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Pyrazoles	Tolpyralate
Inhibition of Hydroxyphenyl Pyruvate Dioxygenase	Isoxazoles	Isoxaflutole
Inhibition of Homogentisate Solanesyltransferase	Phenoxy pyridazine	Cyclopyrimorate
Inhibition of Deoxy-D-Xyulose Phosphate Synthase	Isoxazolidinone	Clomazone
Inhibition of Deoxy-D-Xyulose Phosphate Synthase	Isoxazolidinone	Bixlozone
Inhibition of Enolpyruvyl Shikimate Phosphate Synthase	Glycine	Glyphosate
Inhibition of Glutamine Synthetase	Phosphinic acids	Glufosinate-ammonium
Inhibition of Glutamine Synthetase	Phosphinic acids	Bialaphos/bilanafos
Inhibition of Dihydropteroate Synthase	Carbamate	Asulam
Inhibition of Microtubule Assembly	Dinitroanilines	Benefin = benfluralin
Inhibition of Microtubule Assembly	Dinitroanilines	Butralin
Inhibition of Microtubule Assembly	Dinitroanilines	Dinitramine
Inhibition of Microtubule Assembly	Dinitroanilines	Ethalfuralin
Inhibition of Microtubule Assembly	Dinitroanilines	Fluchloralin
Inhibition of Microtubule Assembly	Dinitroanilines	Isopropalin
Inhibition of Microtubule Assembly	Dinitroanilines	Nitralin
Inhibition of Microtubule Assembly	Dinitroanilines	Prodimamine
Inhibition of Microtubule Assembly	Dinitroanilines	Profluralin
Inhibition of Microtubule Assembly	Dinitroanilines	Oryzalin
Inhibition of Microtubule Assembly	Dinitroanilines	Pendimethalin
Inhibition of Microtubule Assembly	Dinitroanilines	Trifluralin
Inhibition of Microtubule Assembly	Pyridines	Dithiopyr
Inhibition of Microtubule Assembly	Pyridines	Thiazopyr
Inhibition of Microtubule Assembly	Phosphoroamidates	Butamifos
Inhibition of Microtubule Assembly	Phosphoroamidates	DMPA
Inhibition of Microtubule Assembly	Benzoic acid	Chlorthal-dimethyl = DCPA
Inhibition of Microtubule Assembly	Benzamides	Propyzamide = pronamide
Inhibition of Microtubule Organization	Carbamates	Barban
Inhibition of Microtubule Organization	Carbamates	Carbetamide

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Microtubule Organization	Carbamates	Chlorbufam
Inhibition of Microtubule Organization	Carbamates	Chlorpropham
Inhibition of Microtubule Organization	Carbamates	Propham
Inhibition of Microtubule Organization	Carbamates	Swep
Inhibition of Cellulose Synthesis	Triazolocarboxamide	Flupoxam
Inhibition of Cellulose Synthesis	Benzamides	Isoxaben
Inhibition of Cellulose Synthesis	Alkylazines	Triaziflam
Inhibition of Cellulose Synthesis	Alkylazines	Indaziflam
Inhibition of Cellulose Synthesis	Nitriles	Dichlobenil
Inhibition of Cellulose Synthesis	Nitriles	Chlorthiamid
Uncouplers	Dinitrophenols	Dinosam
Uncouplers	Dinitrophenols	Dinoseb
Uncouplers	Dinitrophenols	DNOC
Uncouplers	Dinitrophenols	Dinoterb
Uncouplers	Dinitrophenols	Etinofen
Uncouplers	Dinitrophenols	Medinoterb
Inhibition of Very Long-Chain Fatty Acid Synthesis	Azoly-carboxamides	Cafenstrole
Inhibition of Very Long-Chain Fatty Acid Synthesis	Azoly-carboxamides	Fentrazamide
Inhibition of Very Long-Chain Fatty Acid Synthesis	Azoly-carboxamides	Ipfencarbazone
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Thioacetamides	Anilofos
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Thioacetamides	Piperophos
Inhibition of Very Long-Chain Fatty Acid Synthesis	Isoxazolines	Pyroxasulfone
Inhibition of Very Long-Chain Fatty Acid Synthesis	Isoxazolines	Fenoxasulfone
Inhibition of Very Long-Chain Fatty Acid Synthesis	Oxiranes	Indanofan
Inhibition of Very Long-Chain Fatty Acid Synthesis	Oxiranes	Tridiphane
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Acetochlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Alachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Allidochlor = CDAA
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Butachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Butenachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Delachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Diethatyl-ethyl
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Dimethachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Dimethenamid
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Metazachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Metolachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Pethoxamid
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Pretilachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Propachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Propisochlor

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Prynachlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Chloroacetamides	Thenylchlor
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Oxyacetamides	Mefenacet
Inhibition of Very Long-Chain Fatty Acid Synthesis	α -Oxyacetamides	Flufenacet
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Butylate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Cycloate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Dimepiperate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	EPTC
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Esprocarb
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Molinate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Orbencarb
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Pebulate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Prosulfocarb
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Thiobencarb (=Benthio carb)
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Tiocarbazil
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Tri-allate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Thiocarbamates	Vernolate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Benzo furans	Benfuresate
Inhibition of Very Long-Chain Fatty Acid Synthesis	Benzo furans	Ethofumesate
Auxin Mimics	Pyridine-carboxylates	Picloram
Auxin Mimics	Pyridine-carboxylates	Clopyralid
Auxin Mimics	Pyridine-carboxylates	Aminopyralid
Auxin Mimics	Pyridine-carboxylates	Halauxifen
Auxin Mimics	Pyridine-carboxylates	Florpyrauxifen
Auxin Mimics	Pyridyloxy-carboxylates	Triclopyr
Auxin Mimics	Pyridyloxy-carboxylates	Fluroxypyr
Auxin Mimics	Phenoxy-carboxylates	2,4,5-T
Auxin Mimics	Phenoxy-carboxylates	2,4-D
Auxin Mimics	Phenoxy-carboxylates	2,4-DB
Auxin Mimics	Phenoxy-carboxylates	Clomeprop
Auxin Mimics	Phenoxy-carboxylates	Dichlorprop
Auxin Mimics	Phenoxy-carboxylates	Fenoprop
Auxin Mimics	Phenoxy-carboxylates	Mecoprop
Auxin Mimics	Phenoxy-carboxylates	MCPA
Auxin Mimics	Phenoxy-carboxylates	MCPB
Auxin Mimics	Benzoates	Dicamba
Auxin Mimics	Benzoates	Chloramben
Auxin Mimics	Benzoates	TBA
Auxin Mimics	Quinoline-carboxylates	Quinclorac
Auxin Mimics	Quinoline-carboxylates	Quinmerac
Auxin Mimics	Pyrimidine-carboxylates	Aminocyclopyrachlor
Auxin Mimics	Other	Benazolin-ethyl
Auxin Mimics	Phenyl carboxylates	Chlorfenac = fenac
Auxin Mimics	Phenyl carboxylates	Chlorfenprop
Auxin Transport Inhibitor	Aryl-carboxylates	Naptalam
Auxin Transport Inhibitor	Aryl-carboxylates	Diflufenzopyr-sodium
Inhibition of Fatty Acid Thioesterase	Benzyl ether	Cinnethylin
Inhibition of Fatty Acid Thioesterase	Benzyl ether	Methiozolin
Inhibition of Serine-Threonine Protein Phosphatase	Other	Endothal
Inhibition of Solanesyl Diphosphate Synthase	Diphenyl ether	Aclonifen

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MODE OF ACTION	CHEMICAL CLASSIFICATION	ACTIVE
Inhibition of Lycopene Cyclase	Triazole	Amitrole
Unknown		Bromobutide
Unknown		Cumyluron
Unknown		Difenzoquat
Unknown		DSMA
Unknown		Dymron = Daimuron
Unknown		Etobenzanid
Unknown	Arylamino propionic acid	Flamprop-m
Unknown		Fosamine
Unknown		Methyldymron
Unknown		Monalide
Unknown		MSMA
Unknown		Oleic acid
Unknown		Oxaziclomefone
Unknown		Pelargonic acid
Unknown		Pyributicarb
Unknown		Quinoclamine
Unknown	Acetamides	Diphenamid
Unknown	Acetamides	Naproanilide
Unknown	Acetamides	Napropamide
Unknown	Benzamide	Tebutam
Unknown	Phosphorodithioate	Bensulide
Unknown	Chlorocarbonic acids	Dalapon
Unknown	Chlorocarbonic acids	Flupropanate
Unknown	Chlorocarbonic acids	TCA
Unknown	Trifluoromethanesulfonanilides	Mefluidide
Unknown	Trifluoromethanesulfonanilides	Perfluidone
Unknown		CAMA
Unknown		Cacodylic acid

Appendix 3

[0199] Shown below is Appendix 3.

Main Group and Primary Site of Action	Sub-group, class or exemplifying Active Ingredient	Active Ingredients
1 Acetylcholinesterase (AChE) inhibitors Nerve action {Strong evidence that action at this protein is responsible for insecticidal effects}	1A Carbamates	Alanycarb, Aldicarb, Bendiocarb, Benfuracarb, Butocarboxim, Butoxycarboxim, Carbaryl, Carbofuran, Carbosulfan, Ethiofencarb, Fenobucarb, Fommetanate, Furathiocarb, Isoprocab, Methiocarb, Methomyl, Metolcarb, Oxamyl, Pirimicarb, Propoxur, Thiodicarb, Thiofanox, Triazamate, Trimethacarb, XMC, Xylcarb
	1B Organophosphates	Acephate, Azamethiphos, Azinphos-ethyl, Azinphos-methyl, Cadusafos, Chlorethoxyfos, Chlorfenvinphos, Chlormephos, Chlorpyrifos, Chlorpyrifos-methyl, Coumaphos, Cyanophos, Demeton-S-methyl, Diazinon, Dichlorvos/ DDVP, Dicrotophos, Dimethoate, Dimethylvinphos, Disulfoton, EPN, Ethion, Ethoprophos, Famphur, Fenamiphos, Fenitrothion, Fenthion, Fosthiazate, Heptenophos, Imicyafos, Isofenphos, Isopropyl O-(methoxyaminothio- phosphoryl) salicylate, Isoxathion, Malathion, Mecarbam, Methamidophos, Methidathion, Mevinphos, Monocrotophos, Naled, Omethoate, Oxydemeton-methyl, Parathion, Parathion-methyl, Phenthoate, Phorate, Phosalone, Phosmet, Phosphamidon, Phoxim, Pirimiphos- methyl, Profenofos, Propetamphos, Prothiofos, Pyraclofos, Pyridaphenthion, Quinalphos, Sulfotep, Tebupirimfos, Temephos, Terbufos, Tetrachlorvinphos, Thiometon, Triazophos, Trichlorfon, Vamidothion
2 GABA-gated chloride channel blockers Nerve action {Strong evidence that action at this protein is responsible for insecticidal effects}	2A Cyclodiene Organochlorines	Chlordane, Endosulfan
	2B Phenylpyrazoles (Fiproles)	Ethiprole, Fipronil
3 Sodium channel modulators Nerve action	3A Pyrethroids Pyrethrins	Acrinathrin, Allethrin, d-cis-trans Allethrin, d- trans Allethrin, Bifenthrin, Bioallethrin, Bioallethrin S-cyclopentenyl isomer, Bioresmethrin, Cycloprothrin, Cyfluthrin, beta- Cyfluthrin,

-continued

Main Group and Primary Site of Action	Sub-group, class or exemplifying Active Ingredient	Active Ingredients
{Strong evidence that action at this protein is responsible for insecticidal effects}		Cyhalothrin, lambda-Cyhalothrin, gamma-Cyhalothrin, Cypermethrin, alpha- Cypermethrin, beta-Cypermethrin, theta- cypermethrin, zeta-Cypermethrin, Cyphenothrin, (1R)-trans- isomers], Deltamethrin, Empenthrin (EZ)- (1R)- isomers], Esfenvalerate, Etofenprox, Fenpropathrin, Fenvalerate, Flucythrinate, Flumethrin, tau-Fluvalinate, Halfenprox, Imiprothrin, Kadethrin, Permethrin, Phenothrin [(1R)-trans- isomer], Prallethrin, Pyrethrins (pyrethrum), Resmethrin, Silafluofen, Tefluthrin, Tetramethrin, Tetramethrin [(1R)-isomers], Tralomethrin, Transfluthrin, DDT
	3B	Methoxychlor
4	DDT	
Nicotinic acetylcholine receptor (nAChR) competitive modulators	4A	Acetamiprid, Clothianidin, Dinotefuran, Imidacloprid, Nitenpyram, Thiachloprid, Thiamethoxam, Nicotine
Nerve action	4B	
{Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects}	Nicotine	
	4C	Sulfoxaflo
	Sulfoximines	
	4D	Flupyradifurone
	Butenolides	
	4E	Triflumezopyrim
	Mesoionics	
	4F	Flupyrimin
	Pyridylidenes	
5	Spinosyns	Spinetoram, Spinosad
Nicotinic acetylcholine receptor (nAChR) allosteric modulators - Site I		
Nerve action		
{Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects}		
6	Avermectins, Milbemycins	Abamectin, Emamectin benzoate, Lepimectin, Milbemectin
Glutamate-gated chloride channel (GluCl) allosteric modulators		
Nerve and muscle action		
{Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects}		
7	7A	Hydroprene, Kinoprene, Methoprene
Juvenile hormone mimics	Juvenile hormone analogues	
Growth regulation	7B	Fenoxycarb
{Target protein responsible for biological activity is unknown, or uncharacterized}	Fenoxycarb	
	7C	Pyriproxyfen
	Pyriproxyfen	
8 *	8A	Methyl bromide and other alkyl halides
Miscellaneous non-specific (multi-site) inhibitors	Alkyl halides	
	8B	Chloropicrin
	Chloropicrin	
	8C	Cryolite (Sodium aluminum fluoride), Sulfuryl fluoride
	Fluorides	
	8D	Borax, Boric acid, Disodium octaborate, Sodium borate, Sodium metaborate
	Borates	
	8E	Tartar emetic
	Tartar emetic	
	8F	Dazomet, Metam
	Methyl isothiocyanate generators	
9	9B	Pymetrozine, Pyrifluquinazon
Chordotonal organ TRPV channel modulators Nerve action	Pyridine azomethine derivatives	
{Strong evidence that action at one or more of this class of proteins is responsible for insecticidal effects}	9D	Afidopyropen
	Pyropenes	
10	10A	Clofentezine, Diflovidazin, Hexythiazox
Mite growth inhibitors affecting CHS1	Clofentezine Diflovidazin	
Growth regulation	Hexythiazox	
	10B	Etoxazole

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Main Group and Primary Site of Action	Sub-group, class or exemplifying Active Ingredient	Active Ingredients
{Strong evidence that action at one or more of this class of proteins is responsible for insecticidal effects}	Etoxazole	
11 Microbial disruptors of insect midgut membranes (Includes transgenic crops expressing <i>Bacillus thuringiensis</i> toxins, however specific guidance for resistance management of transgenic crops is not based on rotation of modes of action)	11A <i>Bacillus thuringiensis</i> and the insecticidal proteins they produce	<i>Bacillus thuringiensis</i> subsp. <i>israelensis</i> <i>Bacillus thuringiensis</i> subsp. <i>aizawai</i> <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> <i>Bacillus thuringiensis</i> subsp. <i>tenebrionis</i> B.t. crop proteins: (* Please see footnote) Cry1Ab, Cry1Ac, Cry1Fa, Cry1A.105, Cry2Ab, Vip3A, mCry3A, Cry3Ab, Cry3Bb, Cry34Ab1/Cry35Ab1
	11B <i>Bacillus sphaericus</i>	<i>Bacillus sphaericus</i>
12 Inhibitors of mitochondrial ATP synthase Energy metabolism {Compounds affect the function of this protein, but it is not clear that this is what leads to biological activity}	12A Diafenthiuron	Diafenthiuron
	12B Organotin miticides	Azocyclotin, Cyhexatin, Fenbutatin oxide
	12C Propargite	Propargite
	12D Tetradifon	Tetradifon
13 * Uncouplers of oxidative phosphorylation via disruption of the proton gradient Energy metabolism	Pyrroles Dinitrophenols Sulfluramid	Chlorfenapyr DNOC Sulfluramid
14 Nicotinic acetylcholine receptor (nAChR) channel blockers Nerve action {Compounds affect the function of this protein, but it is not clear that this is what leads to biological activity}	Nereistoxin analogues	Bensultap, Cartap hydrochloride, Thiocyclam, Thiosultap-sodium
15 Inhibitors of chitin biosynthesis affecting CHS1 Growth regulation {Strong evidence that action at one or more of this class of proteins is responsible for insecticidal effects}	Benzoylureas	Bistrifluron, Chlorfluazuron, Diflubenzuron, Flucycloxuron, Flufenoxuron, Hexaflumuron, Lufenuron, Novaluron, Noviflumuron, Teflubenzuron, Triflumuron
16 Inhibitors of chitin biosynthesis, type 1 Growth regulation {Target protein responsible for biological activity is unknown, or uncharacterized}	Buprofezin	Buprofezin
17 Moulting disruptors, Dipteran Growth regulation {Target protein responsible for biological activity is unknown, or uncharacterized}	Cyromazine	Cyromazine
18 Ecdysone receptor agonists Growth regulation {Strong evidence that action at this protein is responsible for insecticidal effects}	Diacylhydrazines	Chromafenozide, Halofenozide, Methoxyfenozide, Tebufenozide
19 Octopamine receptor agonists Nerve action {Good evidence that action at one or more of this class of protein is responsible for insecticidal effects}	Amitraz	Amitraz
20 Mitochondrial complex III electron transport inhibitors - Qo site Energy metabolism {Good evidence that action at this protein complex is responsible for insecticidal effects}	20A Hydramethylnon	Hydramethylnon
	20B Acequinocyl	Acequinocyl
	20C Fluacrypyrim	Fluacrypyrim
	20D Bifenazate	Bifenazate

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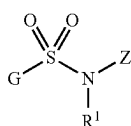
Main Group and Primary Site of Action	Sub-group, class or exemplifying Active Ingredient	Active Ingredients
21 Mitochondrial complex I electron transport inhibitors Energy metabolism {Good evidence that action at this protein complex is responsible for insecticidal effects}	21A METI acaricides and insecticides 21B Rotenone	Fenazaquin, Fenpyroximate, Pyridaben, Pyrimidifen, Tebufenpyrad, Tolfenpyrad Rotenone (Derris)
22 Voltage-dependent sodium channel blockers Nerve action {Good evidence that action at this protein complex is responsible for insecticidal effects}	22A Oxadiazines 22B Semicarbazones	Indoxacarb Metaflumizone
23 Inhibitors of acetyl CoA carboxylase Lipid synthesis, growth regulation {Good evidence that action at this protein is responsible for insecticidal effects}	Tetronic and Tetramic acid derivatives	Spirodiclofen, Spiromesifen, Spiropidion, Spirotetramat
24 Mitochondrial complex IV electron transport inhibitors Energy metabolism {Good evidence that action at this protein complex is responsible for insecticidal effects}	24A Phosphides 24B Cyanides	Aluminium phosphide, Calcium phosphide, Phosphine, Zinc phosphide Calcium cyanide, Potassium cyanide, Sodium cyanide
25 Mitochondrial complex II electron transport inhibitors Energy metabolism {Good evidence that action at this protein complex is responsible for insecticidal effects}	25A Beta-ketonitrile derivatives 25B Carboxanilides	Cyfenopirafen, Cyflumetofen Pyflubumide
28 Ryanodine receptor modulators Nerve and muscle action {Strong evidence that action at this protein complex is responsible for insecticidal effects}	Diamides	Chlorantraniliprole, Cyantraniliprole, Cyclaniliprole Flubendiamide, Tetraniliprole
29 Chordotonal organ modulators - undefined target site Nerve action (Modulation of chordotonal organ function has been clearly demonstrated, but the specific target protein(s) responsible for biological activity are distinct from Group 9 and remain undefined)	Flonicamid	Flonicamid
30 GABA-gated chloride channel allosteric modulators Nerve action {Strong evidence that action at this protein complex is responsible for insecticidal effects}	Meta-diamides Isoxazolines	Broflanilide Fluxametamide, Isocycloseram
31 Baculoviruses Host-specific occluded pathogenic viruses (Midgut epithelial columnar cell membrane target site - undefined)	Granuloviruses (GVs) Nucleopolyhedroviruses (NPVs)	<i>Cydia pomonella</i> GV <i>Thaumetobia leucotreta</i> GV <i>Anticarsia gemmatilis</i> MNPV <i>Helicoverpa armigera</i> NPV
32 Nicotinic Acetylcholine Receptor (nAChR) Allosteric Modulators - Site II Nerve action {Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects}	GS-omega/kappa HXTX-Hv1a peptide	GS-omega/kappa HXTX-Hv1a peptide

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Main Group and Primary Site of Action	Sub-group, class or exemplifying Active Ingredient	Active Ingredients
33 Calcium-activated potassium channel (KCa2) modulators Nerve action {Strong evidence that action at this protein is responsible for insecticidal effects}	Acynonapyr	Acynonapyr
34 Mitochondrial complex III electron transport inhibitors - Qi site Energy metabolism {Modulation of this protein complex has been clearly demonstrated and the specific target site responsible for biological activity is distinct from Group 20}	Flometoquin	Flometoquin
UN* Compounds of unknown or uncertain MoA {Target protein responsible for biological activity is unknown, or uncharacterized}	Azadirachtin Benzoximate Benzpyrimoxan Bromopropylate Chinomethionat Dicofol Lime sulfur Mancozeb Pyridalyl Sulfur	Azadirachtin Benzoximate Benzpyrimoxan Bromopropylate Chinomethionat Dicofol Lime sulfur Mancozeb Pyridalyl Sulfur <i>Burkholderia</i> spp <i>Wolbachia pipientis</i> (Zap)
UNB* Bacterial agents (non-Bt) of unknown or uncertain MoA {Target protein responsible for biological activity is unknown or uncharacterized}		<i>Chenopodium ambrosioides</i> near <i>ambrosioides</i> extract Fatty acid monoesters with glycerol or propanediol Neem oil
UNE* Botanical essence including synthetic, extracts and unrefined oils with unknown or uncertain MoA {Target protein responsible for biological activity is unknown, or uncharacterized}		<i>Beauveria bassiana</i> strains <i>Metarhizium anisopliae</i> strain F52 <i>Paecilomyces fumosoroseus</i> Apopka strain 97
UNF* Fungal agents of unknown or uncertain MoA {Target protein responsible for biological activity is unknown, or uncharacterized}		Diatomaceous earth Mineral oil
UNM* Non-specific mechanical and physical disruptors {Target protein responsible for biological activity is unknown, or uncharacterized}		

[0200] Notwithstanding the appended claims, aspects of the present disclosure are further provided in the following numbered clauses:

[0201] 1. A method for inhibiting apyrase, comprising contacting the apyrase with a compound of formula (I):



(I)

[0202] wherein:

[0203] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

[0204] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl,

bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0205] R¹ is hydrogen or C₁₋₆ alkyl;

[0206] or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclylalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0207] each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅-C₁₀ aryl, C₆-C₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

[0208] each R^a is optionally substituted with one or more groups selected from R^b and R^c;

[0209] each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, =NR^d, =NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂,

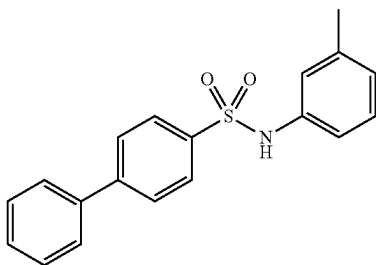
—N₃, —S(O)R^d, —S(O)₂R^d, —S(O)₂OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)NR^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R^a, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^cC(O)]_nR^d, [NHC(O)]_nOR^d, —[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;

[0210] each R^c is independently hydrogen, R³, or, alternatively, two R^e are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclylalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^e;

[0211] each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl;

[0212] each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d;

[0213] provided that the compound does not have the formula

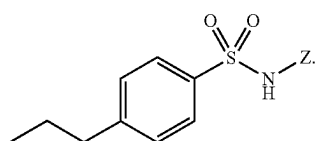


[0214] 2. The method of clause 1, wherein G comprises a six-membered aryl or six-membered heteroaryl ring.

[0215] 3. The method of clause 2, wherein G is an optionally substituted phenyl.

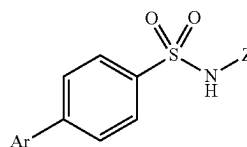
[0216] 4. The method of clause 3, wherein G is substituted phenyl.

[0217] 5. The method of clause 4, wherein the compound has formula (Ia):



(Ia)

[0218] 6. The method of clause 4, wherein the compound has formula (Ib):



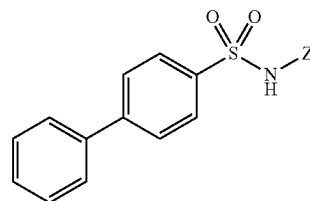
(Ib)

[0219] wherein Ar is optionally substituted aryl or heteroaryl.

[0220] 7. The method of clause 6, wherein Ar in formula (Ib) is unsubstituted, substituted with an R^a group of C₁₋₆ alkyl, or substituted with an R^b group of halogen.

[0221] 8. The method of clause 2, wherein G is an optionally substituted biphenyl group.

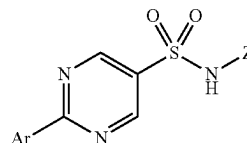
[0222] 9. The method of clause 8, wherein the compound has the formula (Ib1):



(Ib1)

[0223] 10. The method of clause 1, wherein G is optionally substituted pyrimidinyl.

[0224] 11. The method of clause 10, wherein the compound has formula (Ic):



(Ic)

[0225] wherein Ar is optionally substituted aryl or heteroaryl.

[0226] 12. The method of clause 11, wherein Ar in formula (Ic) is unsubstituted, substituted with an R^a group of C₁₋₆ alkyl, or substituted with an R^b group of halogen.

[0227] 13. The method of any one of clauses 1-12, wherein G and its substituents have 6 to 20 non-hydrogen atoms.

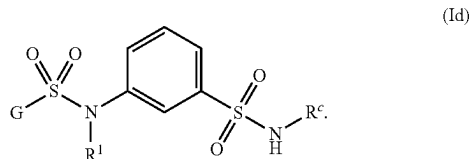
[0228] 14. The method of any one of clauses 1-13, wherein Z is substituted phenyl.

[0229] 15. The method of clause 14, wherein Z is phenyl substituted with an optionally substituted aryl or heteroaryl group.

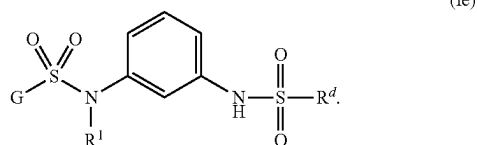
[0230] 16. The method of clause 15, wherein Z is phenyl substituted with an optionally substituted alkyl group.

[0231] 17. The method of clause 16, wherein Z is phenyl substituted with an R^b group.

[0232] 18. The method of clause 17, wherein the compound has the formula (Id):



[0233] 19. The method of clause 17, wherein the compound has the formula (Ie):



[0234] 20. The method of clause 17, wherein Z is phenyl substituted with an R^b group of halogen or $-\text{CF}_3$.

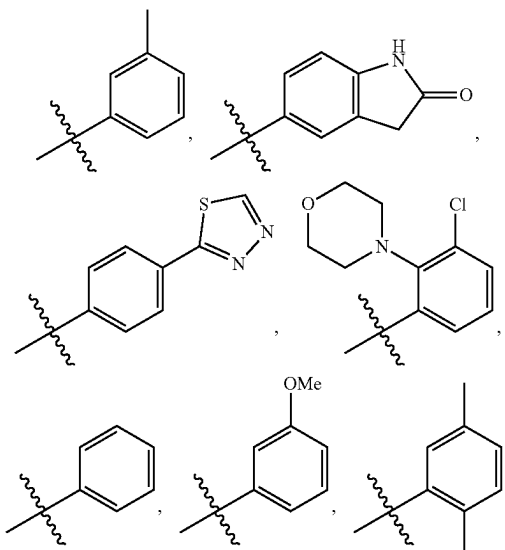
[0235] 21. The method of any one of clauses 1-13, wherein Z is optionally substituted monocyclic heteroaryl.

[0236] 22. The method of clause 21, wherein Z is optionally substituted pyridyl.

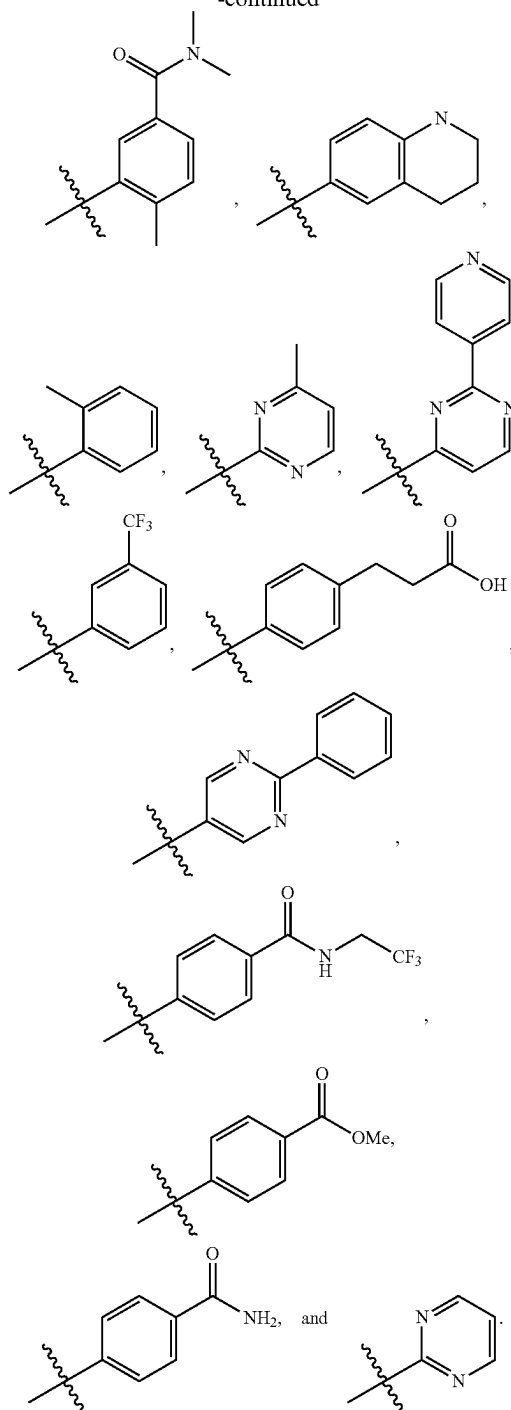
[0237] 23. The method of clause 21, wherein Z is optionally substituted pyrimidyl.

[0238] 24. The method of any one of clauses 1-13, wherein Z and R^1 together form a 5-membered or 6-membered heterocyclalkyl ring.

[0239] 25. The method of any one of clauses 1-13, wherein Z is selected from the group consisting of



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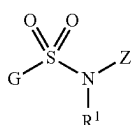


[0240] 26. The method of any one of clauses 1-25, wherein Z and its substituents have 6 to 20 non-hydrogen atoms.

[0241] 27. The method of any one of clauses 1-26, wherein contacting the apyrase comprises treating a crop with the compound.

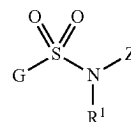
[0242] 28. The method of clause 27, further comprising treating the crop with a pesticide.

- [0243] 29 The method of clause 28, wherein the pesticide is selected from acaricides, fungicides, herbicides, insecticides, molluscicides, nematocides, or a combination thereof.
- [0244] 30. The method of clause 29, wherein the pesticide comprises a fungicide.
- [0245] 31. The method of clause 28, wherein the pesticide comprises an herbicide.
- [0246] 32. The method of clause 31, wherein the herbicide comprises glyphosate.
- [0247] 33. The method of clause 29, further comprising treating the crop with a fungicide selected from selected from benzimidazoles, dicarboximides, phenylpyrroles, anilino pyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors (OSBPI), triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, such as cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof.
- [0248] 34. A composition, comprising
- [0249] a fungicide;
- [0250] a phyto logically acceptable carrier; and
- [0251] a compound of formula (I):



- [0252] wherein:
- [0253] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;
- [0254] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;
- [0255] R¹ is hydrogen or C₁₋₆ alkyl;
- [0256] or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;
- [0257] each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅-C₁₀ aryl, C₆-C₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

- [0258] each R^a is optionally substituted with one or more groups selected from R^b and R^c;
- [0259] each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, =NR^d, =NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^d, —S(O)₂R^d, —S(O)₂OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)N R^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R^d, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^aC(O)]_nR^d, —[NHC(O)]_nOR^d, —[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;
- [0260] each R^c is independently hydrogen, R^a, or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^e;
- [0261] each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl; and
- [0262] each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d.
- [0263] 35. The composition of clause 34, wherein the composition comprises from about 1 to about 80 weight percent of the compound.
- [0264] 36. The composition of clause 35, wherein the composition is a suspension formulation.
- [0265] 37. The composition of clause 36, wherein the composition comprises from about 1 to about 50 weight percent of the compound.
- [0266] 38. The composition of clause 37, further comprising sodium polycarboxylate.
- [0267] 39. The composition of clause 34, further comprising biocides.
- [0268] 40. The composition of clause 35, further comprising organosilicone antifoam emulsion.
- [0269] 41. The composition of clause 38, wherein the composition is a wettable powder.
- [0270] 42. The composition of clause 34, wherein the composition is an emulsifiable concentrate.
- [0271] 43. The composition of clause 40, further comprising tristyrilphenol ethoxylates.
- [0272] 44. The composition of clause 34, wherein the composition is an oil dispersible concentrate.
- [0273] 45. A pesticidal composition, comprising
- [0274] a pesticide;
- [0275] a phyto logically acceptable carrier; and
- [0276] a compound of formula (I):



[0277] wherein:

[0278] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

[0279] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0280] R¹ is hydrogen or C₁₋₆ alkyl;

[0281] or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0282] each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅₋₁₀ aryl, C₆₋₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

[0283] each R^a is optionally substituted with one or more groups selected from R^b and R^e;

[0284] each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, =NR^d, =NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^d, —S(O)₂R^d, —S(O)₂OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)N R^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R^a, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^aC(O)]_nR^d, —[NHC(O)]_nOR^d, —[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;

[0285] each R^c is independently hydrogen, R^a, or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^e;

[0286] each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl; and

[0287] each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d.

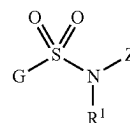
[0288] 46. The pesticidal composition of clause 45, wherein the pesticide comprises an acaricide, fungicide, herbicide, insecticide, molluscicide, nematocide, or a combination thereof.

[0289] 47. A fungicidal composition, comprising

[0290] a fungicide;

[0291] a phylogenically acceptable carrier; and

[0292] a compound of formula (I):



(I)

[0293] wherein:

[0294] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

[0295] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0296] R¹ is hydrogen or C₁₋₆ alkyl;

[0297] or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

[0298] each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅₋₁₀ aryl, C₆₋₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

[0299] each R^a is optionally substituted with one or more groups selected from R^b and R^e;

[0300] each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, =NR^d, =NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^d, —S(O)₂R^d, —S(O)₂OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)N R^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R^a, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^aC(O)]_nR^d, —[NHC(O)]_nOR^d, —[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;

[0301] each R^c is independently hydrogen, R^a, or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^e;

[0302] each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl;

[0303] each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d.

[0304] 48. The composition of clause 47, wherein the fungicide is selected from the group consisting of benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxylanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors, triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation

inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof.

[0305] In view of the many possible embodiments to which the principles of the disclosed invention may be applied, it should be recognized that the illustrated embodiments are only preferred examples of the invention and should not be taken as limiting the scope of the invention. Rather, the scope of the invention is defined by the following claims. We therefore claim as our invention all that comes within the scope and spirit of these claims.

What is claimed is:

1. A method for inhibiting apyrase, comprising contacting the apyrase with a compound of formula (I):



wherein:

G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

R¹ is hydrogen or C₁₋₆ alkyl;

or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclylalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅₋₁₀ aryl, C₆₋₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

each R^a is optionally substituted with one or more groups selected from R^b and R^e;

each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, =NR^d, =NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^d, —S(O)₂R^d, —S(O)₂OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)NR^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R³, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^aC(O)]_nR^d, [NHC(O)]_nOR^d,

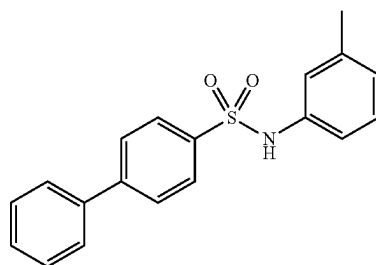
—[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;

each R^c is independently hydrogen, R^a, or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclylalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^e;

each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl;

each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d;

provided that the compound does not have the formula

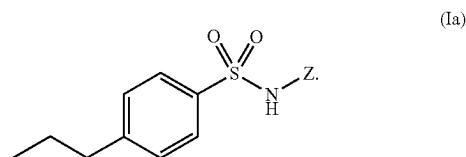


2. The method of claim 1, wherein G comprises a six-membered aryl or six-membered heteroaryl ring.

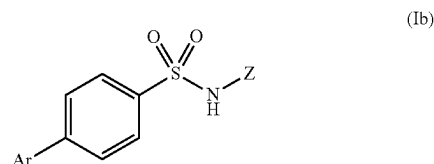
3. The method of claim 2, wherein G is an optionally substituted phenyl.

4. The method of claim 3, wherein G is substituted phenyl.

5. The method of claim 4, wherein the compound has formula (Ia):



6. The method of claim 4, wherein the compound has formula (Ib):

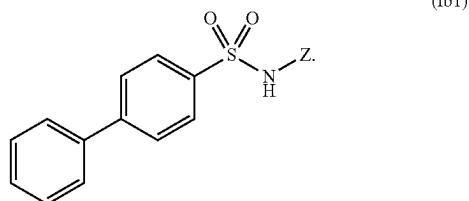


wherein Ar is optionally substituted aryl or heteroaryl.

7. The method of claim 6, wherein Ar in formula (Ib) is unsubstituted, substituted with an R^a group of C₁₋₆ alkyl, or substituted with an R^b group of halogen.

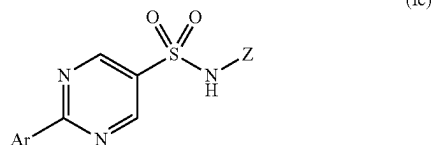
8. The method of claim 2, wherein G is an optionally substituted biphenyl group.

9. The method of claim 8, wherein the compound has the formula (Ib1):



10. The method of claim 1, wherein G is optionally substituted pyrimidinyl.

11. The method of claim 10, wherein the compound has formula (Ic):



wherein Ar is optionally substituted aryl or heteroaryl.

12. The method of claim 11, wherein Ar in formula (Ic) is unsubstituted, substituted with an R^a group of C_{1-6} alkyl, or substituted with an R^b group of halogen.

13. The method of claim 1, wherein G and its substituents have 6 to 20 non-hydrogen atoms.

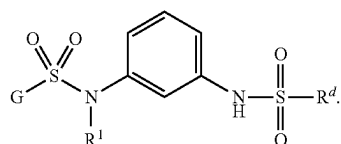
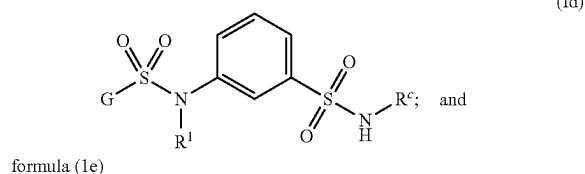
14. The method of claim 1, wherein Z is substituted phenyl.

15. The method of claim 14, wherein Z is phenyl substituted with an optionally substituted aryl or heteroaryl group.

16. The method of claim 15, wherein Z is phenyl substituted with an optionally substituted alkyl group.

17. The method of claim 16, wherein Z is phenyl substituted with an R^b group.

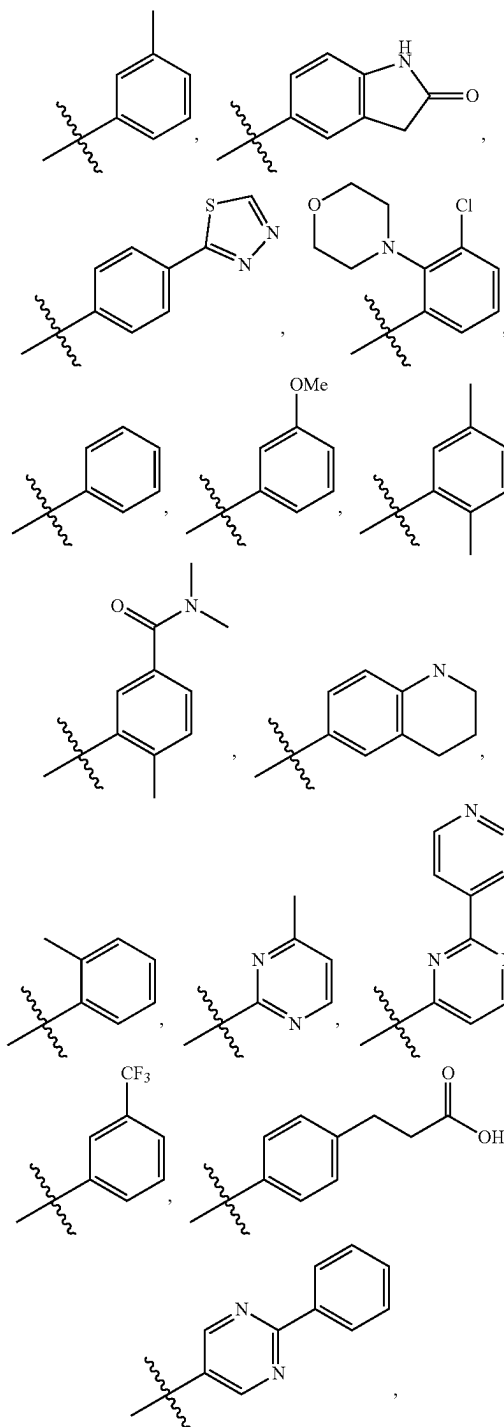
18. The method of claim 17, wherein the compound has a formula selected from formula (Id):

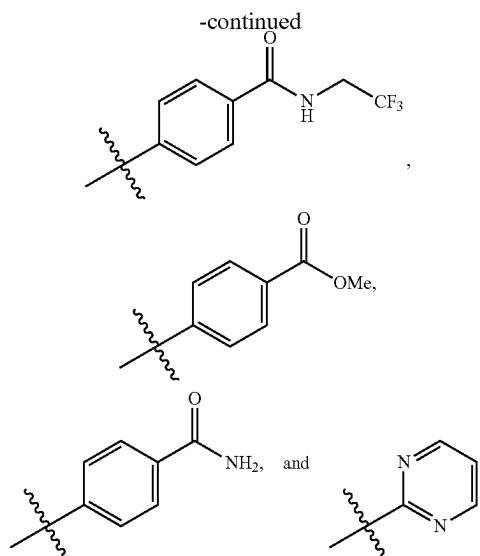


19. The method of claim 17, wherein Z is phenyl substituted with an R^b group that is halogen or $-CF_3$.

20. The method of claim 1, wherein Z is selected from the group consisting of optionally substituted monocyclic heteroaryl; optionally substituted pyridyl; optionally substituted pyrimidyl; and wherein Z and R^1 together form a 5-membered or 6-membered heterocyclylalkyl ring.

21. The method of claim 1, wherein Z is selected from the group consisting of





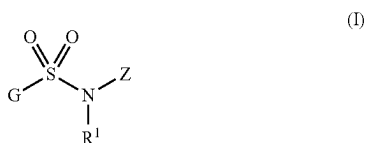
22. The method of claim 1, wherein Z and its substituents have 6 to 20 non-hydrogen atoms.

23. The method of claim 1, wherein contacting the apyrase comprises treating a crop with the compound.

24. The method of claim 23, further comprising treating the crop with a pesticide.

25. The method of claim 24, wherein the pesticide is selected from acaricides, fungicides, herbicides, insecticides, molluscicides, nematocides, or a combination thereof.

26. A pesticidal composition, comprising
a pesticide;
a phytologically acceptable carrier; and
a compound of formula (I):



wherein:

G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

R¹ is hydrogen or C₁₋₆ alkyl;

or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅-C₁₀ aryl, C₆-C₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

each R^a is optionally substituted with one or more groups selected from R^b and R^e;

each R^b is independently selected from the group consisting of =O, —OR^d, —OCF₃, —OCF₂H=S, —SR^d, —NR^d, —NOR^d, —NR^cR^c, halogen, —CF₃, —CF₂H, —CN, —NO₂, —N₃, —S(O)R^d, —S(O)₂R^d, —S(O)OR^d, —S(O)NR^cR^c, —S(O)₂NR^cR^c, —NHS(O)₂R^d, —OS(O)R^d, —OS(O)₂R^d, —OS(O)₂OR^d, —OS(O)₂NR^cR^c, —C(O)R^d, —C(O)OR^d, —C(O)NR^cR^c, —C(NH)NR^cR^c, —C(NR^a)NR^cR^c, —C(NOH)R^a, —C(NOH)NR^cR^c, —OC(O)R^d, —OC(O)OR^d, —OC(O)NR^cR^c, —OC(NH)NR^cR^c, —OC(NR^a)NR^cR^c, —[NHC(O)]_nR^d, —[NR^aC(O)]_nR^d, —[NHC(O)]_nOR^d, —[NR^aC(O)]_nOR^d, —[NHC(O)]_nNR^cR^c, —[NR^aC(O)]_nNR^cR^c, —NHSO₂R^d, —[NHC(NH)]_nNR^cR^c and —[NR^aC(NR^a)]_nNR^cR^c;

each R^c is independently hydrogen, R^a, or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a, =O, halogen, and R^c;

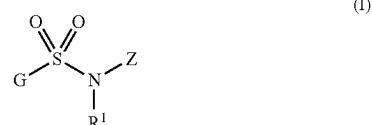
each R^d is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl;

each R^e is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^d, and —C(O)OR^d.

27. The pesticidal composition of claim 26, wherein the pesticide comprises an acaricide, fungicide, herbicide, insecticide, molluscicide, nematocide, or a combination thereof.

28. The pesticidal composition of claim 26, wherein the pesticide comprises an herbicide.

29. A fungicidal composition, comprising
a fungicide;
a phytologically acceptable carrier; and
a compound of formula (I):



wherein:

G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^a and R^b;

Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

R¹ is hydrogen or C₁₋₆ alkyl;

or Z and R¹ together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^a and R^b;

each R^a is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅-C₁₀ aryl, C₆-C₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl;

each R^a is optionally substituted with one or more groups selected from R^b and R^e;

each R^b is independently selected from the group consisting of $=O$, $-OR^d$, $-OCF_3$, $-OCF_2H=S$, $-SR^d$, $=NR^d$, $=NOR^d$, $-NR^cR^c$, halogen, $-CF_3$, $-CF_2H$, $-CN$, $-NO_2$, $=N_2$, $-N_3$, $-S(O)R^d$, $-S(O)_2R^d$, $-S(O)_2OR^d$, $-S(O)NR^cR^c$, $-S(O)_2NR^cR^c$, $-NHS(O)_2R^d$, $-OS(O)R^d$, $-OS(O)_2R^d$, $-OS(O)_2OR^d$, $-OS(O)_2NR^cR^c$, $-C(O)R^d$, $-C(O)OR^d$, $-C(O)NR^cR^c$, $-C(NH)NR^cR^c$, $-C(NR^a)NR^cR^c$, $-C(NOH)R^a$, $-C(NOH)NR^cR^c$, $-OC(O)R^d$, $-OC(O)OR^d$, $-OC(O)NR^cR^c$, $-OC(NH)NR^cR^c$, $-OC(NR^a)NR^cR^c$, $-[NHC(O)]_nR^d$, $-[NR^aC(O)]_nR^d$, $-[NHC(O)]_nOR^d$, $-[NR^aC(O)]_nOR^d$, $-[NHC(O)]_nNR^cR^c$, $-[NR^aC(O)]_nNR^cR^c$, $-NHSO_2R^d$, $-[NHC(NH)]_nNR^cR^c$ and $-[NR^aC(NR^a)]_nNR^cR^c$;

each R^c is independently hydrogen, R^a , or, alternatively, two R^c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^a , $=O$, halogen, and R^c ;

each R^d is independently C_{1-6} alkyl or C_{3-8} cycloalkyl; and

each R^e is independently C_{1-6} alkyl, C_{3-8} cycloalkyl, $-C(O)R^d$, and $-C(O)OR^d$.

30. The composition of claim **29**, wherein the fungicide is selected from the group consisting of benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxylanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors, triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof.

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